DESIGN AND IMPLEMENTATION OF A MACHINE LEARNING-BASED TOOL AND A ROBOTIC PROCESS AUTOMATION WORKFLOW FOR SAP S/4HANA ITEM SETUP

Lappeenranta–Lahti University of Technology LUT

Master's Thesis in Industrial Engineering and Management

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Otto Heinonen

Examiners: Docent, Post-Doctoral Researcher Lasse Metso

Post-Doctoral Researcher Minnu Joseph

ABSTRACT

Lappeenranta–Lahti University of Technology LUT

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Otto Heinonen

Design and Implementation of a Machine Learning-Based Tool and a Robotic Process Automation Workflow for SAP S/4HANA Item Setup

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Keywords: artificial intelligence, machine learning, robotic process automation, SAP S/4HANA, item setup, product data management, intelligent automation

During the study, an intelligent automation solution was developed for SAP S/4HANA item setup. The approach combines machine learning and robotic process automation. The Design Science Research process was used as the method. The objective was to design a feasible architecture and demonstrate its performance relative to the manual baseline.

The core of the solution is a Python-based machine learning application that validates and standardizes the input, predicts fields, and produces an Excel file. The output file serves as an interface to two robots. A user review is required before the robotic stage. The implementation is deliberately modular and file-based. At the time of evaluation, the robot workflows operated in pre-production and passed functional verification. The original objective was to deliver a production-ready robotic solution applying machine learning. Due to time constraints and resource allocation, the scope was refined so that the application was delivered as production-ready and the robotic stage was brought to functional readiness in pre-production. Only large-scale testing remained before moving to production, and the work focused on strengthening version control of the artefacts and the application's installation, user, and update documentation.

The evaluation was conducted in an iterative pilot. Prediction accuracy typically ranged from 96 to 99 percent. Product data quality improved by 6 percent. The robotic stage was measured by straight-through processing, exceptions, and throughput time. When moved to production, the solution replaces manual entry and yields approximately 431 hours of annual savings. The results indicate improvements in efficiency, data quality, and division of labor within the agreed scope. Future work concerns retraining, increasing explainability, full automation, and the use of direct interfaces where possible.

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Koneoppimiseen perustuvan työkalun ja robottiprosessiautomaation työnkulun suunnittelu ja toteutus SAP S/4HANA -nimikkeiden perustamiseen

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Tutkimuksen aikana kehitettiin älykäs automaatio ratkaisu SAP S/4HANA -nimikkeiden perustamiseen. Lähestymistapa yhdistää koneoppimisen ja robottiprosessiautomaation. Menetelmänä käytettiin suunnittelutieteellistä tutkimus -prosessia. Tavoitteena oli suunnitella toteutuskelpoinen arkkitehtuuri ja osoittaa sen suorituskyky suhteessa manuaaliseen lähtötasoon.

Ratkaisun ydin on Python-pohjainen koneoppimis-sovellus, joka validoi ja vakioi syötteen, ennustaa kenttiä ja tuottaa Excel-tiedoston. Tulostiedosto toimii kytkentäpintana kahdelle robotille. Ennen robottivaihetta vaaditaan käyttäjän tarkistus. Toteutus on tietoisesti modulaarinen ja tiedostopohjainen. Arviointihetkellä robotin työnkulut toimivat esituotannossa ja läpäisivät toiminnallisen verifioinnin. Alkuperäinen tavoite oli toimittaa tuotantovalmis koneoppimista soveltava robotti ratkaisu. Aikarajojen ja resurssien kohdennuksen vuoksi laajuutta täsmennettiin siten, että sovellus toimitettiin tuotantokelpoisena ja robottivaihe saatettiin toiminnallisesti valmiiksi esituotantovaiheeseen. Tuotantoon viemistä varten oli enää suoritettava laajamittainen testaus, ja työn painopiste oli artefaktien versionhallinnan sekä sovelluksen asennus-, käyttö- ja päivitysdokumentaation vahvistamisessa.

Arviointi toteutettiin iteratiivisessa pilotissa. Ennusteiden tarkkuus asettui tyypillisesti 96–99 prosentin välille. Tuotedatan laatu parantui noin 6 prosenttia. Robottivaihetta mitattiin suoran läpimenon, poikkeamien ja läpimenoajan perusteella. Tuotantoon siirrettynä ratkaisu korvaa manuaalisen syötön ja tuottaa arviolta noin 431 tuntia vuotuisia säästöjä.

Tulokset osoittavat tehokkuuden, datalaadun ja työnjaon parantumista sovitussa laajuudessa. Jatkokehitys koskee uudelleenkoulutusta, selitettävyyden lisäämistä, täysautomaatiota sekä suorien rajapintojen hyödyntämistä silloin, kun se on mahdollista.

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Abbreviations

AI	Artificial Intelligence
API	Application Programming Interface
BOM	Bill of Materials
BPM	Business Process Management
CNNs	Convolutional Neural Networks
DL	Deep Learning
DSR	Design Science Research
ECO	Engineering Change Orders
ECR	Engineering Change Requests
ERP	Enterprise Resource Planning
ES	Enterprise System
HITL	Human-in-the-Loop
IDP	Intelligent Document Processing
IoT	Internet of Things
IPA	Intelligent Process Automation
IT	Information Technology
LIME	Local Interpretable Model-Agnostic Explanations
M2M	Machine-to-Machine
MDM	Master Data Management
MES	Manufacturing Execution Systems
ML	Machine Learning
NLP	Natural Language Processing

PDM	Product Data Management
PLM	Product Lifecycle Management
PoC	Proof of Concept
UI	User Interface
QMS	Quality Management Systems
RNNs	Recurrent Neural Networks
RPA	Robotic Process Automation
SCM	Supply Chain Management
SHAP	Shapley Additive Explanations
STS	Sociotechnical Systems
XAI	Explainable AI

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Abstract

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1. Introduction

This chapter introduces the background of the study and explains its relevance, particularly in the context of Product Data Management (PDM) and the SAP S/4HANA environment. It defines the research objectives, questions, and delimitations that focus the work on combining Machine Learning (ML) and Robotic Process Automation (RPA) in the item setup process. The chapter also outlines the methodological approach adopted to address these objectives and presents the overall structure of the thesis. In doing so, it provides the reader with a clear starting point for the subsequent chapters covering the literature review, methodological framework, implementation, and final conclusions.

1.1 Background

In contemporary business contexts, the exponential growth of data presents opportunities and challenges for organizations aiming to achieve operational efficiency and facilitate decision-making within their workforces. Specifically, for case company, this data growth is evidenced by the increased number of variants being introduced through products and services. As PDM processes generate increasingly large and complex datasets, manually keeping pace with the demand for accuracy, speed, and scaling has proven to be increasingly difficult. (Owusu-Berko 2024) A concrete example of a workflow in ERP systems, such as SAP S/4HANA, would be the process for setting up an item for order. The item setup process typically involves repetitive data entry, parameter configuration, and validation activities. These steps are time-consuming and prone to human error, which contributes to operational inefficiencies. Adebayo (2024) directly highlight. If data management and processes were not automated, these manual workloads can cause backlogs and lead to operational inefficiencies, increased costs, and less agility to react to changing market conditions within an organization (Solan & Gold 2024). For instance, if the item setup process does not occur in a timely manner, production plans will be disrupted, thereby leading to efficiencies in supply chain management and product launches.

There are important implications for operational processes as advanced technologies such as ML and RPA have the potential to facilitate significant transformation to legacy PDM

processes. ML and RPA can automate the handling of data, generate predictive insights, and create integrations to facilitate interoperability between systems that can help mitigate some complexities associated with the volume of data. (Taofeek et al. 2025) Artificial intelligence (AI) technologies such as predictive analytics enhances pattern recognition and provides organizations with the ability to allocate resources more efficiently and perform data-driven decisions. The research has demonstrated that successful applications of AI technologies into ERP systems can significantly reduce processing times, eliminate human errors, and improve overall authoring efficiency. (Christopher & Blessing 2024). In particular, supervised ML models such as decision trees, random forests, and artificial neural networks have shown strong performance in predicting item parameters in ERP contexts (Jawad & Balázs 2024). Likewise, RPA tools such as UiPath and Automation Anywhere are well-suited for automating structured SAP S/4HANA transactions (van der Aalst et al. 2018).

1.2 Research Objectives and Limitations

The main aim of this thesis is to create a sustainable and scalable solution using ML and RPA to streamline and automate the item setup process for SAP S/4HANA in case company PDM. The solution is implemented as a standalone Python-based ML application, integrated with SAP S/4HANA via UiPath RPA. Through an ML-based tool, the research aims to increase accuracy in data processing, reduced manual effort, and greater efficiency in the process, ultimately producing a functioning tool that automates the repetition of work while providing actionable benefits, such as shorter processing time, lower error rate, and greater operational efficiency.

The original goal defined a production-ready ML+RPA solution for item setup in SAP S/4HANA. During the thesis, the scope was redefined: the ML component became the first production-ready deliverable, while the RPA flows reached a functional pre-production stage, with only testing phase remaining before production rollout. Three factors shaped this terminus: a fixed thesis term, the uneven distribution of the teams' capabilities (the team was more proficient in RPA skills than Python), and change negotiations that prevented an extension. If an extension was possible, documented artifacts could have been planned after RPA layer work was designated production ready, and the thesis term would then naturally become the phase to finish that work. In the time available, it seemed wise to also advance

the company governance priorities: establish version control over ML, RPA, and other artefacts (repository structure, release tagging, change logs), prepare installation, operation, and update documentation for the packaged ML application.

To guide the work, the thesis addresses the following research questions:

RQ1. How to design an intelligent automation solution that combines ML and RPA for SAP S/4HANA item setup?

RQ2. What performance and data-quality gains does it deliver versus the manual baseline?

Although this research intends to provide a holistic solution to some of the challenges of manual, error-prone processes in PDM, there are limitations to the applicability of the work to the broader generalizability and scalability of findings. First, while the experience of implementing the tool will be tested with the item setup process at case company, the applicability of the work being applied to other types of products or processes are not assessed in this research. In addition, the first version of the tool allows the user to validate its required parameters, which is also more human dependent and could compromise automation for efficiency purposes.

The study considers the use of ML and RPA technologies that are deployed in the SAP S/4HANA environment and ultimately does not assess AI applications or other automation technologies. This theoretical delimitation provides focus but there may be fair consideration of other technologies that have value in the research environment. Lastly, while organizational change, user engagement and training are critical to the successful deployment of new technologies, they are not of primary consideration in the study.

Nevertheless, the goal of the study is to provide practical insights and a practical tool that can build on the implications of PDM automation. Addressing the challenges described and providing limits to what was considered in the research helps advance PDM automation and to an extent, academic knowledge in addressing, if not solving, ERP process problems in the field of industrial engineering.

1.3 Research Methodologies

This thesis employs a Design Science Research (DSR) framework as a means of addressing issues related to the automation of the SAP S/4HANA item setup process as applied within case company PDM system. The DSR approach is appropriate for this research as it enables the design and experimentation of a practical tool while combining theoretical notions with real-life application. This research adopts an iterative process to identify challenges, design an intervention, develop a prototype, and ultimately evaluate the prototype. (Edwards et al. 2024)

The research began with an extensive review of the current manual process in order to understand not only existing inefficiencies but also where improvements could be made. Qualitative data was collected through interviews with 10 key stakeholders and quantitative data was collected using historical process logs and documentation. Additionally, a survey was conducted with stakeholders to gather wider contextual understanding about real-world challenges across the PDM and expectations surrounding PDM.

Given the intention to preserve confidentiality for the case company while adhering to data privacy provisions, the thesis does not disclose interview transcripts in full, nor survey and meta-data. Inclusion of this type of sensitive data is required to protect the confidentiality of proprietary business information and anonymity for all participants involved in the research.

In accordance with LUT University's academic standards, the AI-assisted tool ChatGPT was used throughout the research process, supporting the understanding of complex technical topics and improving both the clarity and preciseness of the academic writing style and the final document.

1.4 Structure of Research

This thesis is organized into six key chapters to give a thorough understanding of how ML and RPA can improve the SAP S/4HANA creation of records in the PDM function at case company. The research moves logically from defining the problem to proposing a practical solution and evaluating that solution. In the Introduction chapter, the background, aims, and limitations of the research are provided, allowing the reader to appreciate the necessity for

automation, the research approach, and the structure of the thesis. The Literature Review chapter highlights previous research on PDM, automation technologies, and the application of artificial intelligence to ERP systems, demonstrating where this research is situated academically. In the chapter called Research Methodologies, the methods for DSR approaches are provided, outlining data collection methods including interviews and surveys, and discussing confidentiality. In Development and Implementation, the automation tool was developed including design, iteration in prototype, and how the solution was implemented and enhanced through user tests of the tool. The Evaluation and Results chapter provided evidence of the automated tool's performance using metric-based measures including processing speed, accuracy, and general usability incorporating qualitative and quantitative aspects. Lastly, the Conclusions chapter summarizes the main outcomes of research and discusses significant practical implications for case company, recommendations for further research into the automated tool, and recommendations for companies in similar contexts. Overall, this academic thesis demonstrated the potential of innovative artefacts to turn traditional process workflows into efficient and automated repeatable automated processes aligning to the objectives of the organization using the academic field. An overview of the research, using an input-output table, is shown in Table 1 below.

Table 1. Structure of Thesis

Input	**Chapter**	**Output**
General information about the research topic, objectives, and structure.	1. Introduction	Defines research background, objectives, scope, research questions, and outlines the research methodology and structure.
Research and scientific literature on PDM, PLM and master-data processes. Theoretical frameworks on Industry 4.0 pillars and process-automation maturity.	2. Digital Transformation in Product Data Management	Positions the role of PDM within the Industry 4.0 transformation. Describes current manual challenges in the SAP environment, identifies data-flow and data-quality gaps.
Research and scientific literature on ML and RPA applications in ERP systems → Intelligent Automation	3. Intelligent Automation in ERP	Synthesises the benefits and limitations of ML and RPA solutions for the PDM process. Develops a conceptual framework that combines ML predictions with RPA orchestration
DSR literature and a six-phase process model	4. Methodology	DSR rationale and process model, defined problem and objectives, design and evaluation approach, rigor and validity
Design and evaluation findings	5. Discussion	Results interpreted vs RQs and literature, benefits and limits, generalizability, managerial and technical implications, future work
Synthesis of evidence	6. Conclusions	Answers to RQ1 and RQ2, theoretical and practical contributions, adoption roadmap, research priorities

Input pertains to the key information sources and methods used in each chapter as shown in table 1, Chapter describes the main sections of the thesis, and output depicts the intended outcomes of each chapter.

2. Digital Transformation in Product Data Management

This chapter is the first part of the literature review and outlines how digital transformation affects Product Data Management. It starts by contextualizing Product Data Management to Industry 4.0 and enabling technologies such as IoT, artificial intelligence, cloud computing, and big data analytics. The chapter goes on to summarize common Product Data Management practices like versioning, change management, and bill-of-materials management, and describes why Product Data Management must be tightly coupled with enterprise systems such as ERP, PLM, MES, SCM, and QMS to preserve consistency in the product's single source of truth. SAP S/4HANA is discussed as a contemporary ERP example of real-time product data processing and then the chapter briefly reviews human processes and process automation, new opportunities for RPA and AI processes.

2.1 Industry 4.0

Industry 4.0, referred to as the fourth industrial revolution, is a major shift in manufacturing by utilizing digitalization to a greater extent than previous methods. The term was first used in Germany around 2011 as "Industrie 4.0" to state an idea of the smart factory characterizing cyber-physical systems and machine-to-machine (M2M) technology. (Wahlster 2022) Industry 4.0 is represented by many modern technologies that are used for (or enhanced) industrial processes, such as intelligent automation, the Internet of Things (IoT), AI, and big data analytics, to enable decentralized organization and self-organizing production with improved decision-making. (Mourtzis et al. 2022) RPA automates routine, rule-based digital tasks and reduces manual workload and error rates. In Industry 4.0 transformations it complements enterprise systems by streamlining data handling. In ERP contexts such as SAP S/4HANA this support enables timelier and higher-quality records. More broadly, Industry 4.0 adoption relies on facilitating systems like ERP and benefits from technologies that enable real-time data exchange. (Ribeiro et al. 2021)

Industry 4.0 relies on nine cornerstone pillars that drive change towards new highly automated and connected industrial environments: blockchain, cloud computing, cybersecurity, the IoT, AI, big data analytics, horizontal and vertical system integration, and

simulations, along with additive manufacturing (such as 3D printing) (Leong et al. 2024). Each of these pillars is essential to all aspects of applying modern manufacturing and enterprise management. More specifically, super smart and connected IoT, AI, big data and, cloud computing are critical for PDM and ERP systems (SAP S/4HANA) as there are increasingly more real-time and interactive data collection, intelligent automation, and decision-making processes (Jagatheesaperumal et al. 2021). If you think about the definition of Industry 4.0, it spans the entire the product lifecycle, unlike previous industrial eras which were either defined by mechanization, electrification, or basic automation. Intelligent products that communicate with the producer during production, and in-service highlight the product lifecycle and real-time operating characteristics.

This historical evolution prepares the ability to consider changes in how product data is created and managed in industry. A key facet of Industry 4.0 is the effect it has on PDM practices, traditional PDM systems are expected to understand and manage what now consists of much larger datasets outside the traditional focus upon datasets in the design phase (CAD models, drawings, bills of materials, etc.). Industry 4.0 adds another level of complexity to the design through operation of the "smart" or connected product. Now, smart products or systems produce and communicate many datasets from the initial design to in-use, and through maintenance and have the capability of sending rich and interesting datasets back to the developer. (Khang et al. 2023) So instead of being managed as designed or as manufactured the modern PDM system must handle an explosion of datasets on lifecycle that includes reuse and consideration for future design iterations. Researchers have identified the use of feedback from the product operational life cycle, or what is also sometimes called "technical inheritance" is essential to the next generation of product development. (Ryalat et al. 2023) The challenge for PDM in the Industry 4.0 here is to control all of the relevant datasets across many disciplines and product generations while creating one integrated and coherent dataset for the engineer (Grosse et al. 2023). The goal is for PDM will increasingly meld with Product Lifecycle Management (PLM) and real time data streams to ensure knowledge is integrated rapidly into design and production and is most relevant from IoT sensors and user evaluations.

Numerous advanced technologies collectively defined under the banner of Industry 4.0 translate directly to improving PDM processes in an industrial context. Industry 4.0 relies on automation to a large extent, whether that be robots as your means of automation on the

factory floor or software to automate the flow of information (Brecher et al. 2021). Regarding PDM, automation suggests that many routine data management tasks can be automated, to an extent or replaced by other automated processes that do not involve human intervention. System integration establishes connectivity between various information data stages (e.g., releasing a CAD model, correcting inventory, triggering process plans, etc.) and automates other mundane tasks while reducing manual data entry (Raptis et al. 2019). This degree of automation provides the OPD manager with comfort that their product data is being updated uniformly by systems while the engineer has been relieved of the tedious work of document control.

AI and ML are being integrated into Industry 4.0 to leverage the numerous applications for product data. Intelligent algorithms can analyze large design and field data sets to identify patterns or anomalies not easily detected by humans (Achouch et al. 2022). AI-powered analytics in the real world can enable predictive maintenance, quality predictions, and even possible design recommendations by learning from past product data. The modern PDM processes can utilize AI to classify data, organize data, provide the best workflows for data management, and improve accuracy. That is, AI can provide some degree of decision support on top of PDM. AI can take "raw" product data and derive conclusions for planners, managers, and engineers alike (Enfuse-Solutions 2023).

Another Industry 4.0 discipline that is key now that sensor technologies (IoT) and connected machines can produce massive amounts of data is big data analytics (Di Maria & De Marchi 2022). Advanced analytics tools are used to process and analyze quite large amounts of data at one time, helping the user uncover attributes about their product they may not be able to see, or may miss when dealing with very large amounts of data (Nagy et al. 2023). With respect to PDM, big data analytics can analyze whether the products users are using as intended/specified in the design capacity, or how the products behave over time, feedback that data to the design process, through identifying failure rates of components, or, most impactful, significance of features that are most often used by customers (Ren et al. 2022). By leveraging big data, organizations can increase product quality and shape desired attributes of future products based on setting, while all the while properly managing the data through PDM systems. This means that the big data analytics feature adds another layer on top of traditional PDM in that this additional discipline provides an evidence-based loop for product development.

In conclusion, and from the lens of PDM, Industry 4.0 technologies provide a means to shift the notion of PDM from a passive repository (or database of files) to an active, intelligent, backbone of the product lifecycle. Automation ensures that data streams quickly through the organization, AI represents smart ways to think through product data, and big data analytics offers richer levels of intelligence to aid decision making, thereby raising the level of PDM activity to new heights to meet the rapidly changing environment of smart manufacturing (Wang et al. 2018). In sum, the transition to Industry 4.0 has created a more connected and data rich environment where managing Product Data efficiently will be crucial to leverage complexity to function in a continuous improvement framework (Di Maria & De Marchi 2022).

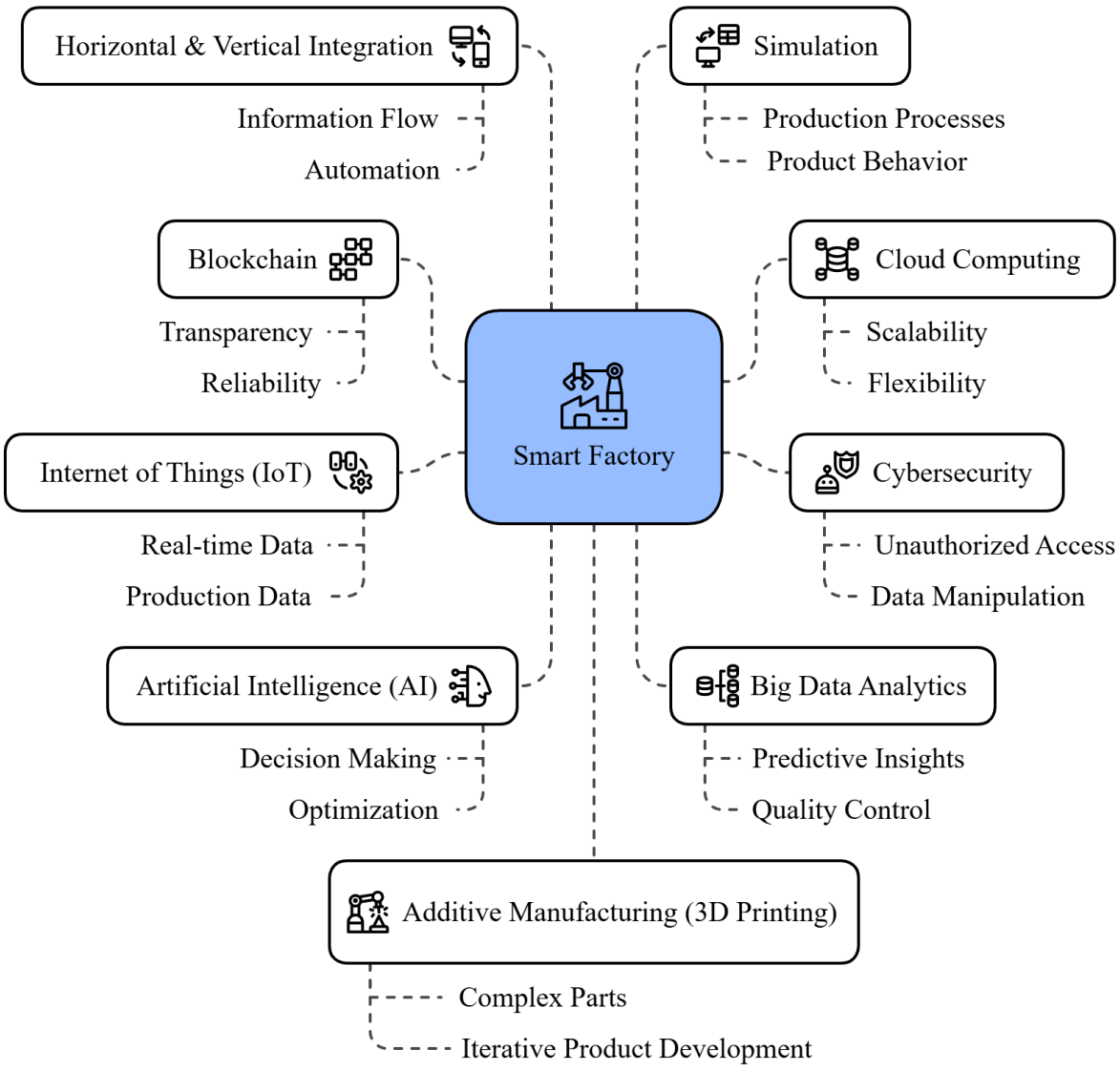


Figure 1. The Smart Factory and the Pillars of Industry 4.0

The figure 1 illustrates the key technological enablers of Industry 4.0 and their relationship to the smart factory. Core pillars such as the IoT, AI, big data analytics, cloud computing, cybersecurity, simulation, horizontal and vertical integration, blockchain, and additive manufacturing provide the foundation for real-time data exchange, intelligent automation, and flexible production environments. (Mourtzis et al. 2022)

2.2 Product Data Management

Product Data Management (PDM) defines an organized approach to managing product data throughout its whole lifecycle, from conception to disposal (Mahato 2023). It includes all processes, tools, and technologies to create, store, capture, and distribute product data within an organization (Fuerst, Camba & Mazumdar 2024). PDM is often mentioned alongside Product Lifecycle Management (PLM), but it focuses on organizing and controlling structured product data. PLM extends to a larger business process and has a more strategic view on product development, production, or market performance (Shaheen and Németh 2022). Today's enterprises have realized the value of PDM; with organizations wanting to manage their data better, improve collaboration, and make better decisions (Kulkarni & Kottursheettar 2022).

A well-defined PDM system assures the integrity, accessibility, and consistency of data is of utmost importance; all these factors are critical to developing products efficiently as well as operational excellence (Porsev & Abramov 2023). Centralized data with version control, workflow automation, product lifecycle management tracking, and collaborative tools that provide communication across departments are keys to managing structured data. These tools can support engineering teams who monitor proper product specifications and the decision process about tracking design changes, and vendors and operators from manufacturing and procurement can always provide access to the approved product data (Mpofu 2023). For example, in a business full of complicated product configurations like case company, PDM would encapsulate and manage the product data to link all the involved parties in a global supply chain and ensure compliance across industry standards.

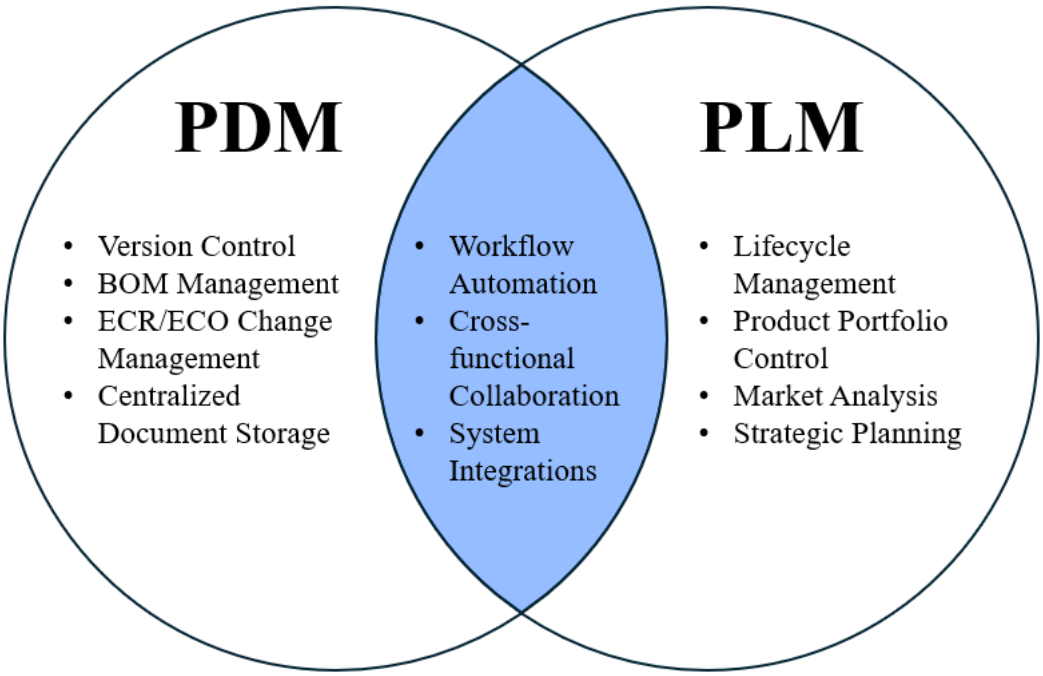


Figure 2. Comparison of PDM and PLM Functions

The figure 2 illustrates the relationship between PDM and PLM systems. PDM focuses on operational aspects such as version control, bill of materials (BOM) management, engineering change management, and centralized document storage. PLM extends the scope to strategic and lifecycle-oriented functions, including product portfolio control, market analysis, and long-term planning. The overlapping area highlights shared capabilities, particularly workflow automation, cross-functional collaboration, and system integrations, which act as a bridge between technical data management and broader lifecycle and business objectives. (Shaheen and Németh 2022)

A major component of PDM is version control. Its value is identifying design changes and tracking them with methods to refer to previous versions should the need arise (Krivykh & Khvan 2023). Engineering change management is important to this process as it allows business to make changes or modifications efficiently while managing product quality and consistency. Two of the major methods of managing these changes include Engineering Change Requests (ECR) and Engineering Change Orders (ECO). An ECR is considered a formal process and is typically opened any time a change is requested or proposed to a particular product design, documenting the reason for the change, items affected, and anticipated effect on production and procurement. Once an ECR has been formally approved, an ECO is formally issued to implement the change as described in the ECR

usually with the appropriate actions requested from each department. The ECO process provides good communication to maintain consistent engineering changes in the design, or manufacturing. Designed in PDM systems, organizations can utilize structured change management capabilities that keeps control over design changes, minimizes product errors, and ensures each team has the most relevant product information they have at their most up to date (Ciancio et al. (2024).

In addition, PDM systems provide BOM management, alongside version control and change management, by outlining a hierarchical structure for the components of a product and defines the configuration of a product. (Gunjal & Belokar 2023). BOM is a standard reference point utilized by engineering, manufacturing, and procurement, that provides data consistency and accuracy for all stakeholders involved throughout the lifecycle of the product. PDM can facilitate BOM management by providing a controlled environment where multiple BOM versions can be maintained and linked to revisions existing within the design model; changes in the engineered product can trigger the automatic update to the BOM, while automatically cascading updates to all business processes connected to components listed on a BOM. (Oleszek & Santarek 2022) In an integrated system a design update can trigger an update to the BOM, which means that the component list sent to manufacturing teams can be consistent, as can the adjustments to procurement processes. This provides a level of consistency that will reduce production errors and obsolete material being used and can optimize our supply chain. (Rahman 2023)

Having structured, accurate, and timely product data is part of productivity gains, many organizations have found ways to improve efficiencies through either single-source BOM's or multi-variant/bill of material in the product data that they manage (Baroud et al 2024). If well managed, PDM systems will reduce errors with the engineering change, aid collaboration across functions of the business and provide enhanced traceability of product changes. As regulatory compliance relies on maintaining some record against a product and ensuring that product record based on quality and safety (Kulkarni et al. 2023). Within case company PDM is more crucial, given the amount of customizing done in our product portfolio and the connections to enterprise solutions such as SAP S/4HANA, when PDM is not structured correctly it raises issues such as product specifications inconsistencies and inefficiencies through manual data handling which impacts our ability to maintain acceptable

timelines for production schedules. Ultimately this will add costs through wasted efforts, diminished quality and product reliability (El-Sayed, Eladwi & Ashour 2022).

Despite the advantages of effectively managing your product data there continues to be significant challenges for organizations. Data accuracy continues to be a challenge and consistency of data across multiple environments/systems remains an issue, especially where product information exists in isolated and fragmented databases or use of manual input, while benefiting from digital transformation, product data is more complex than in the traditional environment. Complexity is related to increased product variations, more embedded software, and hardening of compliance (Ji & Abdoli 2023). The PDM process can become overly cumbersome because of the challenges of integration of systems like PDM, ERP and other enterprise applications which creates silos and redundancy across workflows. Research suggests that traditional PDM implementations continue to lag with issues with scalability, since legacy databases may not keep up with the rapid data volume growth that occurs with digital transformation. (Stark 2022).

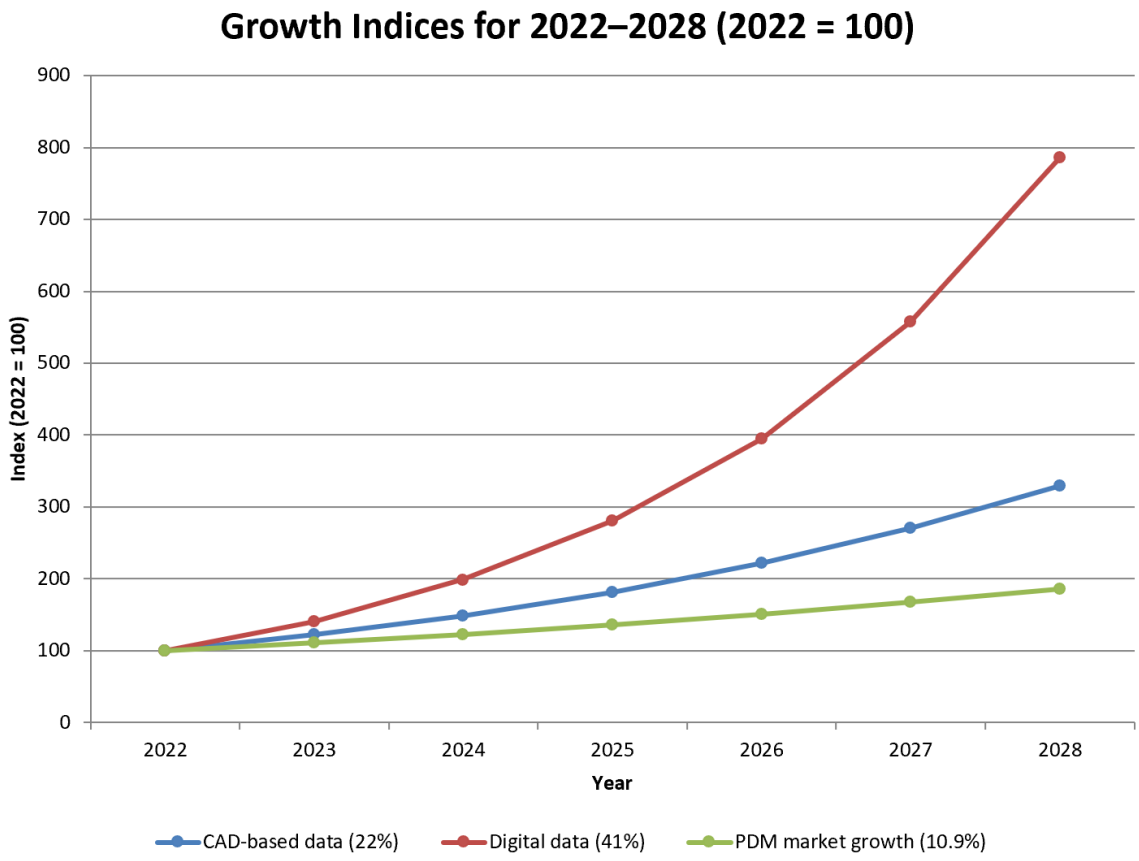


Figure 3. Data Growth Indices for 2022–2028 (2022 = 100)

In figure 3 the line chart shows compounded growth using constant annual rates: CAD-based product data 22 %/year, Digital Universe data 41 %/year, and the PDM software market 10.9 %/year. Indexing to 100 in 2022 enables comparison independent of absolute volumes. The widening gap by 2028 illustrates the effect of compounding at different rates. (industrialmachinerydigest 2020; cycloneinteractive n.d; verifiedmarketresearch 2024)

The growing complexity of product data means that organizations are looking to get ahead of these challenges by adopting best practices to leverage automation, replication, and smart data processing power, (Kulkarni et al. 2023). The emerging technology capabilities being brought to market are fundamentally transforming how product data is managed. The use of AI and ML for automation of data classification, identification of discrepancies, and forecasting insights for improved efficiencies in the design and manufacturing of a product is more prevalent (El-Sayed et al. 2022). RPA is primarily utilized to remove human involvement in repetitive tasks like managing data entry & validation, synchronization data in applications like PDM and ERP. Cloud PDM will easily allow for large volumes of data, real-time spectacle of data across operating locations and also ensure that all stakeholders are working with the same version of a single authentic source of data. (Matenga & Mpofu 2023)

PDM integration with enterprise systems, especially ERP systems like SAP S/4HANA, is an important consideration in product data management (Tanpure et al. 2023). SAP S/4HANA's ability to process data in memory means that product data can be treated in real time so that all updates that are made in engineering designs based on a product config unchanged are automatically brought into manufacturing, procurement, and supply chain planning. Furthermore, workflows can be automated to improve approval timelines and help keep product data aligned across different business functions. For example, upon a change's approval to the product config in PDM, SAP S/4HANA will recognize the change and update the BOM automatically, notify impacted departments, and initiate changes to the supply chain. These capabilities came back to the values and principles of Industry 4.0 in which processing data and making decisions based on that data occur real-time to improve the management of resources for operations in the industrial sector (Shaheen & Németh 2022).

A robustly implemented PDM system, with integrations into other enterprise applications, and taking advantage of modern automation would enable your company's scalability and will set the groundwork for intelligent product data management (Baroud et al. 2024). As

case company digital advancement unfolds, we will examine the role of PDM in enterprise systems, SAP S/4HANA integration, and processing automation in the next sections. These elements are paramount to transforming your PDM from just a stagnant data repository to a dynamic and intelligent system that can facilitate agile product development, real-time decision-making and continued process optimization throughout the product data management process (Rahman 2023).

2.3 Enterprise Systems in PDM

Enterprise Systems (ES) are integrated software applications that help organizations manage, run and optimize business processes across functional perspectives. An ES is capable of ensuring that information flows appropriately across Departments, thus facilitating the organization's ability to use resources efficiently and collaboratively, enhancing decision making (Atkar 2024). ES's are also an important part of PDM, as they include required features to align engineering data with business and operational processes (Dash 2025).

Of the ES's, ERP solutions specialize in integrating product-related data across business functions. The ERP is pivotal to ensuring that design and engineering, production, procurement, and logistics all function from the same data set. ERP systems manage and control important aspects of business, including material requirements planning, supply-chain interfacing, accounting/financial reporting, and production scheduling. This makes ERP a critical bridge between engineering data, and execution by the enterprise (El Haouat et al. 2024). In PDM, the role of ERP serves to ensure that BOMs, engineering changes, product specifications, and manufacturing instructions are properly transferred throughout design, production, procurement, inventory management (Zhang et al. 2023,). By integrating ERP with PDM, organizations can have confidence that they do not need to manually transfer data, which lessens the likelihood of error, and that operations at all levels are relying on the most current product data (Dash 2025).

ERP is not the only system that interacts with PDM. There are several enterprise systems, including PLM systems, Manufacturing Execution Systems (MES), Supply Chain Management (SCM), and Quality Management Systems (QMS), all of which have key roles in ensuring that engineering data is leveraged effectively through the lifecycle (El Haouat et

al. 2024). PLM systems also focus on product development, including elements such as design version control, regulatory compliance, working jointly through the product development lifecycle with other organizations such as R&D and production teams. (Park & Huh 2022). MES systems play an important role, by managing the data loaded into production from PDM systems and ensuring that production delivers the requirements specified by engineering on the shop-floor (Vukadinovic et al. 2022). SCM systems manage the transfer of engineering changes to supplier and logistics businesses that are affected by changes in engineering data thus, improving the effectiveness of procurement planning, ordering, and supplier collaboration (Morozova & Kharchenko 2023). QMS systems specifically manage the integration of quality control procedures and processes with the product data, in order to comply with relevant industry standards, and to manage and track defect reports and measures of product performance for continuous improvement (Chiarini & Kumar 2022). These systems should ideally be very tightly coupled and therefore be able to exchange data with each other in a seamless manner, and to and from engineering, production and other business processes. A change to a design in PDM needs to not only call for an update of an ERP system, but it should also update the change status within MES, and perhaps notify the SCM related to the procurement activity to accommodate the need for changes caused by the design change. The QMS could be messed up without the proper actions being stepped through to ensure that previous quality definitions were now updated and relevant to a new product definition (Dash 2025). When ESs are not properly integrated the inconsistencies in product data will create similar operational inefficiencies at various stages, such as loss of production run-time and delays to production schedules, over-ordering materials or under-ordering materials which creates inventory shortages, or simply introducing unknown defects into the product stream and creating quality issues (Elangovan 2022).

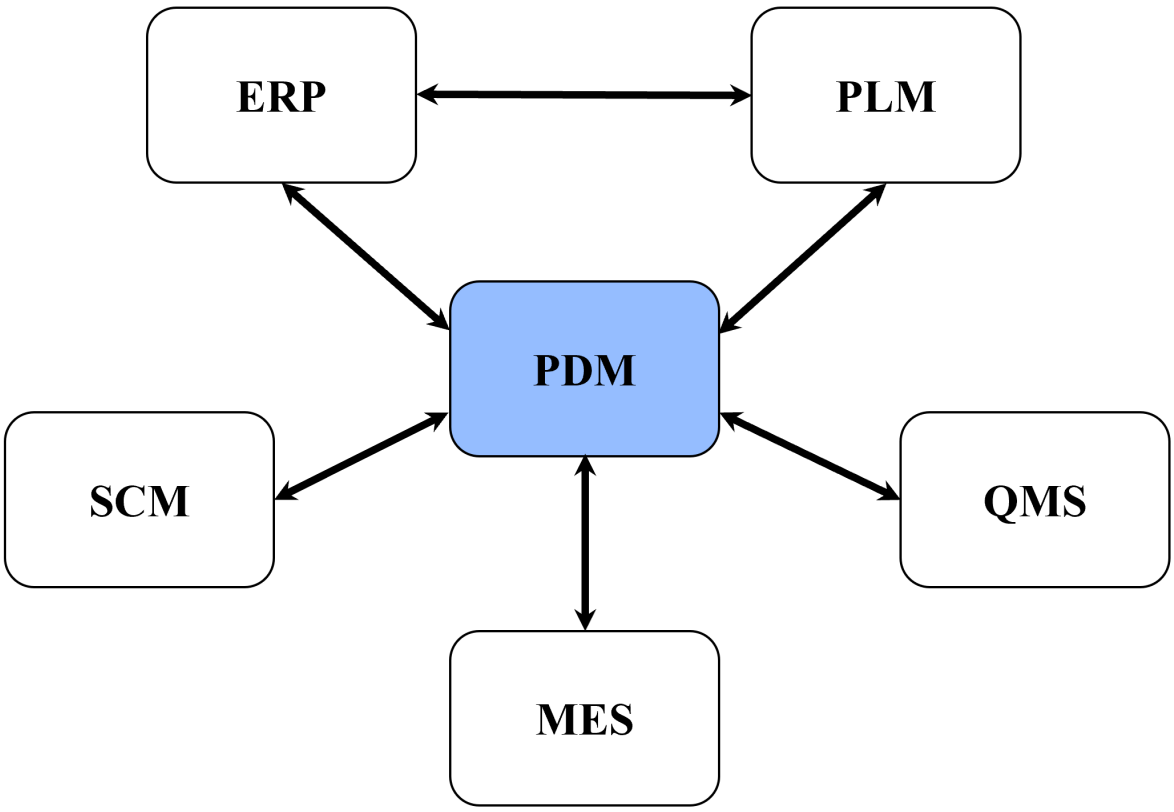


Figure 4. PDM-centric enterprise system architecture and data flows

The figure 4 shows PDM as the product-data hub, bidirectionally integrated with ERP, PLM, MES, SCM, and QMS. PDM distributes and receives key artefacts such as BOMs, engineering changes, and product specifications. ERP links this data to operational processes (MRP, finance, scheduling), PLM manages development and version control, MES executes shop-floor control, SCM coordinates procurement and logistics, and QMS integrates quality procedures and nonconformance records. These integrations ensure consistent, up-to-date product information across the lifecycle and reduce manual transfers and errors. (Penaranda et al. 2010)

A principal hurdle of enterprise system integration with PDM is data harmonization. Different systems establish data structures, classification schemes, and versioning in inconsistent ways, making it difficult to ensure product information is synchronized across the enterprise. (Elangovan 2022) For example, ERP and PDM systems may number part numbers using different conventions, leading to a misalignment between design specifications and procurement orders. To address these problems, organizations need to apply comprehensive master data management (MDM) and computerized validation tools to assure consistency. (McKendry et al. 2021). Compatibility represents another major

challenge of enterprise systems. Many organizations still use legacy versions of either ERP or PDM that were never designed to be compatible. (Djoric et al. 2024). Incorporating legacies with contemporary enterprise applications, may require middle ware conformable to a custom application programming interface (API), or cloud-based integration environments (Ma et al. 2022). Projects with this level of complexity can spiral into excessive implementation costs and necessitate high levels of Information Technology (IT) expertise (Ma et al. 2022). Change management is another important factor to consider in enterprise systems adoption. With engineers, production planners, and procurement specialists being kinesthetically trained to operate in separate systems, they may resist engaging fully integrated work processes based on new workflows changing to meet existing operational habits, Often, successful enterprise system integration requires a more user friendly articulation of benefits, user training, and continual improvement of the system align with business needs (El Haouat et al. 2024).

Modern enterprise platforms as SAP S/4HANA are addressing many challenges of traditional enterprise systems, such as provision for established integration frameworks to connect PDM, ERP, MES, and SCM in one system architecture (Park & Huh 2022). Contemporary ERP platform implementations do not require the high degree of customization required for establishing cross system connectivity. Also, SAP S/4HANA includes real-time data synchronization and analytics features, and partial integration with existing PLM and MES systems (Dash 2025).

2.4 SAP S/4HANA in PDM

SAP S/4HANA is SAP's next-generation ERP suite, designed to integrate and streamline business processes in real-time. It is built on SAP HANA, an in-memory computing platform, which gives organizations better data processing routines and allows instantaneous analytics, seamless process automation, and optimized decision-making. (Antwi & Avickson 2024) The architecture of S/4HANA removes unnecessary data storage, reduces the complexity of databases, and allows for operating enterprise-wide in real-time (Kulkarni 2024). These features confer significant benefits for PDM, as the ability for various teams to access real time reliable product information is critical to ensuring efficient operation (Bhatia 2024).

In PDM, SAP S/4HANA acts as the main system to manage and distribute product-related information to all respective teams, establishing consistency and reliability for product information throughout the product lifecycle (Vaka 2024). The Material Master serves as the primary attribute repository to manage each product. The Material Master has several views, each meant for a specific area of business functionality. (Juturi 2024) Table 2 shows the SAP material views and their descriptions.

Table 2. SAP Material Views (Basic Data, Purchasing, Sales, Production, WM, Accounting)

View	**Description**
Basic Data	Contains fundamental material attributes such as name, type, and classification.
Purchasing	Includes procurement-related data such as vendor details and purchasing conditions.
Sales	Stores sales-specific parameters including pricing, sales units, and customer configurations.
Production	Manages manufacturing-related details such as BOM and routings.
Warehouse Management	Handles inventory-related information, including storage locations and handling instructions.
Accounting	Provides financial attributes, including valuation and cost accounting data.

SAP S/4HANA provides structured material data, providing all business functions with one source of truth. This creates an enterprise to rely on rather than comply with data silos and acknowledges that workers use unsafe data in fragmented environments causing errors. (Sarferaz 2022)

One of the main advantages of SAP S/4HANA in PDM is the integrated enterprise system. SAP S/4HANA has allowed for automation of integration to work with its ERP systems making sure that engineering, production, procurement, and SCM functions are running together. (Banks-Grasedyck et al. 2022) PDM provides engineering data to ERP, ensuring that the engineering of product designs, specifications of components, and the structures of BOM's are the most accurate past the point of procurement and into manufacturing. The integration of ERP to the SCM systems creates real-time automated optimization of the procurement process while supporting logistics from current product data based on previous

design history, and material purchase forecasts reflecting current demand profiles. (Shaik 2023) The interaction of ERP to the MES systems ensures production processes operate with the most current engineering specifications, and manufacturing requirements. The ERP to QMS integration creates tracked quality control data against manufacturing processes, which can monitor verification of product quality in the entire production lifecycle confirming accurate conditions, requirements, and results (Goel, Mangal & Gupta 2023).

In terms of challenges, using SAP S/4HANA in PDM has shortcomings. For example, the main challenge is change management and user adoption. Moving from prior ERP and PDM legacy systems to SAP S/4HANA involves significant process reengineering work, which integrates the governance of structured workflows and accounted data ownership obligations of the trading partner. (Antwi & Avickson 2024) Staff are used to manual data handling rules, and preceding legacy systems to automate their integration into enterprise functions, so the Opportunities for new procedures met with inadequate training support can lead to limited user acceptance orb we will have competent obstacles to user adoption. (Kulkarni 2024) The government would be responsible for the implementation of business frameworks to locate user resistance. Implementing a collaborative change management approach to accommodate change will require early engagement with all stakeholders, supporting multifaceted user training programs, avoiding undue operational expectations with staggered functional go-live activities to enable operational flexibility upon the adoption of the SAP S/4HANA systems (Bhatia 2024). A primary issue noted upon migration to SAP S/4HANA is the system interoperability, and complexity as part of the integration when transitioning from PDM systems while dealing with your historical product data you intend to use (which may contain inconsistencies, duplications, and stale data). (Vaka 2024)

To use Material Master, product data must be rid of poor quality content over cleansing, validating, and periodically forming data standards to in line with new features of SAP S/4HANA platform used at the time of the integration. (Juturi 2024) When PDM or PLM is used alongside SAP S/4HANA, differences in data models are common. Organizations may be overly enthusiastic about the value of ‘real-time’ data, yet fail to consider the challenge of updating product information that changes on a predictable schedule. Moreover, missing content standards (e.g., form/asset lists) and unclear ownership create additional barriers. These issues should be evaluated before integration to justify any IT budget during the SAP S/4HANA implementation phase (Sarferaz 2022). Organizations will also need further to

assume that they are managing a real-time use of systems to develop and/or manage their top-level domain (TLD). It will require expanded IT capacities outsourcing to develop, in some instances, a less developed API-based middleware to link, compare, and engage wherever possible any previous IT architectures, and future IT spending increase by meaningful executive sponsor at built in times to connect and execute to the wider executive goals of the organization for their systems implementing plan (Banks-Grasedyck et al. 2022).

SAP S/4HANA is increasingly utilized as a cloud-based ERP system. The cloud deployment of ERP solutions offers flexibility and scalability as well as integration capabilities. Cloud-based deployments actively enable effective connectivity to third party PDM and PLM platforms, in addition to remote access and collaboration for engineering, purchasing, and manufacturing teams. Increased security and compliance are achieved by implementing automated data back up and disaster recovery services. (Shaik 2023) The role of IoT integration, with SAP S/4HANA, is an impactful contributor to real time product data management. Companies can utilize IoT sensors for many applications such as tracking material consumption, predicting the timing of maintenance requirements, allowing automated product lifecycle tracking, contributing to proactive quality management, and optimizing production activities reducing waste and cost. (Goel, Mangal & Gupta 2023) Regulatory compliance with industry-specific standards is another important function of SAP S/4HANA in PDM (Antwi & Avickson 2024). Built-in compliance management functionality includes ISO quality standards (ISO 9001, ISO 14001), REACH regulations for material traceability, and FDA 21 CFR Part 11 compliance for industries needing to validate electronic records of manufacturing with automated audit trails to confirm traceability and documentation of all engineering changes and updates to manufacturing (Ispas et al. 2025; Legardeur & Ospital 2024; Elmadhoun et al. 2025). By integrating cloud computing, IoT enabled analytics, and compliance management automation, SAP S/4HANA helps PDM processes remain relevant to modern manufacturing, enabling improved operational speed, efficiency, responsiveness, and regulation compliance (Kulkarni 2024).

Real time processing of data, automation, improved data governance and system integrations are significant benefits of utilizing SAP S/4HANA in PDM. However, organizations have to take into account several challenges related to data migrations, system compatibility, user adoption, and compliance management (Bhatia 2024). As more and more organizations adopt cloud-based ERP systems utilizing IoT-driven automation and AI-enhanced analytics,

SAP S/4HANA is positioned as an important enabler of digital transformation with a PDM capabilities (Vaka 2024). Organizations that can successfully integrate SAP S/4HANA PDM solution, PLM, MES, and QMS solutions will achieve enhanced operational agility, regulatory compliance, and sustainable competitive advantages (Juturi 2024).

2.5 Traditional Process Automation in PDM

Traditional process automation in PDM describes the more traditional methodologies for speeding up repetitive administrative activities and increasing efficiency, prior to advanced AI and ML capabilities. Traditional automation relies on a rule-based system, scripting, or non-ML based workflows to carry out repetitive administrative activities with little human involvement or oversight. (Olurin, Okonkwo & Eleogu 2024) Automation of administrative workload, and to some extent data consistency, has fundamentally increased efficiency and productivity in PDM situation (Ravichandran & Machireddy 2023).

Traditional process automation refers to the systematic utilization of scripts, macros, workflow engines, and batch processes to automate the management, validation, and distribution of product-related data in PDM. For many decades traditional methods have performed the somewhat valuable effort needed to maintain consistent data surrounding product-related information, to help organizations remain historically ahead of data versioning as well integrate PDM to ERP solutions. (Boddapati 2025) MPS or traditional automation offers solutions that utilize automation based on rules stated in the software or automating sequences based on a statistical sample of historical transactions, while AI-driven automation employs sophisticated algorithms capable of both learning from historical datasets and adapting to process deviations to help facilitate the outcome. (Mathew, Brintha & Jappes 2023).

Commonly used methods to perform traditional automation are scripting and macros that allow individuals to develop autonomous scripts using computing language commonly recognized by all programming language, such as Visual Basic for Applications (VBA), Python, or licensed PDM scripts automated to perform normal data entry, extraction, or validation. Batch processing schedule predefined operational activities such as data synch, report preparation and database updates/batch processes during period of non-utilization to lessen the load placed on the system. (Chugh, Macht & Hossain 2022) Templates, and

template based automation, reduced customer defined tasking and will reduce inconsistent recordkeeping across platforms and systems. Rule based workflow automation can perform various tasking similar to previous desktop documents by establishing a sequenced workflow that either automates document approvals, or organizes defined change management workflows, plus directed one process step to the next based on corporate predefined business rules. (Fatima et al. 2022) Import/export automation is an established method to transferring organized data between PDM to ERP systems. Import/export methods utilize discrete formats such as XML, JSON, and CSV. Statistically sound databases also include triggers which are controls that automate simple pre-defined database commands once specific events occur, typically may update selected records when the engineering change document has received all evaluations, decision approvals, or status determined at the time of the parties approval list. (Kumar et al. 2023)

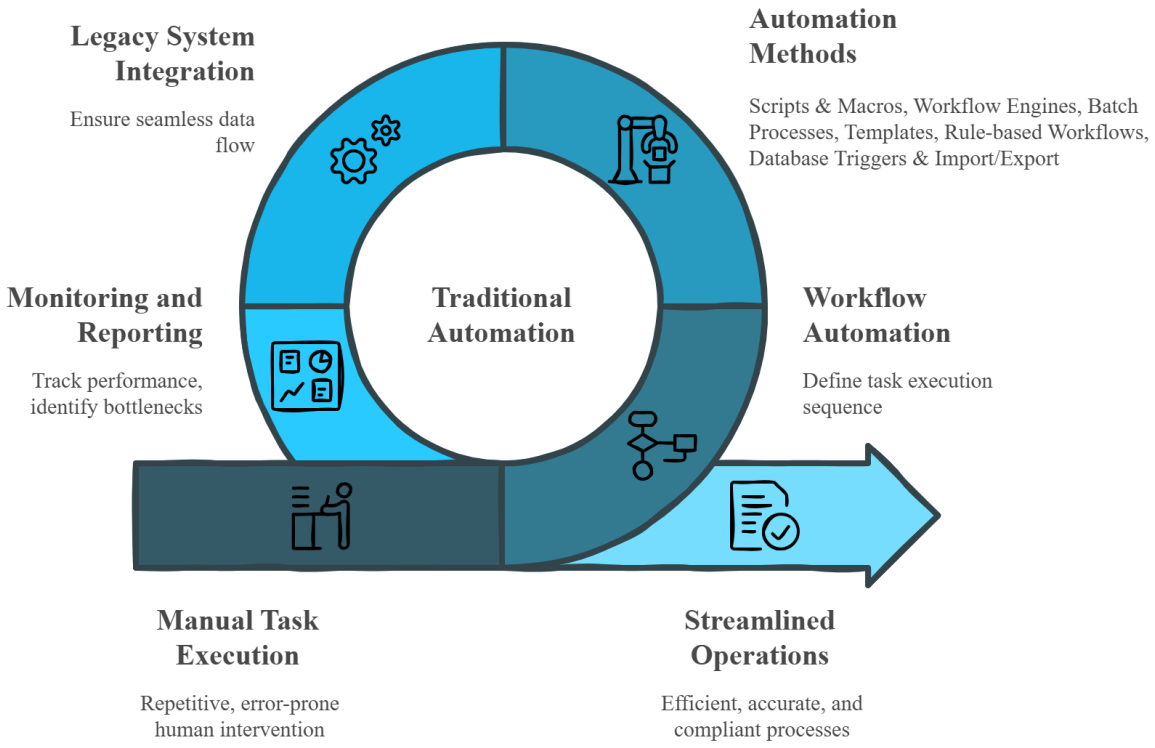


Figure 5. The Process of Traditional Automation in PDM Environments

The figure 5 illustrates the transition from manual, error-prone task execution toward streamlined operations through traditional, rule-based automation. At the core, “Traditional Automation” encompasses methods such as scripts and macros, workflow engines, batch processes, templates, rule-based workflows, database triggers, and import/export functions. The surrounding elements highlight supporting aspects. Legacy system integration ensures

seamless data flow, workflow automation defines the execution sequence of tasks, and monitoring and reporting track performance and identify bottlenecks. Together, these elements enable more efficient, accurate, and compliant processes by reducing manual effort within PDM-ERP environments. (Cao et al. 2016)

Many processes and functions have traditionally automated such as PDM product development associated with the material master data element to assist with the creation of material, specifically with details/update attributes as needed, to ensure all engineering/procurement/manufacturing functions have a materials master that maintains accurate and standard records (Stark 2022). Typically, BOM or bill of manufacture, is managed with multiple automation scripts for each product to perform low-value repetitive calculator tasks prior to the BOM being updated for production or fabrication by validation of the BOM structure, comparing BOM subject back-to, validate the requirement of missing components, and synch-up changes with Production Planning. (Plattfaut et al. 2022) Engineering change process has incorporated automated workflows established for generating a engineering change request ECR forms and engineering change order ECO process for routing through multi-staged or simple predefined group/individual approval processes to assist with compliance and lead-time; ultimately relieving organization of misleading company engagement with intentions to enhance companies compliance. (Olurin et al. 2024). Data synch will ensure consistency of product data across PDM, ERP and other enterprise applications by applying time-defined initiatives to better enforce synch. Document control, and versions, are also aided through automation allowing users to archive records of historical or previous versions while ensuring everyone has equitable and access rights to the last properly approved version of documents after review. Automated reporting and compliance checks to generated standard reports for discussion on product lifecycle data, required compliance requirements, and common design change management modifications. (Ravichandran & Machireddy 2023)

The use of traditional automation in PDM (Product Data Management) limits manual work through automatized repetitive tasks, enabling engineers and data managers to plug themselves into value-add work such as product development and strategic planning (Boddapati 2025). By having standardized workflows and validation rules, organizations can increase data consistency and accuracy, as they remove inconsistencies in product data across multiple departments and applications that can lead to human error (Mathew et al

2023). Since automation can help improve the speed of processing, it also improves the speed of variations in lead times (Chugh et al 2022) and operational efficiency through automating data processing and document approval workflow (Fatima et al 2022). On a much bigger scale, employees also don't have to conduct manual tasks anymore to collaborate with engineering, production, or procurement teams since these will be automated, allowing the teams to work better together through a new automated approval process and document versioning (Kumar et al 2023).

With that said, it is important to note that traditional process automation also comes with its limitations. Rule based automation is rigid, and if there are changes in the process then depending on the automation set up, it may require painstaking manual changes or revisits to the rule-based configurations (Stark 2022). As product complexity increases, scalability is an issue because traditional automation practices will not be able to manage huge volumes of increasingly dynamic data (Plattfaut et al 2022). Additionally, most traditional PDM systems cannot process unstructured data sources like technical documents, sensor generated data, or customer feedback (Olurin et al 2024). The high maintenance costs surrounding traditional automation come from the continued necessity for IT support to keep rule-based scripts and workflows updated (Ravichandran & Machireddy 2023). Integration with older PDM and ERP solutions may also necessitate a custom build that increases system compatibility and process complexity (Boddapati 2025). Lastly, traditional automation represents a point in time of the process, it does not guide decisions and normally does not include predictive or prescriptive analytics to improve workflows (Mathew et al 2023).

It is interesting to highlight why organizations struggle to transition towards automation through AI. Cost uncertainty poses an issue, as they will have to undertake some large investments with AI and RPA-enabled software and hardware infrastructure as well as investment in upfront costs to train personnel (Chugh et al 2022). Because of the upfront costs of building a return on investment (ROI) model, some organizations are still moving along in their existing traditional automation models (Fatima et al 2022). Consequently, the training of existing personnel will need to proceed through this transition, and employees must adapt to a new tool and new workflows (Kumar et al 2023). Many employees are used to traditional automation and will resist AI at the organizational level, fearing either job losses or more complex workflows (Stark 2022). System incompatibility is a further barrier to automation. Organizations using older PDM and ERP solutions struggle with AI and

modern RPA technologies because of apparently unresolvable incompatibility due to the different data models, architectures, and interfaces described in previous sections of the report. These barriers can potentially lead to inefficiencies in the organization, extended implementation timelines, and complications possibly resulting in operational disruptions (Plattfaut et al 2022).

Often, traditional automation systems are difficult to scale, especially for multinational corporations across multiple regions with different regulatory requirements (Olurin et al. 2024). Large enterprises need scalable automation systems to support and promote global product development, supply chain operations and regional regulatory compliance (Ravichandran & Machireddy 2023). Traditional rule-based automation systems often lack the capability to process jurisdictional differences in data, classification, and engineering documentation (Boddapati 2025). Many companies, especially multi-nationals with operations in several markets, suffer from data silos in their organizations with different divisions using different scripts for automation. These are not standardized, nor have the level of integration across corporates leading to inconsistent data, compliance issues and inefficiencies when collaborating across divisions (Mathew et al. 2023). Large-scale manufacturing enterprise which previously invested considerable sums into customized automation frameworks to implement automation to overcome gaps particularly between legacy PDM, PLM and ERP systems, ultimately seeing an increase in long-term maintenance (Chugh et al 2022).

Traditional automation of work processes has enabled PDM to maximize effectiveness by automating the regular routine workflows, increasing data consistency and reducing manual effort. Furthermore, traditional automation systems have limitations in scalability, adaptability and decision-making led researchers to determine transformative automation systems are necessary (Fatima et al. 2022). AI driven automation, RPA and machine-learning based predictive analytics represent the next evolutionary step in Product Data Management providing organizations with improved flexibility and intelligence to manage complex product data challenges (Kumar et al. 2023). For companies which are on the digital transformation journey, deploying automation workflows will be critical to promote agility, accuracy and efficiency to improve product data management (Stark 2022).

3. Intelligent Automation in ERP

This is the second and final literature review and examines how intelligent automation supports enterprise systems and product data processes. It starts with an overview of how artificial intelligence is converting ERP platforms into live environments that provide real-time decision making through optimization. The discussion then outlines use cases for ML in PDM such as automating the creation of material masters in SAP S/4HANA. RPA is introduced as a method to remove repetitive action like updating BOMs, and processing orders without making expensive adjustments to the systems. Finally, the chapter explains how a combination of ML and RPA, provides intelligent automation that integrates predictive analytics to seamless execution, accelerating each phase of product lifecycle management. This overview provides a foundation to evaluate the advantages of intelligent automation and any implementation requirements as the organization embarks on digital transformation.

3.1 AI for Process Optimization in ERP

ERP systems have historically functioned as the digital backbone of organizations, connecting core business functions in a centralized manner. Originally, ERP systems were created to aggregate organizational data and generalize internal processing, while now, ERP has developed into ecosystems that carry functions around finance, logistics, manufacturing, procurement, and human resources (Jawad & Balázs 2024). In this transformative development, AI has emerged as the operation in itself capable of transforming ERP systems into new operational models that significantly contribute to business performance (Pokala 2024).

The implementation of AI into ERP combines a transition from static data pools to dynamic platforms that possess the capability to pursue recognizing patterns, learning from data, adapting to changing conditions and optimizing the decision making in real-time (Hrischev & Shakev 2023). This change is not solely a transition in technology but is an evolution in strategic operations in addition to some emerging technologies will be used in the ERP context. The pressure for organizations to improve operational efficiency and cut costs while

simultaneously increasing agility is a real and emergent risk for operational management. AI-enhanced ERP systems directly contribute to those strategic goals by introducing automated events that minimize and eliminate manual tasks, including faster processing, better quality data management and providing enhanced decision-support options (Bawa 2023).

There are already a slew of emerging AI technologies that organizations are looking to make great advances on to have enhanced decision-making capabilities with their ERP Systems. ML, ML uses historical and real-time data that identifies patterns and processes that enable advanced forecasting, anomaly detection, and adaption to dictating decisions (Godbole & Josyula 2024). Natural Language Processing (NLP) allows individual intra user interactions with the ERP platform through human-like form of dialogue, chatbots, or voice processing, allowing new interactions modalities or simply enriching user experience with the same ERP tool (Sharma & Vaid 2022). RPA, the act of automation for the repetitive tasks which are processing and rule based for example order submission, invoice validations, report generations are all very likely to be possible in RPA. Predictive analytics, put another, utilize historical, statistical modelling and recent data patterns to support decision making in a proactive manner particularly pre-empting demand change or risks in various processes (Gupta & Agarwal 2024).

Along with these technologies there are advanced ML types which are being used in new sophisticated ways in ERP relevant contexts. The most common form of AI in use is Deep Learning (DL). DL is a level beyond ML, as they are made up of many layers of processing from artificial neural networks made up of many layers, so these models are able to provide complex relational links and features from large datasets constructed from unstructured data structures (Hrischev & Shakev 2024). In ERP contexts, deep learning provides significant value or enhanced utility in areas such as image recognition, document classification, and natural language interpretation (Bodepudi et al. 2024). For example, DL has begun to be used to automatically process classifications of scanned invoices from suppliers, product image classification, detect anomalies in sensor data from manufacturing equipment, or enhance conversational AI with pro-active undertone in language processing (Kanaparthi 2023). Use of deep learning, collects and assimilates predictions or executions of document and data into multi-layered datasets. All of this mean ERP systems can operate many more formats of data with many more pragmatic types of approach, quicker and more

autonomously with deep learning (Kunduru 2024). Though it is still early days some ai technologies such as computer vision, and generative AI are still making their way into ERP workflows especially in instances of automated document processing, quality control, and content generation (Nyathani et al. 2024).

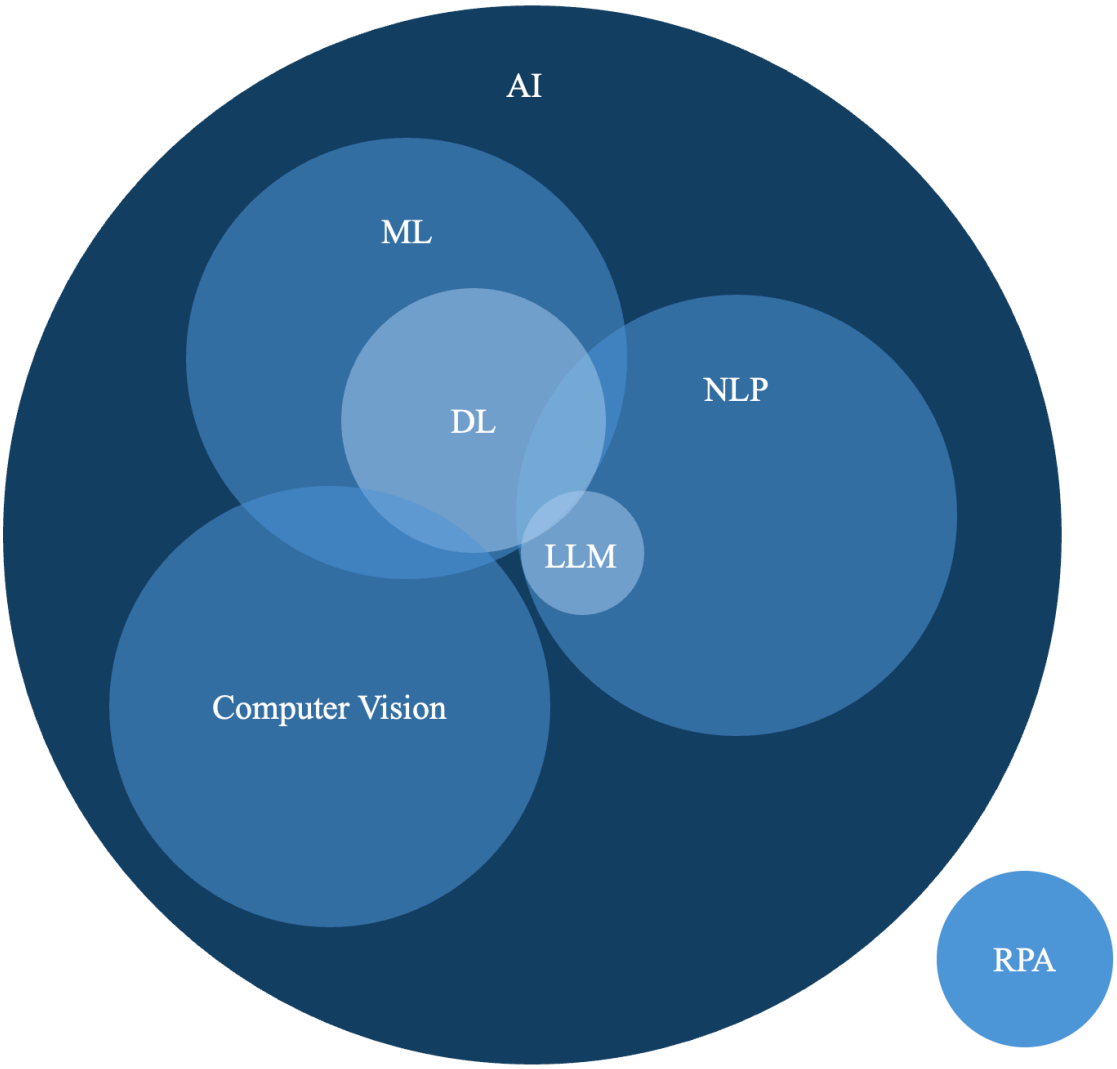


Figure 6. Relationships between AI subfields and RPA.

As shown in figure 6, AI is the umbrella field. ML is a subfield of AI, and DL is a subfield of ML (Goodfellow et al. 2016). NLP is another AI subfield where LLMs belong to NLP and are implemented with DL (transformer networks) (Vaswani et al. 2017). CV is likewise an AI subfield that relies heavily on DL for perception tasks (e.g., detection, classification, segmentation) (LeCun et al. 2015). RPA is not part of AI, it automates rule-based User Interface (UI) tasks but is often integrated with AI in ERP workflows (Aguirre & Rodriguez 2017).

The influence of AI can discover within almost every module of an ERP. In production planning, for example, AI can schedule dynamically, optimize workflow in real time, and can notify in a digital manufacturing environment. Predictive maintenance using IoT sensor-based data is already proven to manage downtimes of equipment and lengthening asset life (Jawad & Balázs 2024). In finance, AI can help automate invoice processing, provide cash flow forecasting, and can deliver alerts based on outliers (Parthasarathy & Padmapriya 2023). In human resources, AI can facilitate smarter candidate matching, and with the help of the ERP system, can create personalized employee experiences. In CRM, AI can support targeted outreach communication, lead scoring, and predictive future engagement models. (Jhurani 2022) In inventory, AI can provide and suggest demand predictive forecasting, optimized safety stock level, and reorder point calculations, while in logistics, AI can assist with time management for delivery routes with real time traffic changes and help with delivery constraints (Bawa 2023). The following table 3 outlines primary AI use cases within ERP functions, and the recommended optimization solution.

Table 3. AI Use Cases in ERP Modules and Their Benefits

ERP Function	**AI Application**	**Optimization Outcome**
Production Planning	ML-based dynamic scheduling	Shorter production cycles, reduced waste
Maintenance	Predictive analytics using sensor data	Fewer unexpected breakdowns, extended equipment lifecycle
Finance	RPA and NLP	Faster invoice processing, more accurate financial forecasting
Human Resources	AI-driven recruitment and personalization	Improved candidate experience, more efficient HR workflows
Customer Management	NLP and ML for personalization	Tailored communication, improved sales pipeline performance
Inventory Management	ML-based demand forecasting	Optimized stock levels, reduced inventory costs
Logistics	Real-time route optimization	Lower transport costs, improved delivery performance

Adopting AI across these areas may provide apparent operational and strategic advantages. However, executing AI successfully requires specific conditions. For instance, high quality structured data is vital to train AI models (Biolcheva & Molhova 2022). As well,

compatibility of AI tools with ERP modules must be ensured, so that intelligence is embedded, and businesses leverage the layer of intelligence directly in their operations (Chen 2025). Furthermore, an organization's readiness, including user training, change management, and alignment with strategic objectives, is essential for adoption and sustained impact.

The benefits of AI-enhanced ERP systems extend beyond operational efficiency. They improve decision quality and thus enhance the organization's risk management skills, while dramatically increasing the potential for fast-moving, agile business (Macron 2025). Organizations that fully adapt AI to their ERP capabilities possess an advantage for deploying strategies that respond to market disruptions, overcome complexity, and provide lasting, sustainable competitive advantage (Paul 2022). Additionally, as the pace of transition to autonomous, self-optimizing platforms accelerates, AI will play an increasingly prominent role in the intelligent organization and make rapid actions adaptive across the organization (Pinto & Pirola 2025).

3.2 ML in PDM

The continuous growth of industrial products and their associated data present ever-increasing demands on PDM systems for precision, consistency and scalability. Manual means of item creation and data maintenance have traditionally been sufficient but moved into reactive approaches using rules and manual methods to ensure data persistence is proving insufficient as products become more complex, the pace of product development increases and expectations surrounding data governance become more stringent. ML, a subset of AI, enables ERP integrated PDM systems to move away from rules-based applied actively to evaluate items and classify them at their source without explicitly writing in every relevant level in the product hierarchy would be beneficial to all levels of in potentially every business process, (Jawad & Balázs 2024; Sarferaz 2024a).

In the context of this thesis, ML in PDM is used for automating item setup in SAP S/4HANA. ML can predict relevant material parameters to handle master data intelligently, classify items similar to historical analogs and check data entered actively when transferring historical data for reduced errors. ML automates knowledge-based practice processes, thus enhances the quality of all the processes and reduces the substantial effort required to manage

product data across the lifecycle concurrently when setting up a new product. All of these activities relate to obtaining greater and scalable automation in enterprise systems, which is the overall aim of this thesis work. (Kocaoglu 2024; Ji & Abdoli 2023).

Put simply, PDM refers to supervised ML with predictive analytics whereby the supervised processes or treatments will take pixels, attributes, or conditions as inputs (e.g., product hierarchy, material type, purchasing group) and manage and produce corresponding standard output variables (arbitrary or constrained targets) example valuation class, MRP type, tax classification. (Sadeeq 2024). However, in general linear regression, decision trees, random forests, and artificial neural networks were commonly acknowledged algorithms to process relevant predictive analytics, for instance, a trained regression nominal model can make educated guesses for expected delivery times based upon material features, and supplier characteristics, whereas classification could recommend an MRP procedure associated with previous similar items (Dolgui et al. 2024).

DL can be defined as ML using artificial neural networks with additional underlying complexity layers, rich features, and relevant heuristics relevant to classification across the range of products in the given product domain. Deep learning models will commonly represent the non-linear relationship, multi-level hierarchy, various degrees of representative probability as attributes of material features, apply heuristics, program or process recursively to all undertaken characteristics of product identifiers. Deep learning models are relevant for PDM applications where automatic various tasks such as attribute parsing from unstructured product descriptions, assigning semantic relationships between items, and classification between products into finer grain categories based on representative description. Deep learning is also amenable to images where they may classify items from attributes of neutral yielding outputs, for example, using any combination, recurrent neural networks (RNN, or transformer) and applying heuristics to process sequential or textual representation for automated technical documentation. (Choudhuri 2024). Noting some of the typical further requirements for relevant deep learning data requirements include additional observations and larger computing power, however following current trends in information management with respect to physical systems/products/developments there is merit to applying recent models with already implemented deep neural networks in numerous methodologies for distinguishing subtle relational data patterns or correlating outcomes representative across

many associated data input types employed for seemingly advanced PDM use cases. (Taghipour 2024)

Table 4. ML Algorithm and their use case in PDM

Algorithm	**Type**	**Application in PDM**
Linear Regression	Supervised	Predicting numerical parameters such as lead time
Decision Trees	Supervised	Classifying item categories or procurement types
Random Forests	Supervised	Handling noisy data to improve classification accuracy
Artificial Neural Networks	Supervised	Complex parameter prediction with multi-variable inputs
Deep Learning (DNN, CNN)	Supervised	High-dimensional classification, attribute extraction
K-Means Clustering	Unsupervised	Segmenting similar items for BOM reuse or cataloging
Autoencoders	Unsupervised	Detecting anomalies or inconsistencies in master data

The table 4 summarizes common supervised and unsupervised methods and typical applications such as prediction, classification, clustering, and anomaly detection. The success of these models is dependent on strong data pipelines and good discipline in the preparation of the data. The historical material master data, within the ERP systems like SAP S/4HANA, has to cleaned, normalized, and structured prior to training ML models (Shaik & Siddque 2023). Feature engineering is important in taking the raw data like material descriptors, and units of measure into much more useful input features that reflect the domain-specific nuances which allow for predictions features to be generated (Sarferaz 2024a). When assessing the ability of ML models to generalize, validation approaches k-fold, train-test splits, precision, recall, RMSE and others can be used to evaluate (Sharma et al. 2024).

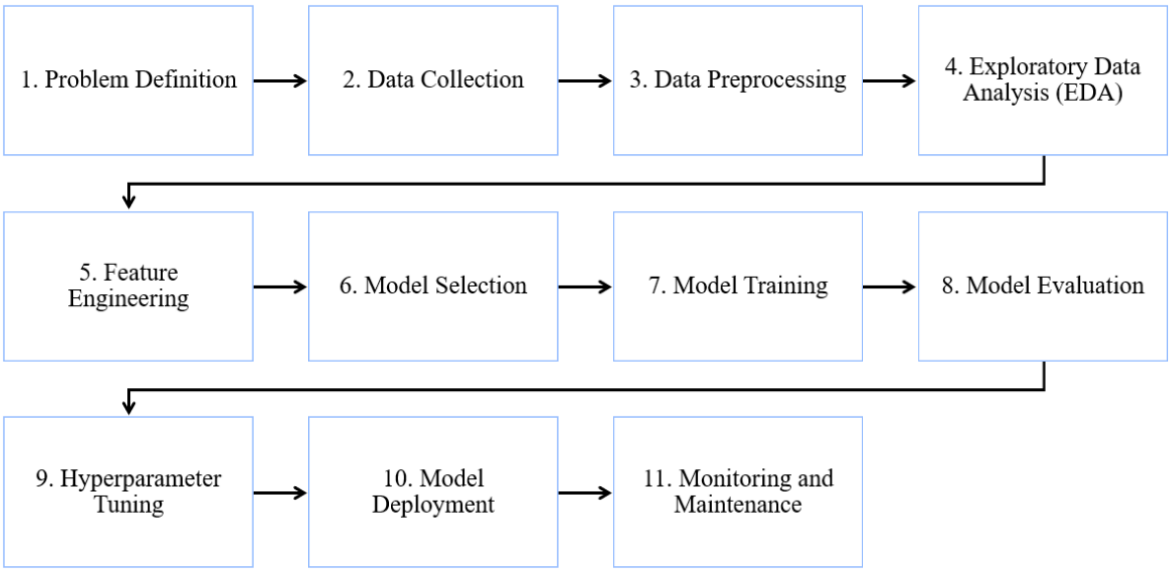


Figure 7. ML Workflow for SAP S/4HANA Item Setup

Figure 7 shows an abstracted view of the ML pipeline for the item setup process in SAP S/4HANA, as evidenced in recent frameworks that incorporate ML workflows in ERP contexts (Sharma et al. 2024; Sarferaz 2024).

This iterative process ensures that the ML model updates to reflect changes in business logic or evolving data dynamics. For example, if a business inserts new materials or changes rules, the model enables retraining at intervals to maintain accuracy and relevance. (Bhatt 2024)

Model transparency is a key enabler of trust and adoption of ML based enterprise solutions. In heavily regulated environments, stakeholders must have a clear understanding of how and why the model is outputting the responses it does. (Adenekan 2025) There are a variety of explainable AI (XAI) techniques such as Shapley Additive Explanations (SHAP), and Local Interpretable Model Agnostic Explanations (LIME) that can be incorporated into the process that clarify, the level of contribution of each feature to the model's predictive output (Höhn & Faradouris 2021). As an example, if an ML system predicted a specific MRP type for a new item, the XAI approach can show which of the attributes (e.g. product hierarchy, procurement type) were most heavily weighted in the final output, thereby building trust and promoting audibility.

Despite these potential advantages, organizations need to overcome common challenges related to this implementation of ML mechanized PDM. These include how to manage data quality and accuracy, even with legacy systems in place, managing the amount of labeled data that may be available for supervised training, and assimilating ML outputs into an

operationally managed ERP process (Shaik & Siddque 2023). In addition, organizations will need to build trust with users, especially where ML driven recommendations deviate from historical (validated) workflows. Overcoming these limitations will require careful change management, especially with technology changes, phased implementation, and hybrid architectures that promote use of ML mechanized automation, but still substantively, include the human-in-the-loop aspect of the process (Sharma et al. 2024).

In reference to establishing intelligent defaults, proactively identifying errors, and accelerating master data onboarding, ML builds an intelligent future prototypical enterprise PDM system that can be reactive, predictive, and accountable in terms of master data compliance processes. As outlined in the above sub-section, aligning the ML piece of the 'item setup' process in SAP S/4HANA fits well with what this thesis intends to deliver in terms of reducing manual workload, enabling reliable, high quality master data, and allowing scaled automation. RPA will form an operational backbone that carries out many of the structured actions that would develop or derive from ML generated outputs.

While ML provides predictive intelligence, RPA enables a direction to be delivered to the Base ERP, which forms synergistic elements to work in tandem increasing the overall automation of a process a task is being undertaken example forecasting or purchasing should involve a framework; projective measurement and accounting within the ERP while the role of ML encourages a competent management approach (Choudhuri 2024).

3.3 RPA in Industrial Workflows

RPA has rapidly become a forerunner technology for management of industrial workflows by enabling optimal use of the workflow technology ecosystem while closing gaps of current automation requirements from legacy enterprise systems not designed for robotic automation or repetitive process automation. RPA extends beyond physical robotics, with its "bots", that mimic human behaviors digitally within systems to perform structured, repetitive tasks (Madakam et al. 2019). With RPA, the digitization of manual entry of data, validation of processes and the agreement between systems offers efficiency and accuracy gains and future scalability of operations in industrial spaces (Hallikainen et al. 2018).

In the context of industrial workflows, especially those that use ERP systems like SAP S/4HANA indicate that many of the workflows, and ERP processes often include administrative repetitive tasks such as invoicing, inventory, compliance and product data entry tasks which have been traditionally enabled by humans (Bhardwaj et al. 2024). Workflow automations involving human operators can lead to material inefficiencies and potentially introduce faults that misrepresent the integrity of data and reduce productivity. RPA systematically mimics human behaviors to execute work at faster rates of speed which will lower processing times and decrease faults. In addition, RPA also allows resources to focus on higher order, analytical work or to leverage creative, problem-solving approaches to action in the workplace. (La Torre et al. 2020)

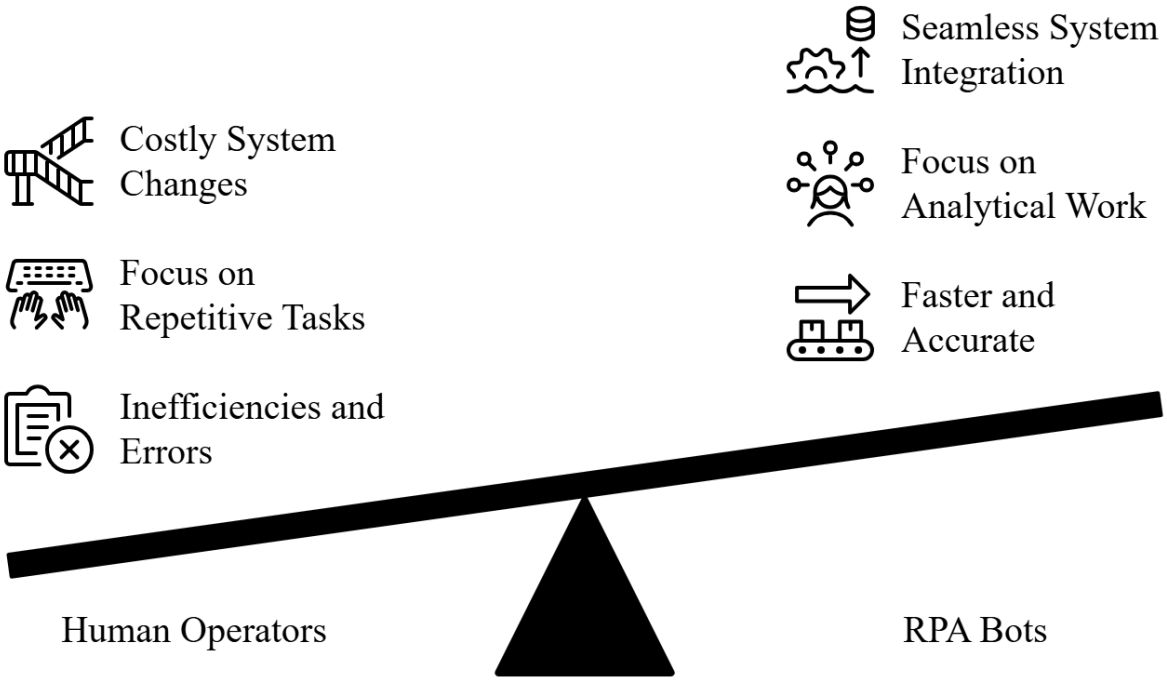


Figure 8. Human Operators vs. RPA Bots

Figure 8 illustrates the contrast between human operators and RPA bots in workflow execution. Human work is often constrained by repetitive tasks, inefficiencies, and costly system changes, whereas RPA bots enable seamless integration, deliver faster and more accurate results, and allow employees to focus on analytical and value-adding tasks. (La Torre et al. 2020).

RPA provides value in industry because of the integration of technology. RPA solutions allow communication with systems without the need for integration, or the need for API

(Czarnecki & Fettke 2021). This feature is appealing for organizations with complex IT ecosystems, and the use of RPA solutions allow organizations to automate technically challenging processes that cross multiple applications and departments without requiring additional costly changes to the technology or existing systems (Nalgozhina et al. 2023).

In the case of PDM, RPA can automate structured work that could include tasks such as material master data entry/maintenance, BOMs, ordering, and data validation (Patrício et al. 2025). For example, in work through SAP S/4HANA, a bot integrated with workflows could carry out work defined into a series of tasks, such as extracting product data from standard input files or from supplier's databases, complete views for material masters, validate chosen element against business rules, and change records in view across all relevant ERP modules. The organizational processes above would improve data consistency, lower change cycle times and complete organization controls against regulation and operation. (El-Gharib & Amyot 2022)

RPA Bots can be programmed to execute transactions such as to create material, to create BOM, to change in mass for SAP systems of record in a structured repeatable manner. Bots can interact with the SAP GUI or SAP Fiori as the medium of activity and format, allowing accurate replication of user activities. Therefore, organizations could include RPA bots in their workflow for activities such as item setup, pricing changes and product statuses without needing either to replicate the manual data entry or apply costly system programming modifications. (Viswanathan 2024) RPA has significant improvements in consistency and compliance. Automating process through RPA for workflows, is executed in accordance with predetermined rules which reduces variations and inconsistencies as dictated by human activity. This is important particularly in regulated businesses where decisions based on product outcomes must be documented according to strict procedural standards. RPA solutions log Produced Audit Trials that reinforce transparency and assist compliance reporting. (Farinha 2023)

RPA implementations usually take place along a maturity continuum for industrial organizations, where simple automation of work tasks is the first step in moving towards integrated process-level automation. Many organizations begin with small, isolated task automation, for example, simply automating a specific data entry routine or performing validation checks (Rautenstrauch & Moser 2023). Once organizations have experience with this technology, the organization's RPA implementation will typically grow incrementally,

moving toward automating complete end-to-end workflows. Moving in a gradual manner allows organizations to incorporate the use of RPA into their current working ways, realize automation benefits incrementally, and manage change more effectively (Costa et al. 2022).

Recent case studies highlight the tangible benefits that SAP-centric organizations are already realizing with RPA. Bot deployments by Zoetis to create service orders offline and automatically inject into SAP ERP resulted in 13.4 minutes saved per order, dramatically eliminating 2.5 FTEs of manual data entry, and still achieving an 82% straight-through automation rate (Viswanathan 2024). Richemont North America implemented UiPath bots to drive the logistical modules of SAP instead of repetitive keystrokes at the shipping desk enabling lights-out automation of order-to-shipment cycles and providing managers with real-time, disposal analytics-ready data to improve decision making (Percherla 2024). In the public-sector domain, National Taiwan University Hospital integrated Lean Six Sigma with RPA and reduced medical-claims processes by 380 minutes while increasing the cycle efficiency from 69% to 96% (Huang et al. 2024). Together, these examples illustrate that RPA can provide organizations with a rapid, measurable improvement in speed, accuracy and capacity, without expensive redevelopment of the SAP landscape.

Given their growing popularity an increase in available RPA platforms has occurred, and no two RPA platforms are the same. The case study discussed in this thesis focuses on UiPath because of its ability to integrate with SAP and it's a low-code application, with a user-friendly interface widely used in industrial settings towards the automation of ERP workflows. (Axmann & Harmoko 2022) Other leading RPA tools include Blue Prism known for governance and scalability, Automation Anywhere that is designed for intelligent document processing (IDP) and cloud-native deployments, and Microsoft Power Automate that provides an integrated experience within Microsoft 365 and Dynamics environments. Collectively, these tools enable organizations to create RPA implementations that are acceptable and consistent with their respective enterprise architecture and whether or not they are ready for the overall digital transformation change. (Abildtrup 2024)

3.4 Integration of ML and RPA for Process Optimization

The combination of ML and RPA provides an innovative opportunity to establish greater levels of automation and improved processes in business processes for ERP systems. ML and RPA are independently beneficial, with ML providing predictive analytics, flexible decision-making and identifying patterns, while RPA provides fast, efficient task execution for structured, repetitive tasks. The true value arises from integrating ML into business processes that are driven by RPA or IPA, or Intelligent Process Automation. With IPA, organizations can automate business processes that are both cognitive and operationally executable. (Myerson 2023) The integration of ML and RPA may be positioned theoretically within Intelligent Automation, which uses multiple technologies to automate end-to-end processes comprised of completed tasks. Intelligent Automation is organized in the progressive, yet interconnected framework consisting of three levels, with each level of automation providing a unique function when taken as a whole. (Taghipour 2024)

At the base of the framework in figure 9 below is the execution level, where RPA is to automate repetitive activities that follow a clear set of process-based rules. In this environment, RPA has provided organizations with the ability to execute tasks in a structured data environment in a high speed, error-free manner, thus laying the operational base for the automation of tasks. (Boas et al. 2025) The cognitive layer is the second layer of the framework, where ML is a component of AI. The cognitive layer adds the capabilities of decision-making, flexible and adaptable work, inference and learning from data. (Ashi et al. 2024). The business process management (BPM) layer of the framework is the top layer and is a governance level through BPM functionality. The governance layer provides organizations with accountability that automation initiatives follow the organization's strategies, compliance policies, and end-to-end visibility over processes. (Lakkhongkha 2024)

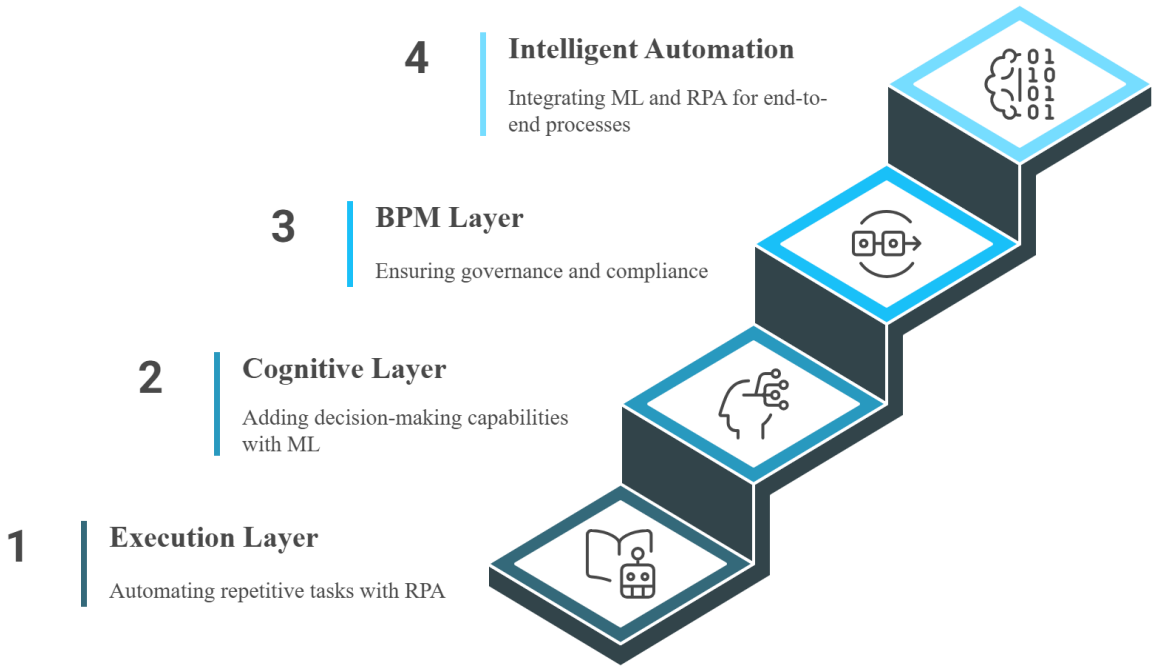


Figure 9. Intelligent Automation Framework

In many frameworks using integration, the cognitive layer consists of ML for prediction, analysis, and intelligent or informed decision-making, while RPA provides the operational execution layer, executing structured tasks and transactions reliably. ML augments the capabilities of traditional RPA by adding an element of learning and adaptability to manage the decisions made in a supporting way to the static, rule-based automation associated with RPA. (Tanase et al. 2025). The RPA layer also allows the operationalization of the insights produced by ML - proposing execution across PDM and ERP system. The two integrate to add a significant number of scopes to automate processes and allow the organization to manage business processes from an end-to-end perspective while reducing the involvement of manual individuals. (Metzker 2025)

Typically, the implementation of ML with RPA follows a stage-wise and modular architecture. The SAP S/4HANA item set up processes are a clear example of this architecture. The first stage is where ML models are trained on historical product data and predict important material parameters example material groups, procurement types, MRP controllers, valuation classes, storage conditions based on item characteristics and previous item set up (Singh 2025). The outputs from this stage are presented in standard formats like JSON which provide a structure for data interfaces. The next stage is where RPA bots consume the structured, ML-generated data and automatically enter information into fields of the relevant SAP (Khalimonov & Sotnik 2024). Bots are also responsible for validating

data entries against business rules, handling exceptions, and reporting results to users. The bot's activity will ensure acceptable data quality, reduce data entry errors by operators, and ultimately automates the process (Tanase et al. 2025).

At first as in figure 10 below, the historical data from the SAP S/4HANA material master's is pre-processed, and an ML model, such as a decision tree, random forest or neural network, would predict the most appropriate parameters for "new" items (Janiesch et al. 2022). Then an RPA bot built with UiPath fetches the predicted parameters and validates them, then executes the relevant transactions in SAP S/4HANA and creates new material master data (Willcocks et al. 2022). The entire integrated process will eliminate the highly manual, not to mention a slow process of setting up items with the highest degree of "artificial intelligence," reduces time from weeks/days to hours/minutes, increase data integrity and data compliance (Jain et al. 2024).

Process can be viewed in both perspectives, one is the layered architecture, and the other view is to consider the level of automation, particularly the interaction with the human and machine. In the automation model defined by Parasuraman et al. (2000), the levels of automation vary from low levels- systems that aide humans complete their tasks, to higher levels, where the machines make autonomous decisions and actions. The ML-RPA is generally operated in higher levels of automation, where the ML system directed human-based decisions (intelligence-based recommendations), or where the RPA could execute decisions autonomously (Muthusamy et al. 2020).



Figure 10. High Level End-to-End Workflow of ML–RPA Integration for SAP S/4HANA

However, in high-risk areas or those with various ethical ramifications, total automation is inappropriate, and human-in-the-loop (HITL) is still needed to maintain human accountability, transparency, and trust. Intelligent automation systems should be designed to augment rather than replace human judgement and foster responsible and productive collaborations between humans and machines (Fischer 2024).

These advantages can also be viewed through Herbert Simon's (1960) classic model of the decision process, which separates the decisions into four sequential phases: intelligence, design, choice, and implementation. The intelligence phase includes their information

gathering, problem identification, and opportunity recognition. The design phase involves the generation and analysis of alternative solutions. The choice phase focuses on selecting the right decision and the implementation phase concludes the decision-making process by translating the decision into action. Within this model, ML will help with the intelligence and design phases, as ML can sift through large amounts of data to identify patterns, detect anomalies, and provide predictive information which aids in finding relevant problems and generating alternative solutions to improve the quality of decisions. (Janiesch et al. 2022) While RPA will connect primarily to the choice and implementation stages of decision making, RPA bots represent the operational actions of implementing those decisions once the decision has been made whether a human, a rule-based system, or a ML model made the choice. The RPA Bots do the operations work of entering, validating, configuring parameters, completion of tasks, data interfacing between systems etc. The RPA & ML mapping to the decision-making process and Simon's model provide an excellent conceptual map for understanding how RPA and ML are necessary to provide an efficient automation link from insight to action and highlight the cognitive aspects of decisions from ML and operationalize it with RPA (Janiesch et al. 2022; Willcocks et al. 2022). There are industry examples that illustrate these advantages. For example, predictive maintenance workflows use ML to predict equipment breakdown by analyzing sensor data, while RPA automatically generates the SAP maintenance order, which is clearly good for reducing downtime costs. ML, and RPA can also work together using predictive pricing and dynamic attribute selection within a material set-up to allow real-time adjustment to the market - seamlessly executed with automated SAP transactions (Jain et al. 2024)

With a range of clear benefits to using ML with RPA, the integration of ML with RPA has specific technical and organizational issues to consider. Technically, proper interface design and careful consideration of interoperability between predictive models and ERP transaction structures facilitate the transfer of data between ML predictions and RPA actions. Monitoring of system performance and feedback loops for both ML and RPA are necessary to keep up with changes and ensure continuing accuracy of ML predictions over time and RPA execution processes. Organizationally, change management is important in relation to end user acceptance, understanding of automation values, and continued training of users who interact with the automated solutions. (Dolata & de Lange 2021) The governance of ML models and RPA scripts, including version control, needs to ensure effective systems and service sustainability, responsibility, accountability, and maintainability in line with

compliance guidelines and standards (Willcocks et al. 2022). Seen through the lens of Sociotechnical Systems (STS) theory we see that using ML with RPA is not only technical but is a reshaping of work systems. STS recognizes the interdependence of social systems like people, culture, workflows and technical systems example ML, RPA and ERP, so the effective integration of these two domains involves reconciling social and technical ways of working, engaging with users and re-negotiating roles, and ensuring trust not only in AI-driven outputs but also with RPA technologies (Dolata & de Lange 2021; Müller et al. 2025).

The quality of the data which underpins ML-RPA integration will be fundamental to its success. ML models use data that is correct, relevant and comprehensible to deliver predictive output of value. On the other hand, poor quality data (inaccurate or incomplete) leads to worthless predictive output causing flawed action by the RPA workflows. Developing robust data governance frameworks to include procedures for data validation, data standards and data anomaly detection is hence a necessary precursor to the effective integration and continuous development of integrated ML-RPA workflows (De Silva & Alahakoon 2021). Ongoing developments will continue to advance the applications for more effective ML-RPA integrations. The rise of intelligent automation, which integrates AI, ML, RPA, and advanced analytics, has already led to advanced automation that integrates complex decisions of this automation and decision-making process in IoT contexts, cloud environments, and with edge computing technology. Concurrently, there is burgeoning interest in using generative AI technologies, which can blend data, automatically write scripts and learn adaptively, that may support deeper integrated automation in ERPs, thereby improving process responsiveness and the quality of decisions (Rajput & Gupta 2023).

While the convenience of increasing automation reduces complexities associated with decision-making, for decision-makers, an increased reliance on automated decision-making raises distinct challenges regarding elements of trust, transparency, and ethics. Key safeguards to build trust in intelligent systems include explainable ML predictions, transparent automation logic or process, and oversight by a human being. Importantly, ethics associated with ML-RPA integration also must consider issues of bias, accountability, and consent from the user, which should be included in the deployment and development processes, especially in the case of critical systems including ERP (Munoz et al. 2023). With the combination of a layered framework, article on automation-level theory, and structured decision-making models, we see that integrating ML-RPA is not just a technical pathway

but a strategic information infrastructure enabler for an adaptive, intelligent, and ethically informed business processes transformation (Petkovic 2022).

The tactical combination of ML and RPA affords transformational possibilities for enterprise process improvement most commonly in ERP contexts like SAP S/4HANA. The combination of intelligent predictive abilities and providing reliable execution through RPA improves process efficiency, accuracy, scalability, and flexibility by facilitating the integration of ML-RPA. The use of ML-RPA specifically for item setup processes will have measurable benefits for organizations that require more manual labor to perform the task, less good data from poor records, and will provide organizations with an ability to respond to changing activity situations in a timely manner. Overall, ML-reliant RPA represents an important step towards fully IPA, specifically, toward organizations attaining Industry 4.0 related benefits and overall industry competitiveness in their digital transformation initiatives. (Wu et al. 2025)

4. Methodology

This chapter establishes the methodological foundation of the study and the approach for addressing the research questions. The overarching framework is Design Science Research (DSR), which is well suited for designing and evaluating an artefact in a real organizational environment. Section 4.1 explains the rationale for selecting DSR, while section 4.2 presents the applied process model. Sections 4.3-4.4 address the problem context and define the solution objectives. Sections 4.5 and 4.6 focus on the design, development, and demonstration of the artefact, forming the basis for the discussion in Chapter 5 and the conclusions in Chapter 6.

4.1 Design Science Research

DSR as research methodology aims to develop a concrete solution to an identified business problem while simultaneously generating new knowledge through design and development. DSR differs from traditional descriptive or explanatory research methods in that its primary focus lies in creating and evaluating a novel artifact. This artifact can take the form of a software application, model, process, or method designed to solve a practical problem while also contributing generalizable knowledge. Such an approach is particularly well suited to technology-driven development initiatives that seek to deliver both practical utility and academic contribution. (Hevner & vom Brocke 2023)

DSR is particularly appropriate when the goal of the research is to build something new rather than merely understand existing phenomena. It is also applicable in cases where the research problem is tightly linked to an organizational context and cannot be abstracted without losing essential characteristics. At the core of DSR is an iterative development process, where the relationship between the problem and the solution is explored step by step, and where design decisions are refined based on empirical observations. The methodology is not purely deductive or inductive, but rather abductive, meaning it continuously moves between theory and practice in search of the most plausible and effective solution. (Stroet 2022)

The DSR methodology is strongly grounded in the philosophy of pragmatism, where the value of knowledge is defined by its practical utility. Pragmatism emphasizes applicability and functionality, suggesting that knowledge has value when it provides solutions to real-world problems. This aligns well with the principles of DSR, as artifact development is driven by practical needs, and the success of the research is evaluated based on how well the developed solution performs in a real-world environment. (Goldkuhl 2011)

In this thesis, DSR provides a systematic and structured framework for designing, implementing, and evaluating an intelligent automation tool. The objective is to develop a solution that leverages ML and RPA to automate the item setup process in the SAP S/4HANA system. In the case company, the item setup process is manual, rule-based, and repetitive. It is time-consuming and prone to human error, making it a typical business problem suitable for DSR-based development.

The tool being developed aims to integrate two complementary technologies: ML is used to predict item-related parameters, while RPA handles the automatic input of these predicted fields into the SAP system. This combination allows for both improved intelligence and performance in the process. The DSR methodology supports the step-by-step development of such a solution, enabling the evaluation of both the artifact's functionality and its business value at each stage (March & Storey 2008).

The methodological foundation of this thesis is the six-phase DSR process presented by Peffers et al. (2007). The process proceeds through the following phases: (1) problem identification and motivation, (2) definition of the solution objectives, (3) design and development of the artifact, (4) demonstration of the artifact, (5) evaluation, and (6) communication of the results. This model emphasizes logical and systematic progress and encourages critical evaluation of the artifact from multiple perspectives. It also supports an iterative approach in which phases may be revisited and refined based on insights gained during development. (Hevner & vom Brocke 2023) The DSR process is visualized in Figure 11.

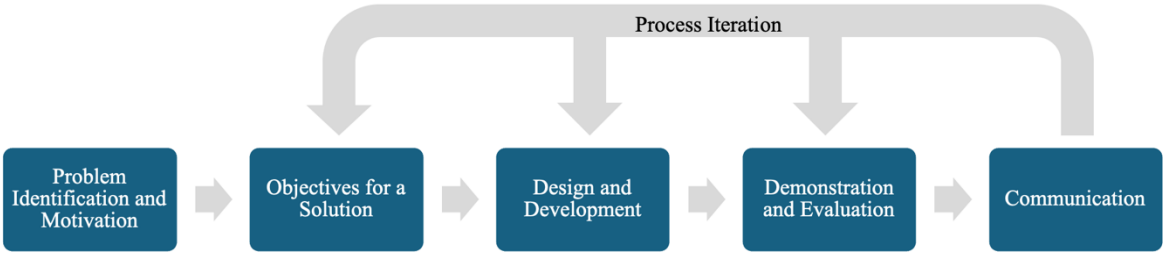


Figure 11. The phases of the DSR methodology.

One of the strengths of DSR lies in its ability to combine design-oriented development with scientific research. The artifact is not developed in isolation from the theoretical framework; rather, choices related to technologies, methods, and architectures are justified through literature and prior research. This enables the generation of scientifically relevant knowledge about which solutions work in a given context and why (Kroop 2025).

DSR has been selected for this thesis primarily because it enables a practical yet scientifically rigorous approach to solving the research problem (Hevner & vom Brocke 2023). The goal of the thesis is not only to demonstrate that combining ML and RPA is technically feasible but also to show that such a solution creates concrete business value. DSR provides a means to both construct this solution and evaluate its impact from technical, business, and academic perspectives (Kroop 2025).

4.2 Process Model and Application

This chapter outlines the practical implementation process used to develop the automation solution for item setup in SAP S/4HANA. The work was structured according to the phases of DSR, which guided the identification of the problem, the definition of solution objectives, the development of the artifact, and its subsequent testing and evaluation. The need for the solution arose from an inefficiency in the company's material master data process. Item setup, despite being essential to material management, was performed manually, leading to frequent errors, inconsistent data quality, and time-consuming operations. The manual nature of the task also made the process difficult to scale and prone to user-specific variability.

Table 5. DSR phases, project activities, and evidence mapped to the research questions

DSR phase	**Concrete activities in this thesis**	**Artifacts and evidence**	**Link to RQs**
1. Problem identification and motivation	Interviews, manual flow, baseline lead time and errors	Process map, baseline metrics, requirements list	Grounds RQ2 and motivates RQ1
2. Objectives of a solution	Define KPIs and acceptance thresholds	KPI table, success criteria	Sets evaluation for RQ2
3. Design and development	Train and package ML exe, design Orchestrator queue, build UiPath robots	ML exe, configs, queue setup, XAML workflows	Answers RQ1
4. Demonstration	Pilot runs on selected items	Run logs, demo datasets, screenshots, sign-off notes	Supports RQ1
5. Evaluation	Measure accuracy, lead time, collect user feedback	Statistical summaries feedback synthesis	Answers RQ2
6. Communication	Document results, prepare handover	Thesis, presentation, ops/update guides, versioned repos	Supports RQ1 and RQ2

The table 5 illustrates how each phase of the DSR process was operationalized in this thesis, what evidence was generated, and how these outputs contribute to answering the research questions. The practical need for the solution was identified already before the official start of the thesis, during a prior role at case company. The initial idea was born through discussions with the team lead on how AI could support the team's processes. To further investigate this, interviews and meetings were conducted with both team members and representatives from other internal units who were actively involved in item-related projects. In total, ten experts were interviewed. The aim was to explore their responsibilities and surface recurring challenges. Based on these insights, a summary of identified problems was compiled and presented in a cross-functional meeting. The outcome of this workshop led to a targeted Microsoft Forms survey, the results of which confirmed that the item setup process in SAP S/4HANA was considered the most impactful area for improvement. This research builds directly upon that foundation.

To gain a more detailed understanding of the process, follow-up interviews were conducted, and selected workflows were recorded. This allowed a structured analysis of the manual process steps without continuous resource involvement. As a result of these early efforts, a

proof-of-concept version of the solution was developed prior to the formal thesis project, forming the basis for the artifact described in this study.

To address the identified need, a two-part artifact was developed. The first component is a ML model implemented in Python, designed to predict multiple item fields simultaneously based on historical master data. The second component is a RPA workflow built using the UiPath platform, responsible for executing the predicted setup values within the SAP S/4HANA system.

The components were designed to be modular and loosely coupled through file-based integration, allowing independent updates and maintenance. The system architecture enables local ML execution and centralized automation, balancing usability and scalability. The ML artifact was iteratively developed and tested using historical item dataset. The RPA artifact was tested using both valid and invalid example items. The tool was evaluated from both technical and business perspectives to ensure it met the intended goals and delivered measurable benefits.

4.3 Problem Identification and Motivation

This thesis is grounded in a well-recognized and recurring challenge within the case company's material master data process, specifically related to the creation of new material items in the SAP S/4HANA environment. Although the creation of item is typically considered a routine administrative task, its impact extends far beyond the initial data entry phase. Inaccuracies or inconsistencies introduced during setup often propagate into downstream functions such as procurement, production, logistics, inventory management, and sales, resulting in inefficiencies that are difficult to trace back to their origin. (Božić 2024)

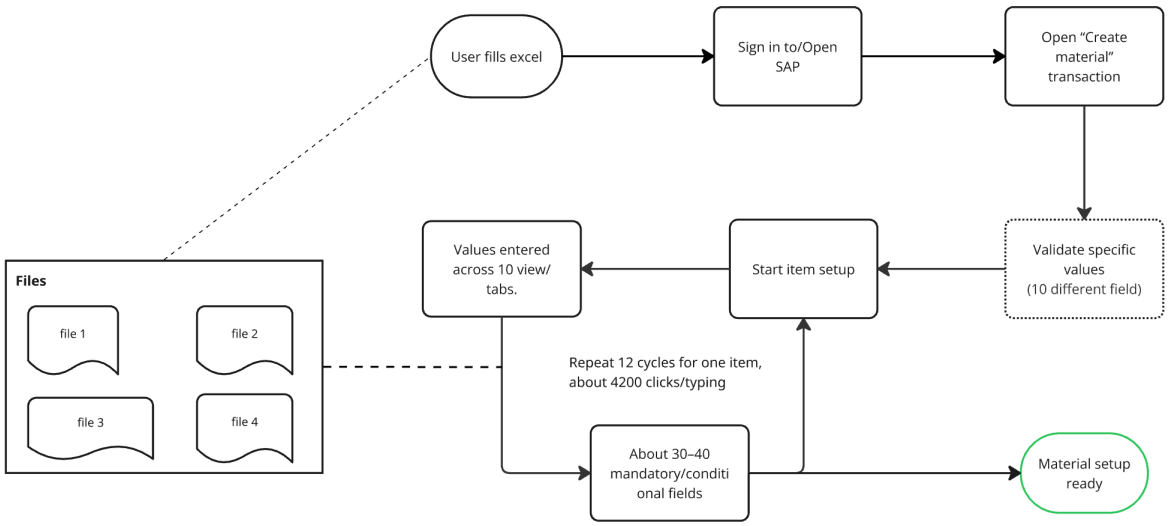


Figure 12. Current Manual Item Setup Process in Case Company

Figure 12 shows in high level current manual item setup process in case company. The user prepares Excel, signs in to SAP, opens “Create Material” transaction, enters values across roughly ten views or tabs with about 30–40 mandatory or conditional fields, performs manual checks for around ten key fields, and completes the setup. Dashed lines indicate supporting files, which user’s use in item setup process. A dashed-outline box indicates a step the user should perform. Although this check is often skipped because it is assumed to be ‘always correct’, it should still be carried out.

In practice, the item setup process was carried out manually using SAP’s graphical user interface. Each item required the accurate completion of several mandatory fields, and the selection of correct values depended on various combinations of material parameters. While some values could theoretically be inferred from existing records in SAP, the lack of automation or recommendation features meant that users had to rely on their own experience, written guidelines, and previous examples. This led to inconsistent interpretations and field entries, resulting in significant variation in data quality and consistency across users and over time. As shown in Figure 13, after product data is transferred to the ERP system, it has an impact on multiple other systems.

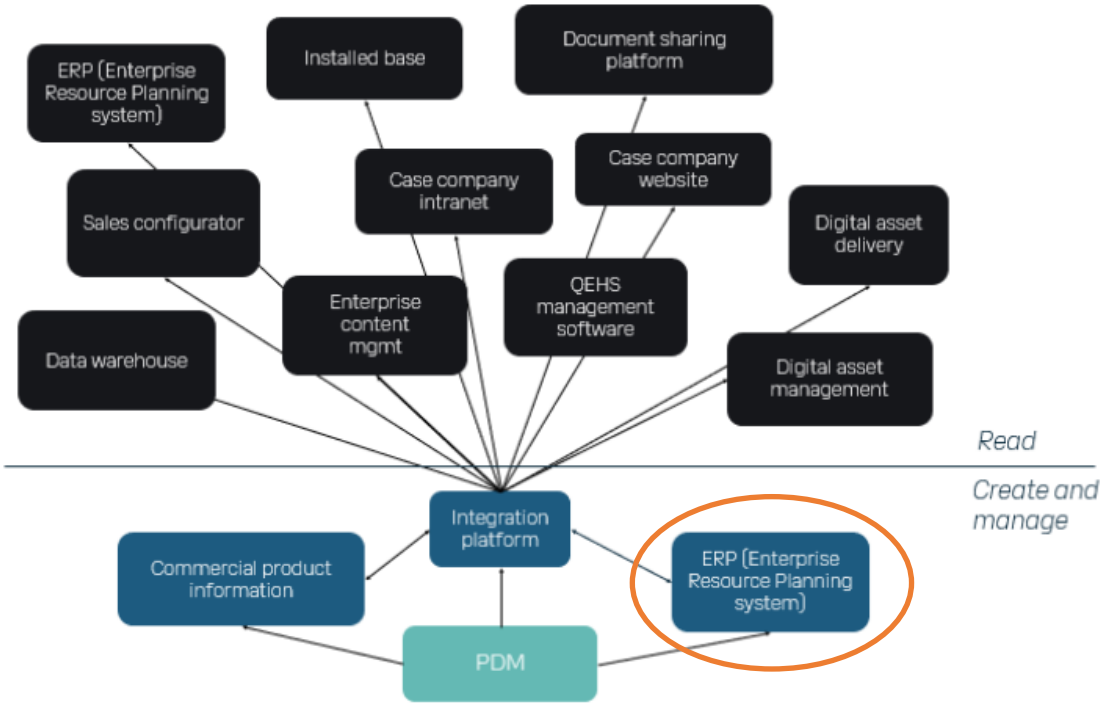


Figure 13. Product Master Data in Case Company, Simplified Architecture.

From a business perspective, the absence of standardization had tangible consequences. Incorrect or incomplete item records caused issues such as failed purchasing transactions, inaccurate inventory reporting, and sales availability errors. Even minor data discrepancies, such as incorrect plant assignments or classification codes, could lead to operational delays, reporting difficulties, or in the worst cases, the need to delete and recreate item. The process was also slow and difficult to scale. New employees required substantial onboarding time, and the lack of centralized logic increased dependency on experienced individuals.

Table 6. Evidence triangulation for problem identification and motivation

Evidence source	**Method & scope**	**Key findings**
User interviews	10 unstructured interviews	• High cognitive load for field selection • Reliance on old instructions and colleagues • Inconsistent logic across users
Stakeholder survey	Microsoft Forms, task burden ranking	• Item creation = 24% of all votes • Parameter checking second together 36% → highest workload and error sensitivity
Baseline observation	Manual item setup process in SAP	• Many mandatory fields • Slow onboarding • Dependence on experienced users
Proof-of-concept	Python, historical item data	• Sufficient signal for prediction Users validated suggestion realism

The table 6 consolidates the main evidence used to justify the problem and define solution requirements. It brings together four sources user interviews, a stakeholder survey, baseline observation of the manual item setup process in SAP, and a Python PoC. For each source, summarizes the method and scope, and the key findings.

Table 7. Annual count by material type

Material Type	**Annual volume, (N)**
Material Type 1	61
Material Type 2	1535
Material Type 3	153
Material Type 4	137
Material Type 5	184
Total	**2070**

Table 7 summarizes the number of new items created by material type during the 12-month period (N = 2,070). This period is used as the annual baseline in this study. For workload estimation, the material types were mapped into two complexity classes, Simple and

Complex item. Combining these volumes with the observed average handling times per class yields the annual workload shown in Table 7. The calculation follows:

A simple item is an item whose setup can be completed without consulting the work instruction or asking a colleague for help. A complex item is the opposite, where the item creator needs to seek assistance from a colleague or follow the work instruction during setup. The annual split between simple and complex items varies considerably, so the annual volume is estimated using a 50/50 mix. Under this assumption as shown in table 8, manual item setup requires about 431 hours per year (≈ 57.5 working days, 11.5 weeks, or 2.9 months of full-time effort).

Table 8. Baseline annual workload of manual item setup by complexity

	Simple Item (10 min), (t)	**Complex Item (15 min), (t)**	Simple and Complex, (t)
Minutes	20700	31035.0	25867.5
Working days	46.0	69.0	57.5
Working weeks	9.2	13.8	11.5
Working months	2.3	3.4	2.9
Hours	**345.0**	**517.3**	**431.1**

Baseline data quality was quantified at the field level using the historical dataset of unique material-type combinations. Errors were detected by comparing SAP master-data field values with a curated reference built from cleaned training data via rule-based validation and expert review. As shown in Table 9, the dataset contains 6743985 field entries, of which 539519 are incorrect, yielding an error rate of 8%.

Table 9. Baseline field-level accuracy from historical data (counts and rates)

	Correct	**Incorrect**	**Overall**
Sum	6204466	539519	6743985
%	92%	8%	

To translate field-level accuracy into annual impact, we map data quality to yearly item volumes. With 51 parameters per item and 2070 items/year, there are 105 570 field entries annually. Applying the baseline error rate gives 8 446 incorrect fields/year. This forms the basis for rework estimation in Table 10.

Table 10. Estimated yearly rework due to baseline field errors (scenario analysis)

Fix time (min)	Hours/year	Working days	Working Weeks	Working Months
1	**140.8**	18.8	3.75	0.94
5	**703.8**	93.8	18.8	4.69
10	**1 407.6**	187.7	37.5	9.38

The table 10 shows that, even at a one-minute fix, baseline field errors cause about 141 hours of rework per year. At five to ten minutes, the burden rises to about 704–1408 hours. These results confirm that item creation errors are a material business problem and justify an automation approach that standardizes field selection and validates critical parameters before save. They also define the baseline for RQ2.

The problem was initially identified informally and later investigated more systematically. During my previous role at case company, I became familiar with the team's daily tasks and noticed the repetitive and error-prone nature of the manual item creation workflow. In everyday discussions with colleagues and the team lead, the idea emerged to explore the potential of using AI to support field selection and decision-making. These early conversations sparked a more structured investigation during the scoping phase of this thesis.

To better understand user pain points, a qualitative data collection effort was carried out. Ten unstructured interviews were conducted with employees from data management, supply chain operations, and production, individuals who were directly responsible for creating or maintaining item records. The interviews explored their daily responsibilities, data entry practices, common challenges, and improvement suggestions. The results consistently highlighted similar issues, such as the cognitive effort required to select correct values, the trial-and-error nature of the process, and the difficulty of maintaining consistent logic. Many participants reported relying on old records, consulting colleagues, or simply guessing values.

To validate and quantify the interview findings, a Microsoft Forms survey was distributed to a broader group of stakeholders. Respondents were asked to choose three of the 17 tasks, identified through the interviews, that they considered most time-consuming and frustrating. The results aligned with the interview observations. The material item creation process received 24 percent of all votes, making it the most burdensome task. The second-highest number of votes went to checking material parameters, an area directly addressed by the solution developed in this thesis. Together, these two tasks accounted for 36 percent of all

responses. Among the many candidate tasks, item creation and data verification ranked highest in terms of workload, error sensitivity, and business impact.

These findings fulfilled the criteria of the first phase of the DSR methodology (Hevner & vom Brocke 2023). The process was well-defined, repetitive, and data-driven, making it a strong candidate for a ML and automation-based solution (Patricio et al. 2024). High user interest and the strategic relevance of improving data quality further reinforced the motivation for this research.

In parallel with the research, an initial Proof of Concept (PoC) was developed to evaluate technical feasibility. The PoC was implemented in Python and used a cleaned dataset of historical item records to test rule-based logic and simple classification algorithms for field prediction. Although the prototype lacked a complete user interface or SAP integration, it served two critical purposes. First, it provided a tangible artifact for stakeholder discussions, allowing users to evaluate the realism of automated field suggestions. Second, it demonstrated that the existing data contained enough structure and signal to support predictive modeling. Encouraged by these results, the project transitioned from the exploration phase into formal artifact development.

In summary, the item setup process in SAP S/4HANA represents a clearly defined, business-critical problem characterized by manual complexity, user-specific variation, and significant downstream consequences. The combination of expert interviews, structured surveys, and PoC experimentation created a strong foundation for this research. These efforts validated the need for a solution and ensured that the problem addressed in this thesis is based on real business needs, aligned with user expectations, and solvable using modern intelligent automation technologies.

4.4 Objectives for a Solution

In the second phase, the objectives for a successful, production-ready solution were defined. The solution covers end to end item setup for selected material types and organizational units in SAP S/4HANA. It follows a modular architecture in which a locally executed ML application is separated from a centrally operated RPA robot and the two components exchange data through a versioned file-based handover. The local application is packaged with PyInstaller in one folder mode, so it runs without a separate Python environment, while the robot operates on the approved remote desktop server under Orchestrator control and schedules. The pilot serves as the controlled path to production for the agreed scope, and the architecture and quality objectives are defined to meet production requirements for reliability, security, maintainability and operability.

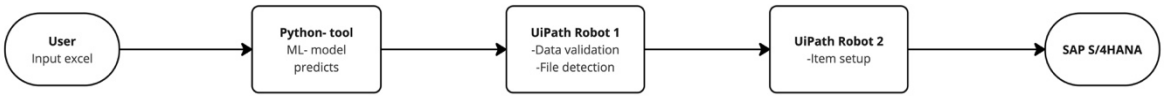


Figure 14. High Level Workflow of Solution

Figure 14 shows the high level workflow process visualization. Functionally the tool must accept user provided item data in a predefined Excel template, validate the presence and plausibility of required fields, and standardize formats and codes needed for prediction and posting. The ML component predicts multiple fields simultaneously from historical product data. The application produces an output file that includes pass-through and predicted values, together with confidence indicators and status flags, so that users can, when necessary, review or override values before triggering RPA posting or performing manual setup. The RPA component reads the output from a predefined folder at scheduled intervals, navigates the SAP interface to complete item creation across the agreed transactions and views, and records identifiers and status for full traceability. Both components generate structured logs with timestamps, file references, outcomes and error details to support auditing and troubleshooting. Robots can run on a schedule outside business hours, which frees the manual effort entirely.

Nonfunctional objectives focus on reliability, performance, maintainability, portability and compliance. The end-to-end workflow includes a mandatory human in the loop step: users must review the generated Excel output and, if needed, make corrections before moving the

file to the designated folder for RPA processing. After this review, valid items are expected to complete automatically without further manual intervention, while invalid inputs fail predictably with informative messages. Prediction for a typical batch must finish in practical time on a standard office workstation, and RPA throughput must meet the daily workload benchmarks agreed with stakeholders. The architecture enables independent updates of the model and the robot through documented retraining and versioning practices on the ML side and parameterized selectors and configurations on the RPA side. Processing follows company policies for data handling and access control, with files handled in controlled folders and credentials managed through approved mechanisms. During the pilot the tool undergoes iterative user driven testing in which each release is exercised by real users through task walkthroughs and a short survey, feedback is incorporated into the next version within a defined turnaround time, and measurable improvement between cycles is required. At least three release, feedback and revision cycles are completed, and iteration continues until the agreed targets for prediction quality, straight through processing and perceived usability are met.

Data and model objectives require that the training dataset be representative of the in scope material types and organizational structures and that known quality issues be mitigated during preprocessing. Prediction quality is reported with per field classification metrics, including accuracy and F1 score. The model lifecycle includes monitoring for distribution shifts, clear triggers for retraining, validation procedures before promotion and documented rollback conditions in case of regression.

Integration and RPA objectives emphasize loose coupling, resilience and observability. The handover between ML and RPA remains file based and versioned to preserve backward compatibility where feasible. The robot tolerates transient SAP states through limited retries, maintains checkpoints between steps and behaves idempotently so that reprocessing does not create duplicate items. Scheduling, status visibility and exception reporting are handled in Orchestrator so operations can track success, and exception counts with sufficient timeliness for support and control.

Usability and adoption objectives ensure that the local application offers a clear workflow with minimal steps for prediction and export and that instructions and validation feedback are written in plain language for non-technical users. Human-in-the-loop review allows gradual adoption and risk control for sensitive fields. Concise user guides and quick start

instructions are provided for end users, while operational runbooks support personnel. Feedback loops continue after deployment using lightweight surveys at the end of each pilot cycle with the expectation of increasing user reported clarity and task success and decreasing manual overrides and rework.

Business objectives align the artefact with measurable outcomes that matter to stakeholders. The solution is expected to reduce manual effort in field entry, lower rework and correction rates, shorten end to end cycle time from input to posted item, increase straight through processing rate at the RPA stage and improve perceived usability. Constraints and assumptions include access to SAP test and production environments with the necessary roles, availability of historical master data for training and validation, execution on standard company workstations for the local application and operation of the robot on the approved remote desktop server. Processing must comply with company information security and data protection requirements, and shared locations used for handover must be governed.

Key risks include data drift and changes to master-data taxonomies, changes to the SAP user interface or process, user adoption challenges, and changes to the file handoff schema between the ML application and the RPA robot. These risks are mitigated through routine monitoring and periodic model retraining, robust UI selector and configuration guidelines backed by regression tests, an optional review mode with clear instructions and training materials, and versioned handoff schemas with validation on both sides to prevent undetected incompatibilities.

Only the chosen material types are in scope, and all other material types are excluded. Integration is limited to the specified file-based handover, and alternative patterns are not considered. Enterprise-wide process re-engineering and SAP customizing changes are also excluded. The solution is considered ready for production when, within the agreed scope, functional coverage is demonstrated for the selected item types and views. ML performance on high impact fields must meet the agreed per field targets. The RPA stage must attain the agreed straight through processing rate on valid inputs. End to end cycle time and error rates must satisfy the agreed improvement thresholds. User feedback must confirm sufficient clarity and usability across the required iteration cycles. Operational prerequisites must be in place, including logging, monitoring, access controls and runbooks.

4.5 Design and Development of Artifact

The artifact was designed as a two-component solution that separates intelligent decision making from transactional execution. A locally executed ML application prepares and predicts item attributes, while a centrally hosted RPA workflow posts the results to SAP S/4HANA. The components communicate through a versioned, file-based handover, which keeps the architecture modular, maintainable and resilient to change. This design balances practicality with rigor: local execution reduces friction for users and avoids workstation-level dependency management, central orchestration ensures controlled and observable posting, and the file-based handover keeps the coupling loose with predictable failure modes. Validation, confidence flags, logging, explicit rules and idempotent processing further ensure that the workflow is transparent, supportable and able to evolve as the scope grows, with retrainable models, versioned schemas and extendable automation.

ML-based Application

The ML application is implemented in Python and performs multi-field prediction on historical item master data using a Random Forest model wrapped in a MultiOutputClassifier. Random Forest was chosen because it effectively models categorical and non-linear dependencies without extensive feature engineering, provides reliable probability estimates for confidence thresholds, and delivers competitive accuracy with minimal hyperparameter tuning and predictable resource usage. In my previous role at the case company, I conducted a comparison of KNN, XGBoost, Random Forest, and multi-output logistic regression models using F1 score and accuracy, where Random Forest offered the most balanced overall performance with the least tuning effort.

Preprocessing standardizes inputs by normalizing codes, harmonizing synonyms, casting types and resolving known data-quality issues. Categorical variables are encoded, and multi-valued categorical inputs are handled with a custom multi-hot encoder so that multiple active categories are represented faithfully rather than forced into a single one-hot choice. Missing values are handled explicitly, and class imbalance is addressed during training. Hyperparameters were tuned through systematic search with cross-validation, and model performance is tracked with per-attribute metrics such as accuracy and F1 score. Training produces a compact validation report that is stored with the model artefacts for traceability. Figure 15 shows the high level process flow of ML application.



Figure 15. High Level Process Flow of ML Application

The application is packaged with PyInstaller in one-folder mode so that users can run it locally without a separate Python environment. All required libraries are bundled, and the build process embeds a semantic version and build timestamp that are written into the prediction file and logs. Operational behavior is driven by yaml configuration files that define settings, column rules, paths and technical parameters. The application detects at runtime whether it is running as a PyInstaller executable or as a script and resolves base directories and path templates accordingly, with documented fallbacks if configuration loading fails. Technical defaults favor single-threaded prediction and memory-mapped model loading for packaged compatibility and predictable resource use.

The user interface follows a short, linear workflow: the user opens the predefined Excel template, completes the required fields and executes the tool. The application then ingests the file, validates its structure and content, standardizes codes, runs the ML prediction, applies the business rules and writes the formatted Excel output with confidence indicators for mandatory review. Validation applies structural checks on required columns, type checks on critical fields and plausibility rules that reflect business constraints. In addition to generic checks, the tool enforces gatekeeper logic by processing only rows that contain a value for a required key field (a specific control field), and it performs dash-split preprocessing on identifier-like inputs so that only the prefix before the dash is used, aligning with canonical training forms. After successful validation, the application standardizes codes and formats and harmonizes synonyms to prepare the data for modelling.

The prediction stage writes both pass-through and predicted values and includes confidence indicators and status flags, after which deterministic business rules are applied on top of the raw predictions to capture policy-driven decisions such as key-field-based mappings and header-row propagation for designated fields. If critical issues are detected, such as a modified input Excel structure, conflicting field values, or input Excel being open during execution, the application halts execution and notifies the user with a clear error message. Non-critical warnings are recorded for review but do not interrupt processing. To support mandatory human review, the Excel output highlights low-confidence suggestions by

shading cells when predicted probability falls below 0.90. The threshold reduces the likelihood of incorrect auto-acceptances and keeps the manual review workload manageable. It can be adjusted during retraining to reflect changes in business priorities. SMOTE was not applied because the dataset is dominated by categorical variables, for which synthetic interpolation does not generate meaningful values. Therefore, we adopted a simpler and more controlled strategy with random oversampling of minority values combined with class weighting and confidence thresholding. Formatting rules keep header-only fields readable, and the output workbook uses a fixed, versioned set of columns that covers inputs, predicted targets and operational metadata.

The development of the ML application progressed iteratively. Four releases were executed, exceeding the minimum requirement of three iterations defined in Section 4.4. The first release established the basic pipeline from file ingestion, validation and standardization to multi-field prediction and Excel export for handover. The second release focused on improving usability and robustness by enhancing validation messages, clarifying field-level feedback, and tightening error handling and logging. At the same time, it improved prediction quality as well as the readability of the output file's user interface. The third release refined the output format, introduced confidence flags to support mandatory review, and further improved prediction quality and output readability. In this release, performance was also optimized by complementing the xlsx-based process with csv-based handling. The fourth release introduced survey-driven enhancements to the user interface. Each release was tested with real users in the intended workflow. Feedback was collected through a Microsoft Forms survey, and agreed changes were prioritized into the next version. Packaging and updates were kept simple so that users could replace the local folder without complex installation steps.

RPA Workflow

The RPA layer consists of two coordinated unattended UiPath robots that run on the company's approved remote desktop server. Both robots are orchestrated centrally and started by UiPath Orchestrator according to defined schedules, and the second robot can also be triggered when new queue items become available. Orchestrator provisions the Windows session on the remote desktop host, applies credential assets, and records job logs for auditability. This centralized and scheduled setup ensures that execution does not depend on

end user workstations and that run windows, maintenance periods and retry policies are enforced consistently.

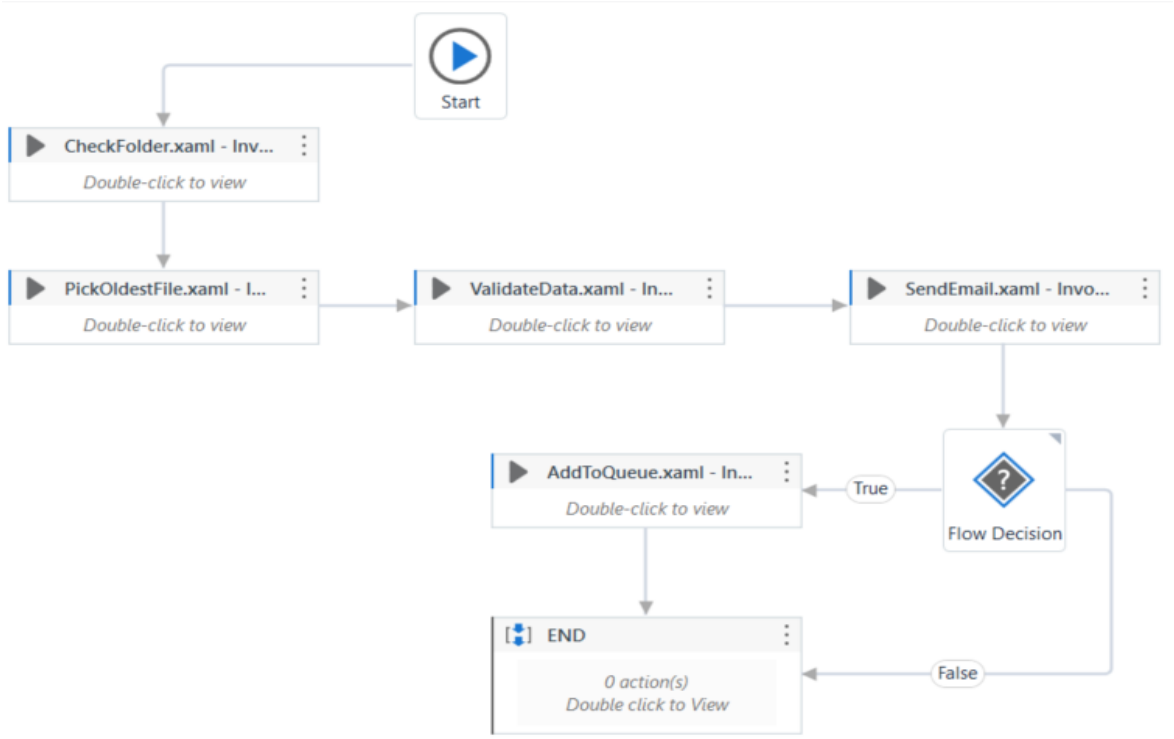


Figure 16. High Level Process Flowchart of UiPath Robot 1 (Data Validation Robot)

As shown in figure 16 the first robot handles ingestion and gating. On each scheduled trigger it checks the input location, moves the oldest candidate to a correct folder to avoid conflicts, and performs schema and business validations. Accepted rows are split into transaction items and added to an Orchestrator queue. Inconsistencies trigger a notification to users and a human validation branch where the file can be corrected and rerun. Cases with conflicting parameters are sent to the user for correction and are not queued for automated posting. Each run is logged with timestamps, file identifiers and validation outcomes to preserve traceability and to support reprocessing where needed.

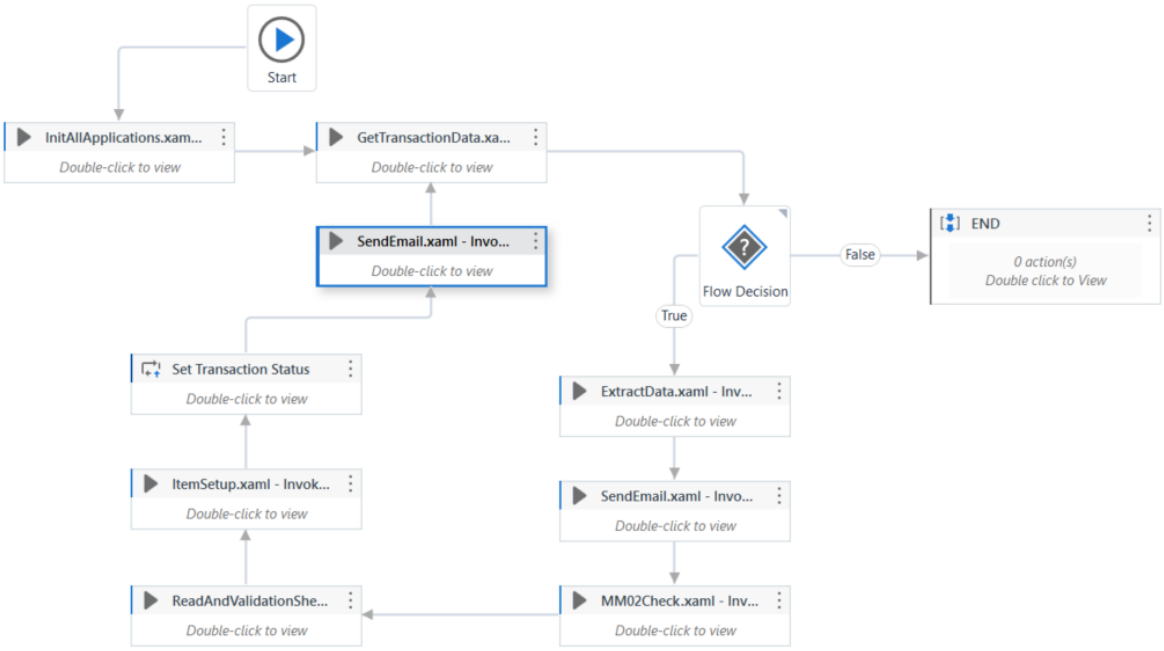


Figure 17. High Level Process Flowchart of UiPath Robot 2 (Material Setup Robot)

As shown in figure 17 the second robot performs the transactional work in SAP on the same remote desktop infrastructure under Orchestrator control. At its scheduled window or when queue items arrive it initializes the required applications, dequeues items from the Orchestrator queue, retrieves any prerequisite transaction data and executes item setup in SAP across the agreed transactions and views. After posting it records the returned identifiers, performs a targeted material master consistency check in SAP when applicable, updates transaction status and sends a completion notification. Selectors and waits are tuned for realistic UI variability, transient states use bounded retries, and checkpoints ensure that partial progress is not lost. If a failure occurs mid-run for example, SAP timeout or unexpected UI state, the robot aborts the current item gracefully, logs the exception in Orchestrator, notifies the user, and leaves the item in a retryable state without partial postings. Idempotent behavior is enforced with unique batch and item identifiers so that reprocessing does not create duplicates. Outcome files and counters are written so that operations can follow success and exception trends directly in Orchestrator, and the schedule can be adjusted without redeploying the robots.

Scope evolution (re-baseline)

The project originally aimed at a full production-ready ML+RPA solution for SAP S/4HANA item setup. During the thesis, the project scope was re-baselined such that the ML component became the initial production-ready deliverable and the RPA layer was in a pre-production functional state, with only testing phase remaining before production rollout. This reflected: a given timeframe for the thesis, the capability distribution of the host team (with greater proficiency in RPA over Python), and change negotiations that began during the author's employment and prevented an extension. With a contract extension, the production hardening of the RPA layer likely would have been completed first and proper documentation for the completion of the thesis could have been scheduled afterwards. Table 11 shows deliverable's purpose and readiness (state).

Table 11. Deliverables table

Deliverables	**Purpose**	**Readiness**
Training data	• Previously created items from SAP • ML application learns dependencies between parameters • Around 130 000 unique rows	Ready
Python code (ML Application)	• ML algorithm + Business logic + Excel automation	Production ready
UiPath robot 1	• Validates the data and announces the start of the item setup process	Pre-production
UiPath robot 2	• SAP item creation	Pre-production
Version control	• To maintain robots and ML Application • To manage four iterative releases and feedback-driven modifications	In-place
User & Developer Guide	• How to download and use • How to update tools code • What code does	In-place

Within the available time, company-governance priorities were advanced: a Git-based workflow was established for the ML codebase and RPA packages, release tagging was aligned with Orchestrator package versions, and installation, operations, and update guides

were authored for the packaged ML application. These priorities advanced maintainability and handover readiness but constrained the time to finalize RPA for production.

Integration and end-to-end process

Integration between the components is deliberately simple. The handover format is a versioned Excel file with a documented schema and three clear groups of fields: inputs, ML-predicted outputs and pass-through fields. The schema contains batch metadata, the model version and per-item identifiers. The ML application writes to a controlled drop folder, and the robot reads from a separate pickup folder to reduce accidental edits. After processing, the robot archives both the input and outcome files for audit and support. Version headers in the file are used to detect incompatibilities and to provide backward compatibility during controlled changes. The figure 18 shows high level swimlane end-to-end process flow of solution.

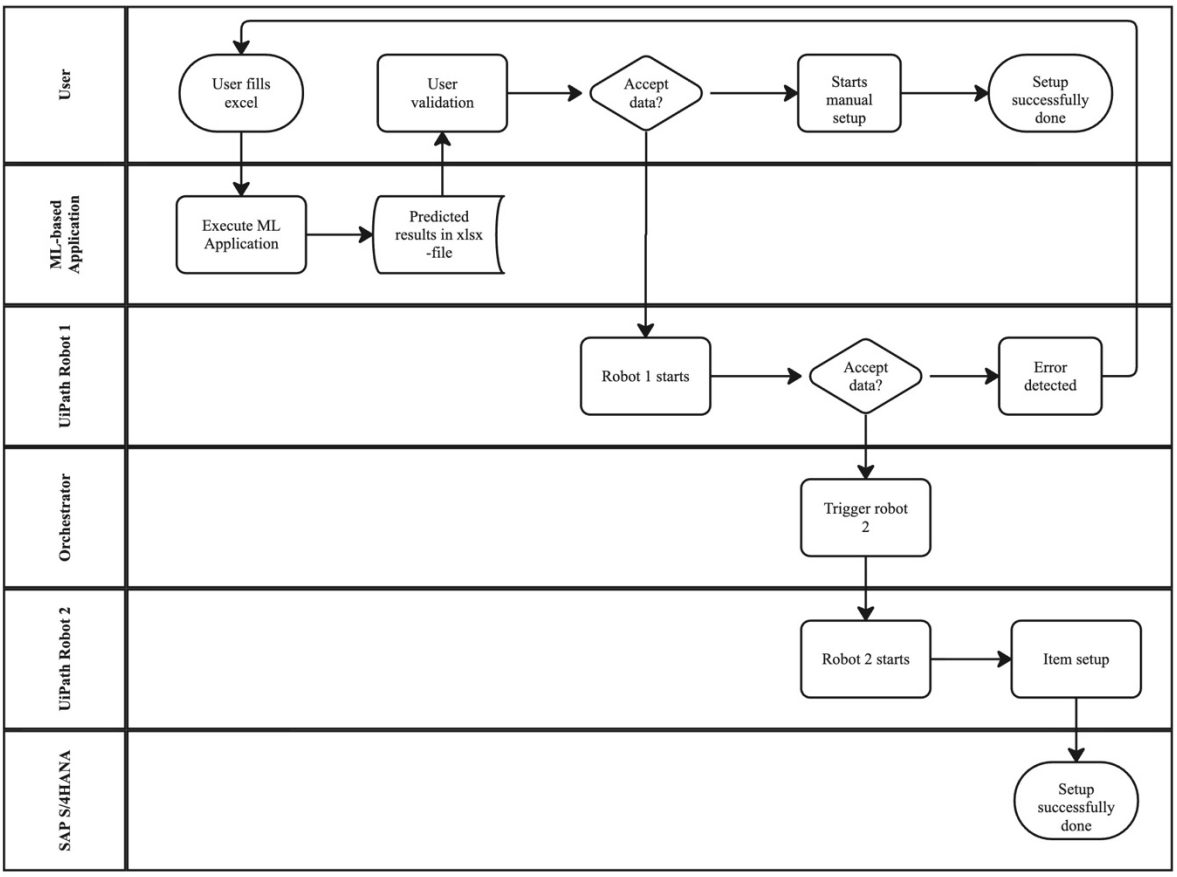


Figure 18. High Level Swimlane End-to-End Process Flow of Solution

The user completes the master Excel and runs the ML application. The ML application generates predictions, and the user validates the data. After validation, the user chooses to perform item setup manually or trigger the robot for automatic execution. If automatic, Robot 1 validates the data. If issues are found it returns the file to the user. If the file passes check it forwards the accepted items to Orchestrator. When Orchestrator receives a clean file, Robot 2 is signaled to start the SAP S/4HANA item setup. If no errors occur during posting, the item setup completes successfully.

Security and Maintenance

To support maintenance, the ML artefact includes a documented retraining procedure. The training data extract, preprocessing steps and model selection are scripted so that a new model can be produced consistently when data distributions change. The documentation includes a module inventory that explains each of the tool's 45 modules, their responsibilities, configuration keys and dependencies, and provides step by step instructions for updating a module when business rules or data structures change. Both the local ML application and the two UiPath robots are under version control with semantic versioning, tagged releases, change logs and a defined rollback path. Schema versions are recorded in the handover file so that ML and RPA remain compatible across upgrades.

Security and operability were considered throughout the design. The local application processes files only from organization-approved locations and does not persist sensitive information beyond what is needed for the task. No secrets are stored in the handover file. SAP credentials are managed on the RPA side using standard mechanisms. Logging is structured so that sensitive fields can be excluded or masked according to policy. Operational runbooks describe installation of the local package, the support flow for common errors and the steps to recover from interrupted runs. Configuration values and transaction-specific parameters are externalized where appropriate so that environment changes do not require code edits.

4.6 Demonstration and Evaluation

During the pilot, the solution ran in the environment and architecture described in Section 4.5. The final project release of the local ML application and the pre-production RPA workflows were executed as unattended, orchestrated runs on the approved remote-desktop infrastructure. Data exchange followed the versioned handover schema. Realistic datasets covering the in-scope material types were processed across several cycles. Test sets included both valid inputs and intentionally invalid cases to exercise validation, error handling, and the human-in-the-loop review.

Each end-to-end demonstration followed the intended operational sequence without reconfiguration. Users prepared the input with the predefined template and executed the local ML application to generate the handover file. Item that passed the user's validation and met the confidence thresholds progressed to the RPA stage in the pilot environment. Items that failed the robot validation were returned to users for correction with structured feedback. Execution and outcomes were traceable through structured logs on the local ML application, which made failures visible and diagnosable. On the RPA side, Orchestrator and workflow logs provided step-level traces and error codes. Because this was a pre-production pilot, exception handling covered only critical cases.

ML quality was reported using the training pipeline's validation protocol with classification metrics such as accuracy and F1 score. Confidence distributions were inspected to confirm that the low-confidence highlighting threshold aligned with observed error patterns. Additionally, prediction quality was assessed using feedback from four test-phase users (n=4), focusing on perceived correctness, clarity, and actionability of the outputs. For targets governed by explicit business rules, checks verified that rule precedence produced predictable outcomes when model suggestions and rules intersected.

Table 12. ML application category-level quality (min–max) across 19 targets

Category	#Targets (fields)	Accuracy (min–max)	F1-macro (min–max)
MRP fields	4	97–99%	96–99%
Accounting	3	96–98%	96–98%
Sales	3	96–98%	96–98%
Logistics/Warehouse	3	96–97%	96–97%
Procurement	2	96–98%	96–98%
Controlling	2	96–97%	96–97%
Classification/Hierarchy	2	88–96%	96–97%

Category-level summary of prediction quality for the RandomForest model with a MultiOutputClassifier on the validation set. For each category, the table 12 reports the minimum and maximum top-1 accuracy and macro-F1 across its targets (19 total). Field names are anonymized. Overall per-field accuracy ranged from 96% to 99% for most target fields, while classification and hierarchy fields achieved 88%–96%. A mandatory user review further reduces residual errors before the RPA stage. Given the baseline data accuracy of 92% in table 9, the overall data quality is estimated to rise to about 98%, which is an improvement of roughly 6 percentage points.

Process performance in the robotic stage was assessed by the straight-through processing rate on valid inputs, exception incidence and throughput within scheduled windows. Idempotency was validated by controlled reprocessing without duplicate creation. Once the RPA phase is promoted to production and scheduled outside working hours (see Section 4.4), manual effort is replaced by automation. The resulting annual time saving is estimated at 431.1 hours, equal to the current manual item-setup effort (Table 8).

Usability and adoption were evaluated through brief surveys and task walkthroughs during the pilot cycles. Users rated clarity of the local workflow, usefulness of validation feedback, confidence in model suggestions and the effort to complete a batch. Changes between cycles were prioritized from free-text feedback and tracked by improvements in survey scores, reduced need for overrides and shorter preparation time.

Maintainability and operability were reviewed against the documented procedures. Evidence covered the retraining steps, configuration management, versioning of the schema and robots, and the completeness of runbooks and logs for support. Small regression checks ensured that critical paths remained intact after updates and that configuration-only adjustments were sufficient for minor environmental changes.

5. Discussion

The developed ML driven automation for SAP S/4HANA item setup produced tangible business benefits in the pilot. Manual effort in field entry decreased as routine data selection and multi-screen navigation were shifted from users to the combined ML and RPA workflow, which in turn shortened end to end processing and reduced rework caused by posting errors. In addition to efficiency, data quality improved measurably. Standardized entries, validation prior to prediction, and consistent rule application reduced user specific variation and downstream corrections. These improvements were visible in the pilot results and in stakeholder comparisons against the previous manual process, which highlighted a lower workload, faster processing, and more consistent item records.

From a technical perspective, the integration worked as intended within the designed constraints. The solution remained deliberately modular, and file based. The local application validates and standardizes the input template, predicts multiple targets with confidence indicators and status flags, and writes Excel handover. Unattended robots, scheduled and orchestrated centrally on the remote desktop infrastructure, consume the handover and complete SAP item setup across the agreed transactions and views. Validation, business rules, confidence highlighting, and structured logging increased transparency and supported user trust and troubleshooting without tight coupling between components. At the time of evaluation, the RPA layer was functionally complete, with only testing phase remained prior to production rollout.

Despite the positive outcomes, several limitations require careful consideration. First, data quality and the risk of concept drift constrain sustained model performance. The model learns from historical master data, which may contain inconsistencies, and taxonomies and policies evolve over time. If distributions or rules shift, prediction quality will decline. The mitigation is a process that combines routine monitoring of distributions and outcomes, periodic retraining on updated data, and clear promotion and rollback criteria for new models, reinforced by versioned handover schemas and validation at both ends of the pipeline. Second, model lifecycle management introduces ongoing maintenance needs. Each retraining yields a new artefact that must be versioned, validated, and, if necessary, rolled back. Documented retraining steps, archived validation reports, semantic versioning of the

local application and the robots, and traceable change logs reduce operational risk but also imply a standing responsibility for stewardship of data, models, and configuration. Third, generalizability is bounded by the scope used during development. The pilot focused on selected material types and organizational structures. Other material types and organizational structures may rely on different defaults or mandatory fields. Transfer to new contexts should therefore be treated as incremental rollout with data exploration, targeted retraining or sub models where warranted, and parameterization of RPA selectors and configurations rather than hard coding local specifics. Fourth, SAP GUI automation remains sensitive to interface changes and environmental variability. Even with element-based selectors, minor layout or naming updates can interrupt automated posting. Bounded retries, checkpoints, exception classification, selector guidelines, configuration driven adjustments, and operation under Orchestrator with scheduled windows and observability all help, but continued coordination with SAP change cycles and quick remediation paths are essential to sustain reliability.

User feedback across the four pilot iterations shaped the artefact materially. Requests for transparency and control translated into confidence indicators in the output and an mandatory human-in-the-loop step before robotic posting, which helped calibrate trust and allowed gradual adoption. Clarity and simplicity concerns led to a linear, minimal interaction in the local application with validation messages written in plain language. Operational resilience comments prompted stronger exception handling, retries, and idempotent behavior in the robots. Survey driven refinements between cycles improved perceived usability and reduced the need for overrides, which aligns with the evaluation approach and the objectives set for user adoption.

Compared to the objectives and acceptance conditions defined earlier, the evaluation focused on per field prediction quality for high impact targets, straight through processing on valid inputs, end to end cycle time, error and rework rates, robustness under realistic conditions, and perceived usability. Results and any deviations were documented against these criteria to guide the pilot exit decision within the agreed scope. This linkage ensures that the discussion of benefits is grounded in the same measures that govern readiness for broader deployment rather than in anecdotal improvements.

The architectural choices taken here support scalability and reuse. Local prediction with confidence governance, a versioned file-based handover, and centrally orchestrated RPA

with clear exception paths form a pattern that can be extended beyond item setup where similar conditions hold. Within SAP master data, the same approach can be applied when historical data exists for learning, rule complements are available for policy constraints, and GUI based execution is acceptable or can be substituted with more stable interfaces when available. Boundary conditions remain explicit. The solution is not a universal template, but a design that travels effectively when accompanied by domain specific data preparation, incremental validation, and the governance mechanisms exercised in the pilot.

6. Conclusions

This thesis set out to develop and evaluate an AI driven automation solution for the SAP S/4HANA item setup process, addressing the inefficiencies and error proneness of manual product data management. The work followed the Design Science Research methodology and was guided by two research questions. RQ1 asked how to design an intelligent automation system that combines ML and RPA for item setup in SAP S/4HANA. RQ2 asked what performance and data quality improvements such a system can achieve compared to the existing manual process. An artefact was engineered and refined through four iterative development cycles with user feedback incorporated at each step, resulting in a functioning ML enhanced RPA workflow that automates item setup while preserving human oversight when needed.

RQ1. *How to design an intelligent automation solution that combines ML and RPA for SAP S/4HANA item setup?*

The research demonstrates a viable design for ML and RPA integration in this context. The final architecture comprises a Python based ML application trained on historical item data and packaged as a standalone executable using PyInstaller, and two unattended UiPath robots that carry out the SAP S/4HANA transactions on the approved remote desktop infrastructure under Orchestrator control. The ML model is a Random Forest multi-output classifier that predicts 19 target fields based on 18 input attributes, while 14 other attributes are copied unchanged from the input excel file to the output excel file. Operationally, the first unattended robot validates the handover file and enqueues approved transactions in Orchestrator, while the second robot consumes the queue to execute the SAP postings. Centralized unattended operation on the approved remote desktop server supports scheduled runs outside business hours. The components are decoupled by a versioned Excel handover that carries inputs, predicted field values, confidence indicators and status flags. This file-based interface proved reliable for synchronizing local prediction with centrally orchestrated posting, and it supports maintainability through explicit versioning and schema checks. The solution leverages ML to propose and validate master data fields, and uses rule governed RPA to enter the data into SAP, combining predictive and procedural capabilities in a cohesive workflow. A mandatory human-in-the-loop review gates the handover, and users

must check, and possibly correct predictions highlighted by confidence thresholds before moving the file to the RPA intake folder. The RPA layer was functionally complete at evaluation, with only testing phase remaining before production rollout.

From a theoretical perspective, the findings refine the principles of intelligent automation in ERP environments. The core of the solution is complementarity. Probabilistic ML generates predictions together with confidence measures, while RPA executes deterministic, rule-based transactions. This supports the claim that predictive and procedural automation jointly enhance quality and throughput in structured data processes. The findings, evidenced through DSR cycles, constitute documented design knowledge for both research and practice.

The study contributes generalizable design principles. A loosely coupled, file-based and versioned interface provides a robust integration mechanism under organizational constraints. Risk-based control with confidence thresholds and exception handling reduces error risk without removing user accountability. The human-in-the-loop gate is positioned at the decision point where ML uncertainty is highest. A versioned schema contract and explicit governance model strengthen maintainability and auditability. Temporal decoupling via queue-based orchestration enables capacity utilization outside business hours. These theory-based findings confirm and extend existing frameworks by demonstrating how intelligent automation principles can be operationalized in a regulated ERP context to improve both data quality and process efficiency, while outlining boundary conditions for transferability across data, rule, and interface environments.

RQ2. *What performance and data-quality gains does it deliver versus the manual baseline?*

Evaluation against the manual baseline shows clear improvements in efficiency, accuracy and workload distribution. Routine data selection and multi-screen navigation are shifted from users to the combined ML and RPA pipeline, which shortens end to end processing and reduces rework caused by posting errors. Standardized entries, validation prior to prediction and consistent business rules reduce user specific variation and improve the consistency of item records. User feedback corroborates these outcomes by reporting lower cognitive load and faster task completion compared to the fully manual procedure. In sum, the intelligent automation matches and, in several respects, surpasses the legacy process on key aspects of speed, accuracy and reliability within the agreed scope.

The practical contribution is a working solution that streamlines item setup, reduces errors and saves time for the case organization. The implementation serves as a reference for organizations seeking to modernize similar processes without invasive changes to core systems. Key practices include separating prediction from execution, using a versioned handover for loose coupling, embedding validation and confidence governance for transparency and trust, and operating robots centrally under schedules with observability and defined exception paths. These practices are transferable to comparable contexts, if data preparation, user training and maintenance responsibilities are in place.

Table 13. Results table

Metric	Manual baseline	With ML+RPA	Improvement
Field-level accuracy	~ 92%	~ 98%	+ 6%
Manual effort for item creation	~ 431.1 h/year	~ 0 h/year (RPA production-ready)	~ 431.1 h/year

Table 13 summarizes the primary quantitative outcomes used to answer RQ2. The baseline figures reflect historical field-level accuracy and the measured manual effort for item creation. The ML+RPA figures reflect the validated model accuracy and a production-ready RPA workflow that removes manual item-creation effort. Improvements are reported as percentage points and hours per year.

Implications, Limitations and Future Directions

At thesis closure, the RPA workflows had passed functional verification and production rollout depended solely on executing the testing phase. The broader implications concern how organizations manage and govern master data. Delegating routine decision making and data entry to an AI supported system can improve efficiency and consistency while allowing staff to focus on exceptions and analysis. Realizing these benefits requires alignment with governance and user expectations, along with clear ownership of data, models and robotic operations. The results illustrate that intelligent automation can be introduced incrementally within existing ERP workflows when the artefact is designed for transparency, versioned compatibility and operational sustainability.

The study has limitations that frame the boundary conditions for interpretation. The ML component depends on historical data and is exposed to concept drift as business rules and data distributions evolve. Sustained performance therefore requires monitoring, periodic

retraining and disciplined version control, synchronized with the robots through the versioned handover schema. Generalizability is bounded by the scope of material types and organizational settings used during development, which implies that transfer to other categories or plants requires data exploration, targeted retraining or sub models, and parameterized RPA logic. Reliance on SAP's graphical user interface introduces fragility because even minor interface changes can interrupt automation; robust selectors, exception handling, scheduled orchestration and coordination with change cycles mitigate but do not eliminate this risk.

Grounded in the discussion and evaluation results, identified follow-up avenues that can improve the artifact's reliability, scalability, and generalizability. The key lines of action and inquiry are as follows:

- Proactive concept-drift management and automated monitoring to keep the model reliable as data and business rules evolve.
- Add XAI features to strengthen user trust by exposing feature attributions, confidence calibration and error analysis at field level.
- Move toward fully automated item setup process. After the user submits the input Excel, with the robot launching the ML application automatically.
- Test transferability by applying the approach to other master data domains and to other organizations, documenting domain-specific adaptations.
- Prefer direct interfaces or APIs over GUI automation where available to reduce selector fragility and enable idempotent retries.
- Extend scope to include the four excluded material types, which are approximately 28% of annual items (2070) to achieve an additional 165 hour savings per year.
- Run a longitudinal production study to assess long-term performance, user engagement, exception rates, and maintenance effort.

Conclusions and communication of results are closely linked in this work. The artefact, its versioned handover schema and configuration examples, together with user guides, runbooks and change logs, constitute the replication and dissemination package that documents the design for reuse and evaluation. Ownership of data, model retraining and robot maintenance is defined, and the governance practices exercised during the pilot provide a template for continued operation. The thesis records the design principles, decision

rules and evaluation evidence so that researchers and practitioners can assess transferability to comparable ERP contexts.

Ultimately, the thesis answers RQ1 by presenting a practical and maintainable design for combining ML and RPA in SAP item setup, and answers RQ2 by showing that the resulting system delivers measurable benefits in efficiency, data quality and user workload within the agreed scope. These findings contribute both a tangible solution for industry and a documented design knowledge item for the research community, while outlining clear opportunities for further exploration as organizations continue to digitalize data intensive workflows.

References

Abildtrup A. 2024. The Rise of Robotic Process Automation in the Banking Sector: Streamlining Operations and Improving Efficiency. Journal of Computing and Natural Science 4(1): 31–40. Available: https://anapub.co.ke/journals/jcns/jcns_pdf/2024/jcns_volume_4-issue_1/JCNS202404004.pdf

Achouch, M., Dimitrova, M., Ziane, K., Sattarpanah Karganroudi, S., Dhouib, R., Ibrahim, H. & Adda, M. 2022. On Predictive Maintenance in Industry 4.0: Overview, Models, and Challenges. Applied Sciences, 12(16):8081. [Referred 15 Jun 2025]. Available: https://doi.org/10.3390/app12168081

Adebayo, O. 2024. AI in ERP and Supply Chain Management: Revolutionizing Business Operations. [Referred 25 Feb 2025]. Available: https://www.researchgate.net/profile/Jummy-Johnson/publication/389166956_AI_in_ERP_and_Supply_Chain_Management_Revolutionizing_Business_Operations/links/67b7323a8311ce680c6b2fde/AI-in-ERP-and-Supply-Chain-Management-Revolutionizing-Business-Operations.pdf.

Adenekan, T.K. 2025. Integrating Generative AI with SAP S/4HANA: Revolutionizing Enterprise Predictive Maintenance and Decision-Making. ResearchGate. [Referred 26 Mar 2025]. Available: https://www.researchgate.net/publication/388359847

March, S & Storey, V. 2008. Design science in the information systems discipline: An introduction to the special issue on design science research. [Referred 25 Jun 2025]. Available: https://www.researchgate.net/publication/228380933_Design_science_in_the_information_systems_discipline_An_introduction_to_the_special_issue_on_design_science_research

Anderl, R. 2022. Simulation Data Management in the Digital Twin (SDM-DT)–Evolution of Simulation Data Management Along the Product Life Cycle. Procedia CIRP. [Referred 17 Mar 2025]. Available: https://www.sciencedirect.com/science/article/pii/S221282712200141X.

Antwi B.O. & Avickson E.K. 2024. Integrating SAP, AI, and Data Analytics for Advanced Enterprise Management. International Journal of Research. [Referred 19 Mar 2025]. Available: https://www.researchgate.net/publication/384692321_Integrating_Sap_AI_And_Data_Analytics_for_Advanced_Enterprise_Management

Antwi, B.O. & Avickson, E.K. 2024. Integrating SAP, AI, and Data Analytics for Advanced Enterprise Management: automating routine SAP-based business processes to boost efficiency and agility. *International Journal of Research Publication and Reviews*, 5(10), 621–636. [Referred 15 Jun 2025]. Available: https://www.researchgate.net/publication/384692321_Integrating_Sap_AI_And_Data_Analytics_for_Advanced_Enterprise_Management

Arowosegbe, O.B., Israel, A., Fadojutimi, B. & Ashi, T.A. 2024. Future-Proofing Supply Chains: Leveraging ERP Platforms for Advanced Automation and Interoperability. [Referred 31 Mar 2025]. Available: https://www.researchgate.net/publication/384264698

Ashi, T.A., Agbelusi, J. & Chukwunweike, S.O. 2024. Breaking Down Silos: Enhancing Supply Chain Efficiency through ERP Integration and Automation. [Referred 31 Mar 2025]. Available: https://www.researchgate.net/publication/385139027

Atkar A. 2024. ERP SYSTEM (Enterprise Resource Planning) Integrated Software Solution for Business Management. Gurukul International Multidisciplinary Research Journal. [Referred 18 Mar 2025]. Available: https://doi.org/10.69758/GIMRJ2406I8V12P061

Axmann B. & Harmoko H. 2022. Process & Software Selection for Robotic Process Automation (RPA). Tehnički Glasnik 16(3): 412–419. [Referred 29 Mar 2025]. Available: https://hrcak.srce.hr/file/404732

Banks-Grasedyck D., Lippke E., Oelfin H. & Schwaiger R. 2022. Successfully Managing S/4HANA Projects. Springer. [Referred 19 Mar 2025]. Available: https://link.springer.com/book/10.1007/978-3-030-86084-4

Baroud, S. Y., Yahaya, N. A., & Elzamly, A. M. 2024. Cutting-Edge AI Approaches with MAS for PdM in Industry 4.0: Challenges and Future Directions. Journal of Applied Data Science. [Referred 17 Mar 2025]. Available: http://www.bright-journal.org/Journal/index.php/JADS/article/view/196.

Bawa, S.S. 2023. How Business Can Use ERP and AI to Become Intelligent Enterprise. ResearchGate. [Referred 26 Mar 2025]. Available: https://www.researchgate.net/publication/372946571_How_Business_can_use_ERP_and_AI_to_become_Intelligent_Enterprise

Bhardwaj, V., Noonia, A., Chaurasia, S., Kumar, M., Rashid, A.R. & Ben Othman, M.T. 2024. Optimizing Structured Data Processing through Robotic Process Automation. arXiv preprint. [Referred 15 Jun 2025]. Available: https://arxiv.org/abs/2408.14791

Bhatia R. 2024. Transforming Industries with SAP S/4HANA. [Referred 19 Mar 2025]. Available: https://www.researchgate.net/publication/387806695_Transforming_Industries_with_SAP_S4HANA

Bhatt, S. 2024. Best Practices for SAP ECC to S/4HANA Upgrades in Cloud Environments. ResearchGate. [Referred 27 Mar 2025]. Available: https://www.researchgate.net/publication/388658595

Biolcheva, P. & Molhova, M. 2022. Integration of AI supported risk management in ERP implementation. Computer and Information Science. [Referred 26 Mar 2025]. Available: https://www.academia.edu/download/99478763/50949.pdf

Boas, L.V., Faria, J.M. & Simões, N. 2025. Hybrid AI-Assisted Approach for Unstructured Invoice Data Extraction to Improve Insertion Reliability in ERP Systems. [Referred 31 Mar 2025]. Available: https://www.researchgate.net/publication/389563762

Boddapati, V.N. 2025. Optimizing Production Efficiency in Manufacturing Using Big Data and AI/ML. Computers in Industry. [Referred 20 Mar 2025]. Available: https://papers.ssrn.com/sol3/Delivery.cfm?abstractid=5101316.

Bodepudi, V., Chinta, P.C.R., Katnapally, N. & Ja, K. 2022. Exploring the Role of Neural Networks in Big Data-Driven ERP Systems for Proactive Cybersecurity Management. ResearchGate. [Referred 26 Mar 2025]. Available: https://www.researchgate.net/publication/387916314

Bogdanova, A., Imakura, A., Sakurai, T., Fujii, T., Sakamoto, T. & Abe, H. 2022. Achieving Transparency in Distributed Machine Learning with Explainable Data Collaboration. [Referred 31 Mar 2025]. Available: https://arxiv.org/abs/2212.03373

Börjeson N. & Ågerstrand M. 2025. “The Problems that we Have Today, are Yesterday’s Solutions”: Enabling Circular Non-Toxic Supply Chains. Circular Economy and Sustainability 5(1). [Referred 19 Mar 2025]. Available: https://link.springer.com/article/10.1007/s43615-025-00501-x

Božić D, Živičnjak M, Stanković R, Ignjatić A. 2024. Impact of the Product Master Data Quality on the Logistics Process Performance. Logistics. [Referred 30 Jun 2025]. Available: https://www.mdpi.com/2305-6290/8/2/43

Brecher, C., Müller, A., Dassen, Y. & Storms, S. 2021. Automation technology as a key component of the Industry 4.0 production development path. International Journal of Advanced Manufacturing Technology. [Referred 15 Jun 2025]. Available: https://www.researchgate.net/publication/351733482

Cao K, Liu L, Cheng Z. 2016. Research on Configurable Workflow Technology in PDM System. ITM Web of Conferences, 12:09012. [Referred 20 Mar 2025]. Available: https://doi.org/10.1051/itmconf/20160709012

Chen, B. 2025. The Comprehensive Evolution of ERP Systems: Classification, Financial Integration, and Future AI Integration. Preprints. [Referred 26 Mar 2025]. Available: https://www.preprints.org/frontend/manuscript/a485d1e6538cc494d03d7448be553b45/download_pub

Chen, J., Storchan, V. & Kurshan, E. 2021. Beyond Fairness Metrics: Roadblocks and Challenges for Ethical AI in Practice. [Referred 31 Mar 2025]. Available: https://arxiv.org/abs/2108.06217

Chiarini A. & Kumar M. 2022. What is Quality 4.0? An exploratory sequential mixed methods study of Italian manufacturing companies. International Journal of Production Research. [Referred 18 Mar 2025]. Available: https://orca.cardiff.ac.uk/id/eprint/142174/3/Quality40_IJPR_rev3-31st%20May.pdf

Choudhuri, S.S. 2024. AI in ERP and Supply Chain Management. Google Books. [Referred 26 Mar 2025]. Available: https://books.google.com/books?id=nQ0wEQAAQBAJ

Christopher, G. & Blessing, A. 2024. Optimizing Enterprise Resource Planning with SAP and Artificial Intelligence: Boosting Efficiency, Automation, and Decision-Making. [Referred 25 Feb 2025]. Available: https://www.researchgate.net/profile/Adebis-

Samuel/publication/389165142_Optimizing_Enterprise_Resource_Planning_with_SAP_and_Artificial_Intelligence_Boosting_Efficiency_Automation_and_Decision-Making/links/67b71ed18311ce680c6b2d2c/Optimizing-Enterprise-Resource-Planning-with-SAP-and-Artificial-Intelligence-Boosting-Efficiency-Automation-and-Decision-Making.pdf.

Chugh, R., Macht, S. & Hossain, R. 2022. Robotic Process Automation: A Review of Organizational Grey Literature. International Journal of Information Systems and Project Management. [Referred 20 Mar 2025]. Available: https://aisel.aisnet.org/ijispm/vol10/iss1/2/.

Ciancio, V., Homri, L., & Dantan, J. Y. 2024. Development of a Flexible Data Management System to Implement Predictive Maintenance in the Industry 4.0 Context. International Journal of Production Research. [Referred 17 Mar 2025]. Available: https://hal.science/hal-04218626v1/file/LCFC_IJPR_2023_CIANCIO.pdf.

Costa, D.A. da S., Mamede, H.S. and Mira da Silva, M. 2022. Robotic Process Automation (RPA) Adoption: A Systematic Literature Review. [Referred 29 Mar 2025]. Available: https://www.researchgate.net/publication/362035572_Robotic_Process_Automation_RPA_Adoption_A_Systematic_Literature_Review

Cyclone Interactive. n.d. EMC Digital Universe. [Referred 8 May 2025]. Available: https://www.cycloneinteractive.com/our-work/emc-digital-universe/

Czarnecki, C. & Fettke, P. 2021. Robotic Process Automation does not require new APIs or services but uses existing UIs to mimic human behavior. Business & Information Systems Engineering. [Referred 15 Jun 2025]. Available: https://link.springer.com/article/10.1007/s12599-018-0542-4

Dash S. 2025. Green AI: Enhancing Sustainability and Energy Efficiency in AI-Integrated Enterprise Systems. IEEE Access. [Referred 18 Mar 2025]. Available: https://doi.org/10.1109/ACCESS.2025.3532838

De Silva, D. & Alahakoon, D. 2021. An Artificial Intelligence Life Cycle: From Conception to Production. [Referred 31 Mar 2025]. Available: https://arxiv.org/abs/2108.13861

Debbadi, R.K. & Boateng, O. 2025. Optimizing End-to-End Business Processes by Integrating Machine Learning Models with UiPath for Predictive Analytics and Decision

Automation. International Journal of Scientific Research. [Referred 20 Mar 2025]. Available: https://www.researchgate.net/profile/Rama-Krishna-Debbadi/publication/389043018_Optimizing_end-to-end_business_processes_by_integrating_machine_learning_models_with_Uipath_for_predictive_analytics_and_decision_automation/links/67b2250196e7fb48b9c4f871/Optimizing-end-to-end-business-processes-by-integrating-machine-learning-models-with-Uipath-for-predictive-analytics-and-decision-automation.pdf.

Di Maria, E., & De Marchi, V. 2022. Industry 4.0 technologies and circular economy: The mediating role of supply chain integration. Business Strategy and the Environment. [Referred 12 Mar 2025]. Available: https://onlinelibrary.wiley.com/doi/abs/10.1002/bse.2940.

Djoric D. Jordanovic M. Lalic B. & Majstorovic V.D. 2024. On the Way to the Development of Smart Manufacturing Model in Practice. Preprints.org. [Referred 18 Mar 2025]. Available: https://www.preprints.org/frontend/manuscript/391d14b4251a766315478d449dcd9bb6/download_pub

Dolata, M., Feuerriegel, S. & Schwabe, G. 2021. A Sociotechnical View of Algorithmic Fairness. [Referred 31 Mar 2025]. Available: https://arxiv.org/abs/2110.09253

Dolgui, A., Haddou Benderbal, H. & Sgarbossa, F. 2024. Editorial for the special issue: AI and data-driven decisions in manufacturing. Journal of Intelligent Manufacturing. [Referred 27 Mar 2025]. Available: https://link.springer.com/article/10.1007/s10845-024-02499-9

Edwards, J., Nguyen, A., & Lämsä, J. 2024. Human-AI collaboration: Designing artificial agents to facilitate socially shared regulation among learners. British Journal of Educational Technology. [Referred 26 Feb 2025]. Available: https://bera-journals.onlinelibrary.wiley.com/doi/pdfdirect/10.1111/bjet.13534.

El Haouat Z. Essalih S. Bennouna F. & Ramadany M. 2024. Environmental optimization and operational efficiency: Analysing the integration of life cycle assessment (LCA) into ERP systems in Moroccan companies. Results in Engineering, Elsevier. [Referred 18 Mar 2025]. Available: https://www.sciencedirect.com/science/article/pii/S2590123024003852

El-Gharib, N.M. and Amyot, D. 2022. Robotic Process Automation Using Process Mining—A Systematic Literature Review. arXiv. [Referred 29 Mar 2025]. Available: https://arxiv.org/abs/2204.00751

El-Sayed, W. A., Eladwi, M. M. T., & Ashour, N. S. 2022. A Novel Program to Enhance Production Data Management (PDM) System in Apparel Industry. International Design Journal. [Referred 17 Mar 2025]. Available: https://journals.ekb.eg/article_260620.html.

Elangovan U. 2022. Industry 5.0: The future of the industrial economy. Taylor & Francis. [Referred 18 Mar 2025]. Available: https://www.taylorfrancis.com/books/mono/10.1201/9781003190677/industry-5-0-uthayan-elangovan

Elmadhoun B., Alsaidalani R. & Burczynski F. 2025. Quality Risk Management in the Final Operational Stage of Sterile Pharmaceutical Manufacturing: A Case Study Highlighting the Management of Sustainable Related Risks in Product Sterilization, Inspection, Labeling, Packaging, and Storage Processes. Sustainability 17(4): 1670. [Referred 19 Mar 2025]. Available: https://www.mdpi.com/2071-1050/17/4/1670

Enfuse-Solutions 2023. The Role Of Artificial Intelligence In Product Data Management. Enfuse-Solutions white paper. [Referred 15 Jun 2025]. Available: https://www.enfuse-solutions.com/the-role-of-artificial-intelligence-in-product-data-management/

Farinha, D., Pereira, R. and Almeida, R. 2023. A Framework to Support Robotic Process Automation. Journal of Information Technology, 38(1), pp.3–21. [Referred 29 Mar 2025]. Available: https://journals.sagepub.com/doi/10.1177/02683962231165066

Fatima, Z., Tanveer, M.H., Waseemullah & Zardari, S. 2022. Production Plant and Warehouse Automation with IoT and Industry 5.0. Applied Sciences. [Referred 19 Mar 2025]. Available: https://www.mdpi.com/2076-3417/12/4/2053.

Favoretto, C., Mendes, G. H. S., & Filho, M. G. 2022. Digital transformation of business model in manufacturing companies: Challenges and research agenda. Journal of Business & Industrial Marketing. [Referred 12 Mar 2025]. Available: https://www.emerald.com/insight/content/doi/10.1108/jbim-10-2020-0477/full/html.

Ferdaus, M.M., Abdelguerfi, M., Ioup, E., Niles, K.N., Pathak, K. & Sloan, S. 2024. Towards Trustworthy AI: A Review of Ethical and Robust Large Language Models. [Referred 31 Mar 2025]. Available: https://arxiv.org/abs/2407.13934

Fernandez, I.A., Neupane, S., Chakraborty, T., Mitra, S., Mittal, S., Pillai, N., Chen, J. & Rahimi, S. 2024. A Survey on Privacy Attacks Against Digital Twin Systems in AI-Robotics. [Referred 31 Mar 2025]. Available: https://arxiv.org/abs/2406.18812

Fischer, S.W.S. 2024. Questioning AI: Promoting Decision-Making Autonomy Through Reflection. [Referred 31 Mar 2025]. Available: https://arxiv.org/abs/2409.10250

Fuerst, T., Camba, J. D., & Mazumdar, A. 2024. Engineering Graphics Education for the Digital Enterprise. ASEE Annual Conference. [Referred 17 Mar 2025]. Available: https://peer.asee.org/47283.pdf.

Galinska, M. and Stachura, M. 2023. Robotic Process Automation Technology in Supply Chain Management: Practical Applications for Simplifying Workflows. Routledge. [Referred 29 Mar 2025]. Available: https://www.taylorfrancis.com/books/mono/10.4324/9781003541622/robotic-process-automation-technology-supply-chain-management-barbara-gali%C5%84ska-mateusz-stachura

Godbole, M. & Josyula, H.P. 2024. Navigating the Future: A Comprehensive Analysis of AI, ML, ERP, and Oracle Integration in Financial Digital Transformation. International Journal of Computer Engineering & Technology. [Referred 26 Mar 2025]. Available: https://www.academia.edu/download/114058403/IJCET_15_01_007.pdf

Goel O., Mangal A. & Gupta P. 2023. Comparative Analysis of Optimizing SAP S/4HANA in Large Enterprises. SSRN. [Referred 19 Mar 2025]. Available: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4985357

Goldkuhl G. 2011. Design Research in Search for a Paradigm: Pragmatism Is the Answer. Conference Proceedings. [Referred 25 Jun 2025]. Available: https://www.researchgate.net/publication/279442662_Design_Research_in_Search_for_a_Paradigm_Pragmatism_Is_the_Answer

Goodfellow, I., Bengio, Y. & Courville, A. 2016. Deep Learning. MIT Press. [Referred 26 Mar 2025]. Available:

https://www.researchgate.net/publication/320703571_Ian_Goodfellow_Yoshua_Bengio_and_Aaron_Courville_Deep_learning_The_MIT_Press_2016_800_pp_ISBN_0262035618

Grosse, E. H., Sgarbossa, F., & Berlin, C. 2023. Human-centric production and logistics system design and management: Transitioning from Industry 4.0 to Industry 5.0. International Journal of Production Research. [Referred 12 Mar 2025]. Available: https://www.tandfonline.com/doi/abs/10.1080/00207543.2023.2246783.

Gunjal, H. V., & Belokar, R. M. 2023. Systematic Review: Implementation of Product Lifecycle Management in Industries. Springer. [Referred 17 Mar 2025]. Available: https://link.springer.com/chapter/10.1007/978-981-19-6107-6_19.

Gupta, A. & Agarwal, P. 2024. Enhancing Sales Forecasting Accuracy Through Integrated ERP and CRM Using Artificial Intelligence. IEEE. [Referred 26 Mar 2025]. Available: https://ieeexplore.ieee.org/abstract/document/10574785

Hallikainen, P., Bekkhus, R., & Aversa, P. 2018. How does robotic process automation create value for firms? Information Systems Frontiers. [Referred 15 Jun 2025]. Available: https://link.springer.com/article/10.1007/s10257-024-00685-z

Hevner A, vom Brocke J. 2023. A Proficiency Model for Design Science Research Education. Journal of Information Systems Education. [Referred 25 Jun 2025]. Available: https://jise.org/volume34/n3/JISE2023v34n3pp264-278.pdf

Höhn, S. & Faradouris, N. 2021. What does it cost to deploy an XAI system: A case study in legacy systems. In: Explainable, Transparent Autonomous Agents and Multi-Agent Systems. Springer. [Referred 27 Mar 2025]. Available: https://link.springer.com/chapter/10.1007/978-3-030-82017-6_11

Hrischev, R. & Shakev, N. 2022. Artificial Intelligence in ERP Systems. Engineering Sciences. [Referred 26 Mar 2025]. Available: https://e-university.tu-sofia.bg/ETUS/studenti/e-publ/files/11141_Artificial_intelligence_in_ERP_systems.pdf

Hrischev, R. & Shakev, N.G. 2023. Artificial intelligence in enterprise resource planning system. Engineering Sciences. [Referred 26 Mar 2025]. Available: https://www.researchgate.net/publication/369321353_Artificial_IntelligenceinEnterprise_Resource_Planning_Systems

Huang, W-L., Liao, S-L., Huang, H-L., et al. 2024. A Case Study of Lean Digital Transformation Through Robotic Process Automation in Healthcare. Scientific Reports 14, 14626. [Referred 29 Mar 2025]. Available: https://www.nature.com/articles/s41598-024-65715-9

Industrial Machinery Digest. 2020. CADENAS pops the corks: 405 million 3D CAD models downloaded in record year 2019. [Referred 8 May 2025]. Available: https://industrialmachinerydigest.com/quality/cadenas-pops-the-corks-405-million-3d-cad-models-downloaded-in-record-year-2019/

Ispas L., Mironeasa C. & Silvestri A. 2025. A Study on the Emergence and Resilience of Integrated Management Systems in Organizations with an Industrial Profile in Romania. Sustainability 17(6): 2401. [Referred 19 Mar 2025]. Available: https://www.mdpi.com/2071-1050/17/6/2401

Jagatheesaperumal, S.K., Rahouti, M., Ahmad, K., Al-Fuqaha, A. & Guizani, M. 2021. The Duo of Artificial Intelligence and Big Data for Industry 4.0: Review of Applications, Techniques, Challenges and Future Research Directions. IEEE Internet of Things Journal. [Referred 15 Jun 2025]. Available: https://arxiv.org/abs/2104.02425

Jain, A., Paliwal, S., Sharma, M., Vig, L. & Shroff, G. 2024. SmartFlow: Robotic Process Automation using LLMs. [Referred 31 Mar 2025]. Available: https://arxiv.org/abs/2405.12842

Janiesch, C., Zschech, P. & Heinrich, K. 2022. Machine Learning in Business: Opportunities for Enterprise-wide Applications. [Referred 31 Mar 2025]. Available: https://link.springer.com/book/10.1007/978-3-030-91242-9

Jawad, Z.N. & Balázs, V. 2024. Machine Learning-Driven Optimization of Enterprise Resource Planning (ERP) Systems: A Comprehensive Review. Beni-Suef University Journal of Basic and Applied Sciences. [Referred 26 Mar 2025]. Available: https://link.springer.com/article/10.1186/s43088-023-00460-y

Jawad, Z.N. & Balázs, V. 2024. Machine learning-driven optimization of enterprise resource planning (ERP) systems: a comprehensive review. Beni-Suef University Journal of Basic and Applied Sciences. [Referred 27 Mar 2025]. Available: https://link.springer.com/article/10.1186/s43088-023-00460-y

Jhurani, J. 2022. Revolutionizing Enterprise Resource Planning: The Impact of Artificial Intelligence on Efficiency and Decision-Making for Corporate Strategies. International Journal of Computer Engineering and Technology. [Referred 26 Mar 2025]. Available: https://www.academia.edu/download/112788583/IJCET_13_02_019.pdf

Ji, X., & Abdoli, S. 2023. Challenges and Opportunities in Product Life Cycle Management in the Context of Industry 4.0. Procedia CIRP. [Referred 17 Mar 2025]. Available: https://www.sciencedirect.com/science/article/pii/S2212827123004225.

Jones 2024. Iterative Design: a cyclical process of concept development, prototyping, user testing, and refinement improves quality and usability through successive feedback-informed iterations. [Referred 25 Jun 2025]. Available: https://www.researchgate.net/publication/384399410_Iterative_Design_A_cyclical_process_of_design_testing_and_refinement

Juturi V.P.K. 2024. Embedded Architecture of SAP S/4 HANA ERP Application. Universal Journal of Computer Sciences and Technology. [Referred 19 Mar 2025]. Available: https://www.researchgate.net/publication/378847423_Embedded_Architecture_of_SAP_S4_HANA_ERP_Application

Kanaparthi, V.K. 2023. Examining the Plausible Applications of Artificial Intelligence & Machine Learning in Accounts Payable Improvement. FinTech. [Referred 26 Mar 2025]. Available: https://www.mdpi.com/2674-1032/2/3/26

Khalimonov, Y.I. & Sotnik, S.V. 2024. Overview of Computer Vision Areas for ERP-Integrated Inspection and Quality Control. [Referred 31 Mar 2025]. Available: https://openarchive.nure.ua/bitstreams/30ca1337-2fb4-4288-80f8-5a2d295c31b6/download

Kocaoglu, B. 2024. Enterprise Applications in Logistics. In: Information Systems: Digital Transformation and Industry 4.0. Springer. [Referred 27 Mar 2025]. Available: https://link.springer.com/chapter/10.1007/978-3-031-60290-0_5

Krivykh, A., & Khvan, A. 2023. Evaluation of the Results of the Implementation of a Product Data Management System in the Technology Department of a High-Tech Enterprise. E3S Web of Conferences. [Referred 17 Mar 2025]. Available: https://www.e3s-conferences.org/articles/e3sconf/pdf/2023/20/e3sconf_tt21c2023_04038.pdf.

Kroop S. 2025. Artifact Validity in Design Science Research (DSR): A Comparative Analysis of Three Influential Frameworks. DESRIST 2025 Proceedings. [Referred 25 Jun 2025]. Available: https://arxiv.org/abs/2502.11199

Kulkarni A. 2024. Digital Transformation with SAP Hana. International Journal on Recent and Innovation Trends. [Referred 19 Mar 2025]. Available: https://www.researchgate.net/publication/386540281_Digital_Transformation_with_SAP_Hana

Kulkarni, V. N., & Kottursheṭtar, B. B. 2022. Product Lifecycle Management (PLM): A Key Enabler in the Implementation of Industry 4.0. Springer. [Referred 17 Mar 2025]. Available: https://link.springer.com/content/pdf/10.1007/978-3-030-84205-5_14.pdf.

Kulkarni, V. N., Gaitonde, V. N., & Kottursheṭtar, B. B. 2023. A Review on Industry 4.0 with PLM Technologies. AIP Conference Proceedings. [Referred 17 Mar 2025]. Available: https://www.researchgate.net/profile/Vinayak-Kulkarni-5/publication/372428228_A_review_on_industry_40_with_PLM_technologies/links/64c364b90fae1319bfbd4835/A-review-on-industry-40-with-PLM-technologies.pdf.

Kumar, S., Gopi, T., Harikeerthana, N. & Gupta, M.K. 2023. Machine Learning Techniques in Additive Manufacturing: A State of the Art Review on Design, Processes and Production Control. Journal of Intelligent Manufacturing. [Referred 20 Mar 2025]. Available: https://link.springer.com/article/10.1007/S10845-022-02029-5.

Kunduru, A.R. 2023. Effective Usage of Artificial Intelligence in Enterprise Resource Planning Applications. International Journal of Computer Trends and Technology. [Referred 26 Mar 2025]. Available: https://www.academia.edu/download/102928816/IJCTT_V71I4P109.pdf

La Torre, M., Dumay, J., Rea, M.A. and Abhayawansa, S. 2020. Robotic Process Automation and Consequences for Knowledge Workers: Beyond Technicalities. Journal of Knowledge Management, 24(7), pp.1671–1691. [Referred 29 Mar 2025]. Available: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7134300/

Lakkhongkha, K. 2024. Digital Logistics and Retail Automation with ERP, ML, and RPA Integration. [Referred 31 Mar 2025]. Available: https://so03.tci-thaijo.org/index.php/iba/article/download/273615/184390

LeCun, Y., Bengio, Y. & Hinton, G. 2015. Deep learning. Nature, 521(7553), 436–444. [Referred 26 Mar 2025]. Available: https://www.researchgate.net/publication/277411157_Deep_Learning

Leong, A.H.I., Chong, A.Y.L., Tan, B.L., Ooi, K.B. & Hew, T.S. 2024. Internationalization via Industry 4.0 technologies: application areas through nine-pillar lens. Socio-Economic Review. [Referred 15 Jun 2025]. Available: https://link.springer.com/article/10.1007/s11301-024-00459-4

Macron, T. 2025. The Future of AI in ERP: Emerging Trends and Innovations in the Next Decade. ResearchGate. [Referred 26 Mar 2025]. Available: https://www.researchgate.net/publication/389486024

Madakam, S., Holmukhe, R.M. and Jaiswal, D.K. 2019. The Future Digital Work Force: Robotic Process Automation (RPA). JISTEM - Journal of Information Systems and Technology Management, 16, pp.1–19. [Referred 29 Mar 2025]. Available: https://www.redalyc.org/journal/2032/203261541001/html/

Mahato, A. 2023. AI Based PDM in Manufacturing Industry 4.0: A Bibliographic Review. IJERT. [Referred 17 Mar 2025]. Available: https://www.academia.edu/download/114430541/ai_based_pdm_in_manufacturing_industry_40_a_bibliographic_review_IJERTV13IS040310.pdf.

Mathew, D., Brintha, N.C. & Jappes, J.T.W. 2023. Artificial Intelligence Powered Automation for Industry 4.0. New Horizons for Industry 4.0 in Digital Manufacturing. Springer. [Referred 20 Mar 2025].
Available: https://link.springer.com/chapter/10.1007/978-3-031-20443-2_1.

McKendry DA, Whitfield RI, Duffy AHB. 2021. Product lifecycle management implementation for high value engineering to order programmes: an informational perspective. Journal of Industrial Information Integration. [Referred 11 Sep 2025]. Available: https://doi.org/10.1016/j.jii.2021.100264

Metzker, Z. 2025. How Technology Affects the Use of ESG in SMEs in the V4 Countries. [Referred 31 Mar 2025]. Available: https://www.researchgate.net/publication/388679527

Mithas, S., Chen, Z. L., & Saldanha, T. J. V. 2022. How will artificial intelligence and Industry 4.0 emerging technologies transform operations management?. Production and

Operations Management. [Referred 12 Mar 2025]. Available: https://journals.sagepub.com/doi/abs/10.1111/poms.13864.

Mourtzis, D., Angelopoulos, J., & Panopoulos, N. 2022. A Literature Review of the Challenges and Opportunities of the Transition from Industry 4.0 to Society 5.0. Energies. [Referred 12 Mar 2025]. Available: https://www.mdpi.com/1996-1073/15/17/6276.

Mpofu, K. 2023. Blockchain-Based Product Lifecycle Management Using Supply Chain Management for Railcar Remanufacturing. Procedia CIRP. [Referred 17 Mar 2025]. Available: https://www.sciencedirect.com/science/article/pii/S2212827123000872.

Müller, D., Chiodo, M. & Sienknecht, M. 2025. Integrators at War: Mediating in AI-assisted Resort-to-Force Decisions. [Referred 31 Mar 2025]. Available: https://arxiv.org/abs/2501.06861

Munoz, C., da Costa, K., Modenesi, B. & Koshiyama, A. 2023. Evaluating Explainability in Machine Learning Predictions through Explainer-Agnostic Metrics. [Referred 31 Mar 2025]. Available: https://arxiv.org/abs/2302.12094

Muthusamy, V., Unuvar, M., Völzer, H. & Weisz, J.D. 2020. Do's and Don'ts for Human and Digital Worker Integration. [Referred 31 Mar 2025]. Available: https://arxiv.org/abs/2010.07738

Myerson, P. 2023. The Lean, Smart, Digital Supply Chain: How to Enable a Lean and Agile Global Supply Chain with the Help of Technology. [Referred 31 Mar 2025]. Available: https://www.taylorfrancis.com/books/mono/10.4324/9781003372639

Nalgozhina, N., Razaque, A., Raissa, U. and Yoo, J. 2023. Developing Robotic Process Automation to Efficiently Integrate Long-Term Business Process Management. Technologies, 11(6), p.164. [Referred 29 Mar 2025]. Available: https://www.mdpi.com/2227-7080/11/6/164

Nyathani, R., Allam, K., Joseph, S. & Engineer, B.I. 2024. Synergizing AI, Cloud Computing, and Big Data for Enhanced Enterprise Resource Planning (ERP) Systems. ResearchGate. [Referred 26 Mar 2025]. Available: https://www.researchgate.net/publication/377967651

Ojha, S., Narendra, A., Mohapatra, S. & Misra, I. 2023. From Robots to Books: An Introduction to Smart Applications of AI in Education (AIEd). [Referred 31 Mar 2025]. Available: https://arxiv.org/abs/2301.10026

Oleszek, S., & Santarek, K. 2022. Product Lifecycle Management (PLM) in the Context of Industry 4.0. Google Books. [Referred 17 Mar 2025]. Available: https://books.google.com/books?hl=en&lr=&id=tN5mEAAAQBAJ.

Olurin, J.O., Okonkwo, F. & Eleogu, T. 2024. Strategic HR Management in the Manufacturing Industry: Balancing Automation and Workforce Development. International Journal of Research. [Referred 20 Mar 2025]. Available: https://www.researchgate.net/profile/Nsisong-Eyo-Udo/publication/377351878_Strategic_HR_Management_in_the_Manufacturing_Industry_Balancing_Automation_and_Workforce_Development/links/67982b70207c0c20fa61159b/Strategic-HR-Management-in-the-Manufacturing-Industry-Balancing-Automation-and-Workforce-Development.pdf.

Owusu-Berko, L. 2024. Integrating AI-Driven Predictive and Prescriptive Analytics for Enhancing Strategic Decision-Making and Operational Efficiency Across Industries. [Referred 25 Feb 2025]. Available: https://www.researchgate.net/profile/Louis-Owusu-Berko-2/publication/389206374_INTEGRATING_AI-DRIVEN_PREDICTIVE_AND_PRESCRIPTIVE_ANALYTICS_FOR_ENHANCING_STRATEGIC_DECISION-MAKING_AND_OPERATIONAL_EFFICIENCY_ACROSS_INDUSTRIES/links/67b884a08311ce680c6f3e9b/INTEGRATING-AI-DRIVEN-PREDICTIVE-AND-PRESCRIPTIVE-ANALYTICS-FOR-ENHANCING-STRATEGIC-DECISION-MAKING-AND-OPERATIONAL-EFFICIENCY-ACROSS-INDUSTRIES.pdf.

Parthasarathy, S. & Padmapriya, S.T. 2023. Understanding Algorithm Bias in Artificial Intelligence-Enabled ERP Software Customization. Journal of Ethics in Entrepreneurship and Technology. [Referred 26 Mar 2025]. Available: https://www.emerald.com/insight/content/doi/10.1108/jeet-04-2023-0006/full/pdf

Patrício L, Varela L, Silveira Z. 2024. A Framework for Robotic Process Automation and Artificial Intelligence Integration in Business Processes. Applied Sciences. [Referred 27 Jun 2025]. Available: https://www.mdpi.com/2076-3417/14/21/9648

Patrício, L., Varela, L. & Silveira, Z. 2025. Implementation of a Sustainable Framework for Process Optimization Through the Integration of Robotic Process Automation and Big Data in the Evolution of Industry 4.0. Processes, 13(2):536. [Referred 15 Jun 2025]. Available: https://doi.org/10.3390/pr13020536

Paul, C. 2022. The Role of Conversational AI in Automating ERP Workflows and Enhancing Efficiency. ResearchGate. [Referred 26 Mar 2025]. Available: https://www.researchgate.net/publication/387673240

Penaranda N, Mejía-Gutierrez R, Romero D, Molina A. 2010. Implementation of Product Lifecycle Management Tools using Enterprise Integration Engineering and Action-Research. International Journal of Computer Integrated Manufacturing. [Referred 24 Mar 2025]. Available: https://www.researchgate.net/publication/220381551_Implementation_of_Product_Lifecycle_Management_Tools_using_Enterprise_Integration_Engineering_and_Action-Research

Percherla, A.K. 2024. Effortless Logistics Automation – SAP Logistics Process Automation with UiPath RPA. Journal of Artificial Intelligence & Cloud Computing 3(1). [Referred 29 Mar 2025]. Available: https://www.onlinescientificresearch.com/articles/effortless-logistics-automation--sap-logistics-process-automation-with-uipath-rpa.html

Petkovic, D. 2022. It is not "accuracy vs. explainability" — we need both for trustworthy AI systems. [Referred 31 Mar 2025]. Available: https://arxiv.org/abs/2212.11136

Pinto, R. & Pirola, F. 2025. A Preliminary Multidimensional AI Readiness Assessment Model for SMEs. Procedia Computer Science. [Referred 26 Mar 2025]. Available: https://www.sciencedirect.com/science/article/pii/S1877050925001474

Plattfaut, R., Borghoff, V., Godefroid, M. & Koch, J. 2022. The Critical Success Factors for Robotic Process Automation. Computers in Industry. [Referred 20 Mar 2025]. Available: https://www.sciencedirect.com/science/article/pii/S0166361522000434.

Pokala, P. 2024. Leveraging ERP and AI for Business Transformation into Intelligent Enterprises. SSRN. [Referred 26 Mar 2025]. Available: https://papers.ssrn.com/sol3/Delivery.cfm?abstractid=5102243

Pokala, P. 2024. Leveraging ERP and AI for Business Transformation into Intelligent Enterprises. SSRN. [Referred 27 Mar 2025]. Available: https://papers.ssrn.com/sol3/Delivery.cfm?abstractid=5102243

Porsev, K., & Abramov, P. 2023. Product Data Management System as an Element of Information Support of the Standardization Service of a High-Tech Industrial Enterprise. AIP Conference Proceedings. [Referred 17 Mar 2025]. Available: https://pubs.aip.org/aip/acp/article-abstract/2700/1/040019/2880991.

PyInstaller 2025. Bundling to one-folder mode (onedir) creates a directory with the executable and all dependencies, enabling distribution without requiring Python installation. [Referred 25 Jun 2025]. Available: https://pyinstaller.org/en/stable/operating-mode.html

Radke, H. 2020. Using robotic process automation (RPA) to enhance item master data maintenance process: benefits include improved data quality, error reduction, and time savings. *Logforum / Baztech*. [Referred 15 Jun 2025]. Available: https://yadda.icm.edu.pl/baztech/element/bwmeta1.element.baztech-2bd2b0dc-5717-4bc5-8840-669b3c19d8d7/c/Radke_1_2020.pdf

Rahman, M. 2023. Product Data Management Practices in a Bangladeshi Agrochemical Company. OuluRepo. [Referred 17 Mar 2025]. Available: https://oulurepo.oulu.fi/handle/10024/42421.

Rajput, A.S. & Gupta, R. 2023. Hyperautomation—The Next Peripheral for Automation in IT Industries. [Referred 31 Mar 2025]. Available: https://arxiv.org/abs/2305.11896

Raptis, T.P., Passarella, A. & Conti, M. 2019. Data Management in Industry 4.0: State of the Art and Open Challenges. arXiv preprint. [Referred 15 Jun 2025]. Available: https://arxiv.org/abs/1902.06141

Rautenstrauch, T. & Moser, S. 2023. Performance evaluation regarding the implementation of robotic process automation: A proposed maturity model. Business Performance Review, 1(2), 34–47. [Referred 15 Jun 2025]. Available: https://doi.org/10.22495/bprv1i2p3

Ravichandran, P. & Machireddy, J.R. 2023. Data Analytics Automation with AI: A Comparative Study of Traditional and Generative AI Approaches. Journal of Bioinformatics and Intelligent Computing. [Referred 20 Mar 2025]. Available: https://www.researchgate.net/profile/Jeshwanth-Reddy-

Machireddy/publication/389169288_Data_Analytics_Automation_with_AI_A_Comparative_Study_of_Traditional_and_Generative_AI_Approaches/links/67b75c71207c0c20fa8f5e0b/Data-Analytics-Automation-with-AI-A-Comparative-Study-of-Traditional-and-Generative-AI-Approaches.pdf.

Ren, S., Zhang, Y., Liu, Y., Sakao, T. & Huisingh, D. 2022. A comprehensive review of big data analytics throughout product lifecycles to support sustainable smart manufacturing: a framework, challenges and future research directions. Journal of Cleaner Production. [Referred 15 Jun 2025]. Available: https://www.researchgate.net/publication/328804993

Ribeiro, J., Lima, R., Eckhardt, T., & Paiva, S. 2021. Robotic Process Automation and Artificial Intelligence in Industry 4.0 — A Literature Review. Procedia Computer Science, 181: 51–58. [Referred 10 Sep 2025]. Available: https://doi.org/10.1016/j.procs.2021.01.104

Ryalat, M., ElMoaqet, H., & AlFaouri, M. 2023. Design of a smart factory based on cyber-physical systems and Internet of Things towards Industry 4.0. Applied Sciences. [Referred 12 Mar 2025]. Available: https://www.mdpi.com/2076-3417/13/4/2156.

Sadeeq, H. 2024. AI/ML-Driven Business Intelligence Strategies for IoT-Enabled Manufacturing with ERP Cloud Integration. ResearchGate. [Referred 27 Mar 2025]. Available: https://www.researchgate.net/publication/388681968

Sarferaz S. 2022. Compendium on Enterprise Resource Planning: Market, Functional and Conceptual View Based on SAP S/4HANA. Springer. [Referred 19 Mar 2025]. Available: https://link.springer.com/book/10.1007/978-3-030-93856-7

Sarferaz, S. 2024a. Embedding Artificial Intelligence into ERP Software. Springer. [Referred 27 Mar 2025]. Available: https://link.springer.com/book/10.1007/978-3-031-54249-7

Sarferaz, S. 2024b. ERP Reference Artificial Intelligence Technology. In: Embedding Artificial Intelligence into ERP Software. Springer. [Referred 27 Mar 2025]. Available: https://link.springer.com/chapter/10.1007/978-3-031-54249-7_5

Shaheen, B. W., & Németh, I. 2022. Integration of Maintenance Management System Functions with Industry 4.0 Technologies. MDPI. [Referred 17 Mar 2025]. Available: https://www.mdpi.com/2227-9717/10/11/2173.

Shaik M. 2023. SAP-ERP Software's Pivotal Role in Shaping Industry 4.0: Transforming the Future of Enterprise Operations. Computer Science and Engineering. [Referred 19 Mar 2025]. Available: https://www.researchgate.net/publication/376798710_SAP_-_ERP_Software%27s_Pivotal_Role_in_Shaping_Industry_40_Transforming_the_Future_of_Enterprise_Operations

Shaik, M. & Siddque, K.Q. 2023. Predictive Analytics in Supply Chain Management Using SAP and AI. Journal of Computer Sciences and Applications. [Referred 27 Mar 2025]. Available: https://www.researchgate.net/publication/376953951

Sharma, C. & Vaid, A. 2022. Leveraging SAP and Artificial Intelligence for Optimized Enterprise Resource Planning: Enhancing Efficiency, Automation, and Decision-Making. World Journal of Advanced Research and Reviews. [Referred 26 Mar 2025]. Available: https://www.researchgate.net/publication/386534481

Sharma, C., Sharma, R. & Sharma, K. 2024. The Convergence of Intelligent Systems and SAP Solutions: Shaping the Future of ERP. ResearchGate. [Referred 27 Mar 2025]. Available: https://www.researchgate.net/publication/388096794

Singh, V. 2025. AI-Driven ERP Evolution: Enhancing Supply Chain Resilience with Neural Networks and Predictive LSTM Models. [Referred 31 Mar 2025]. Available: https://www.researchgate.net/publication/389894071

Smith, J.J., Deng, W.H., Smith, W.H., Sap, M., DeCario, N. & Dodge, J. 2024. The Generative AI Ethics Playbook. [Referred 31 Mar 2025]. Available: https://arxiv.org/abs/2501.10383

Sola, T. & Gold, J. 2024. The Future of Blockchain in Insurance: Enhancing Security and Transparency. [Referred 25 Feb 2025]. Available: https://www.researchgate.net/profile/Adebis-Samuel/publication/389166399_The_Future_of_Blockchain_in_Insurance_Enhancing_Security_and_Transparency/links/67b72a5e461fb56424e21824/The-Future-of-Blockchain-in-Insurance-Enhancing-Security-and-Transparency.pdf.

Stark, J. 2022. Product Lifecycle Management (Volume 1): 21st Century Paradigm for Product Realization. Springer. [Referred 20 Mar 2025]. Available: https://link.springer.com/chapter/10.1007/978-3-030-98578-3_1.

Stark, R. 2022. Major Technology 5: Product Data Management and Bill of Materials—PDM/BOM. Springer. [Referred 17 Mar 2025]. Available: https://link.springer.com/chapter/10.1007/978-3-662-64301-3_11.

Stroet A. 2022. Design Science Research Framework for Performance Analysis. Electronics. [Referred 25.6.2025]. Available: https://www.mdpi.com/2079-9292/11/16/2504

Taghipour, A. 2024. Applications of New Technology in Operations and Supply Chain Management. [Referred 31 Mar 2025]. Available: https://books.google.com/books?id=PCIeEQAAQBAJ

Taghipour, A. 2024. Applications of New Technology in Operations and Supply Chain Management. Google Books. [Referred 27 Mar 2025]. Available: https://books.google.com/books?id=PCIeEQAAQBAJ

Tanase, G., Adina-Theodora, N. & Andreea, M. 2025. Adoption of Smart Technologies Through the Transition to a New ERP System Within Industry 4.0. [Referred 31 Mar 2025]. Available: https://www.utgjiu.ro/revista/ec/pdf/2025-01/23_Tanase.pdf

Tanpure, G., Yadav, V., Jain, R., & Soni, G. 2023. Framework for Cloud Product Lifecycle Management System: A Case Study of an Automotive Industry. Business Process Management Journal. [Referred 17 Mar 2025]. Available: https://www.emerald.com/insight/content/doi/10.1108/BPMJ-10-2022-0527/full/html.

Taofeek, A., Liang, W., Hamzah, F. & Mary, B.J. 2025. Revolutionizing Business Efficiency: AI Solutions for Operational Optimization. [Referred 25 Feb 2025]. Available: https://www.researchgate.net/profile/Britney-Johnson-Mary/publication/389138431_Revolutionizing_Business_Efficiency_AI_Solutions_for_Operational_Optimization/links/67b69bcc96e7fb48b9c7e2c4/Revolutionizing-Business-Efficiency-AI-Solutions-for-Operational-Optimization.pdf

Vaka D.K. 2024. The SAP S/4HANA Migration Roadmap: From Planning to Execution. Journal of Scientific and Engineering Research. [Referred 19 Mar 2025]. Available: https://www.researchgate.net/publication/382062824_The_SAP_S4HANA_Migration_Roadmap_From_Planning_to_Execution

Vaka, D.K. 2024. SAP S/4HANA: Revolutionizing Supply Chains with Best Implementation Practices. Google Books. [Referred 27 Mar 2025]. Available: https://books.google.com/books?id=IKoPEQAAQBAJ

van der Aalst, W., Bichler, M. & Heinzl, A. 2018. Robotic Process Automation. Business & Information Systems Engineering, 60(4), 269–272. [Referred 18 Jun 2025]. Available: https://www.researchgate.net/publication/325129640_Robotic_Process_Automation

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł. & Polosukhin, I. 2017. Attention is all you need. Advances in Neural Information Processing Systems, 30. [Referred 26 Mar 2025]. Available: https://arxiv.org/abs/1706.03762

Verified Market Research. 2024. Product Data Management (PDM) Software Market Size And Forecast. [Referred 8 May 2025]. Available: https://www.verifiedmarketresearch.com/product/product-data-management-pdm-software-market/

Viswanathan J. 2024. Transforming ERP Transactions Using SAP and Robotic Process Automation (RPA). International Journal of Computer Sciences and Engineering 12(9), 18–24. [Referred 29 Mar 2025]. Available: https://www.academia.edu/124384853/Transforming_ERP_Transactions_Using_SAP_And_Robotic_Process_Automation_RPA

Wahlster, W. 2022. Ten years of Industrie 4.0. Science. [Referred 12 Mar 2025]. Available: https://www.mdpi.com/2413-4155/4/3/26.

Wang, J., Zhang, S., Yin, Y., Zhao, X. & Chen, X. 2018. Industrial Big Data Analytics: Challenges, Methodologies, and Applications. arXiv preprint. [Referred 15 Jun 2025]. Available: https://arxiv.org/abs/1807.01016

Willcocks, L. & Lacity, M. 2016. Robotic Process Automation: The Next Transformation Lever for Shared Services. Journal of Information Technology Teaching Cases, 6(2), 49–57. [Referred 26 Mar 2025]. Available: https://www.umsl.edu/~lacitym/OUWP1601.pdf

Willcocks, L., Lacity, M.C. & Craig, A. 2022. Becoming Strategic with Robotic Process Automation: Stories and Lessons from Experience. [Referred 31 Mar 2025]. Available: https://link.springer.com/book/10.1007/978-3-030-95372-9

Wu, J., You, H. & Du, J. 2025. AI Generations: From AI 1.0 to AI 4.0. [Referred 31 Mar 2025]. Available: https://arxiv.org/abs/2502.11312

Yang, Q. & Parasuraman, R. 2023. A Hierarchical Game-Theoretic Decision-Making for Cooperative Multi-Agent Systems. [Referred 31 Mar 2025]. Available: https://arxiv.org/abs/2303.16641

Zhang X. Ming X. & Tong Q. 2023. Construction of Sustainable Digital Factory for Automated Warehouse Based on Integration of ERP and WMS. MDPI Sustainability. [Referred 18 Mar 2025]. Available: https://www.mdpi.com/2071-1050/15/2/1022/pdf

Zhang, Y., Ma, X., Pang, J., Xing, H., & Wang, J. 2023. The impact of digital transformation of manufacturing on corporate performance—The mediating effect of business model innovation and the moderating effect of strategy. Journal of International Business Studies. [Referred 12 Mar 2025]. Available: https://www.sciencedirect.com/science/article/pii/S0275531923000168.